
DESIGN REPORT

Course: Bridge Theory and Design

Topic: Design of a Prestressed Box Girder Bridge

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Part A: Bridge Characteristics

1. Introduction

This technical report is undertaken as a part of the course of Bridge Theory and Design. The problem of study comprises design of single cell box-girder bridge with the length of 40 m. Based on the length of the bridge, the structure is considered as long-span. The bridge is designed against self-weight and traffic loads, and it is a double lane on a single carriageway type of bridge. The height of the cross-section of the box-girder is 2.2 m. The mid-span cross-section is shown below for better visual understanding of the problem. The design is provided by adopting a simply supported static scheme.

General parameters of bridge girder		
Parameter	Value	Unit
Span length, l	40	m
Cross-sectional height, h	2.2	m
Exposure class	XD3	-
Vehicle class	C	-

Table 1-1: General parameters of bridge girder.

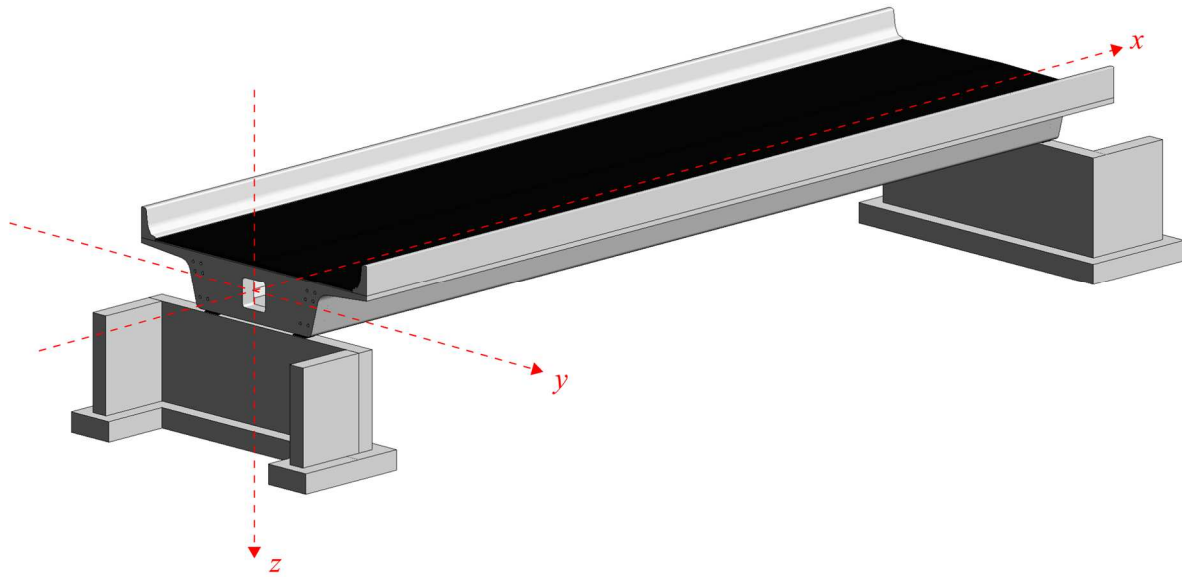


Figure 1-1: Isometric view of bridge girder and global frame of reference.

This part focuses on the description of pre-dimensioning of the cross section, necessary codes and standards that are followed, and definition of material and loading conditions acting on the structure.

2. Bridge description

The preliminary design of the box-girder cross section is carried out by referring to Figure 2-1, which is an illustration taken from [1], which shows recommended values for the geometry of a box girder.

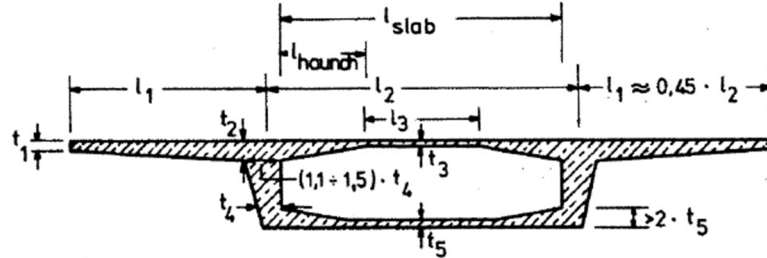


Figure 2-1: Reference design aid [1].

The recommended as well as the adopted values for the given project are summarized in the following table.

Parameter	Recommendations	Adopted value	Unit	
l_1	$0.45 \cdot l_2$	2500	mm	
l_2		7000	mm	
l_3		6000	mm	
$\frac{l_{haunch}}{l_{slab}}$	≤ 0.2	0.07	-	
$\frac{t_1}{t_2}$	1:2 to 1:3	0.5	-	
Deck Slab	t_3	$t_3 > \frac{l_3}{30}$	250	mm
	t_1	$\geq 200 \text{ mm}$	200	mm
	t_2	$\geq 300 \text{ mm}$	400	mm
Lateral webs	t_4	$\geq 300 \text{ mm}$	550	mm
Bottom slab	t_5	$\geq 150 \text{ mm}$	250	mm
Web inclination	4:1 to 3:1	4:1	-	

Table 2-1: Section geometry.

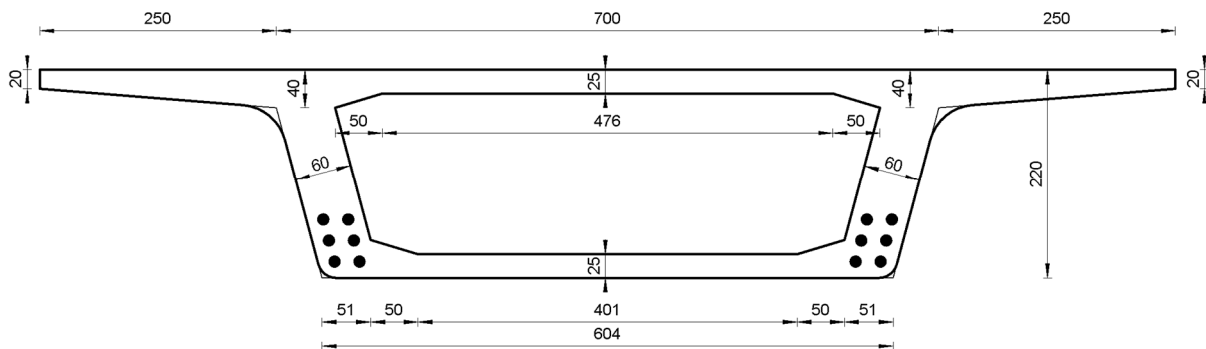


Figure 2-2: Adopted geometry.

3. Codes

Following regulations are adopted to design the box-girder bridge:

- EN1990 Eurocode 0: Basis of structural design
- EN1991-2: Eurocode 1: Actions on structures - Part 2: Traffic loads on bridges
- EN1992-1: Eurocode 2: Design of concrete structures - General rules
- EN1992-2: Eurocode 2: Concrete bridges - Design and detailing rules
- NTC18: Norme tecniche per le costruzioni

Additionally, the following documentation is used to determine post-tensioning cables.

- TENSA, post-tensioning catalogue

4. Materials

In the following section, the safety factors according to EN1990 are used and shown in Table .

Safety factor	Value
Concrete, γ_C	1,5
Steel, γ_S	1,15

Table 4-1: Safety factors for materials.

4.1 Concrete

The compressive strength of the concrete is denoted by concrete strength classes which relate to the characteristic cylinder strength f_{ck} and the recommended minimum strength class for concrete is included in EN206-1 depending on exposure class. The specification of designed concrete to resist identified exposure classes is given in Table A.4 of BS EN206-1. For exposure class of XD3, the recommended value for concrete is chosen as C45/55. Other parameters are found by following Eurocode 2, Table 3.1.

Characteristic strengths

$$\begin{aligned}
 f_{ck} &= 45 \text{ MPa} \\
 f_{ctm} &= 0.3 \cdot f_{ck}^{\frac{2}{3}} = 3.8 \text{ MPa} \\
 f_{ctk,0.05} &= 0.7 \cdot f_{ctm} = 2.7 \text{ MPa} \\
 E_{cm} &= 22 \cdot \left[\frac{f_{cm}}{10} \right]^{0,3} = 36 \text{ GPa} = 36000 \text{ MPa}
 \end{aligned}$$

Design tensile strength

$$f_{ctd} = \alpha_{ct} \cdot \frac{f_{ctk,0.05}}{\gamma_C} = 1 \cdot \frac{2.7 \text{ MPa}}{1,5} = 1,8 \text{ MPa}, \quad \text{where } \alpha_{ct} = 1 \text{ (Recommended in EN1990)}$$

Concrete cover

According to EN1992-1, the concrete cover is defined as a minimum cover plus an allowance in design for deviation.

$$c_{nom} = c_{min} + \Delta c_{dev}$$

The minimum concrete cover c_{min} shall be provided to ensure the safe transmission of bond forces, the protection of the steel against corrosion and an adequate fire resistance. To determine the concrete cover, it needs to be determined the structural class first, which is done according to Table 4.3N of Eurocode 2. The structure has a design working life of 100 years and exposure class of XD3, thus the minimum cover for the bridge is selected as:

$$c_{min} = 45 \text{ mm}$$

Concrete		
Parameter	Value	Unit
γ	25	kN/m ³
f_{ck}	45	MPa
f_{ctm}	3.8	MPa
$f_{ctk,0.05}$	2.7	MPa
f_{cd}	20	MPa
f_{ctd}	1.8	MPa
f_{bd}	2.835	MPa
E_{cm}	36000	MPa
ε_{cu3}	3.5	‰

Table 4-2: Concrete, properties.

4.2 Post-tensioning steel

All the given properties of the post-tension cables are referring to Tensa's product catalogue.

Tensa Multi-strand Tendons		
Parameter	Value	Unit
Tendon type	Y1860S7	-
Diameter of tendon, d	15.7	mm
Cross sectional area, A_s	150	mm ²
Ultimate strength, f_{pk}	1860	MPa
Characteristic yield strength, $f_{p,0.1,k}$	1640	MPa
Design yield strength	1426	MPa
Characteristic value of maximum force, F_{pk}	279	kN
Maximum value of maximum force, $F_{m,max}$	329	kN

Tendons with 27 strands		
No. of tendons	12	-
Area per tendon, A_p	4050	mm ²
Diameter of tendon, d_p	71.81	mm
Diameter of duct, d_{duct}	115	mm
Ultimate force per tendon	7533	kN
Distance of centroid of tendons from bottom edge	360	mm

Table 4-3: Post tensioning steel, properties

4.3 Ordinary reinforcement steel

The steel used for ordinary reinforcement is B450C. Its properties are given below:

Ordinary reinforcement steel		
Parameter	Value	Unit
Characteristic yield strength, f_{yk}	450	MPa
Design yield strength, f_{yd}	391,3	MPa
Young's modulus, E	200000	MPa

Table 4-4: Ordinary reinforcement steel, properties.

4.3.1 Anchorage length for ordinary reinforcement

Anchorage length is the minimum length of the reinforcing bar that must be incorporated into the structural element and must be sufficient to facilitate the failure in bond due to the tensile forces can be prevented. The bond ensures the transfer of shear stresses between two materials in each cross section.

General definition is given in EN1992-1, 8.4.1. Reinforcing bars shall be anchored such that the bond forces are safely transmitted to the concrete avoiding longitudinal cracking or spalling.

Ultimate bond stresses

The ultimate bond strength should be sufficient to prevent bond failure. The design value of the ultimate bond stress, f_{bd} , is computed as follows:

$$f_{bd} = 2,25 \cdot \eta_1 \cdot \eta_2 \cdot f_{ctd} = 3.38 \text{ MPa}$$

Where:

f_{ctd} is design value of concrete tensile strength that is computed in **Section 4.1**.

η_1 is a coefficient related to the quality of the bond condition and the position of the bard during concreting:

$\eta_1 = 1,0$ (good conditions)

$\eta_1 = 0,7$ (all other conditions)

η_2 is related to the bar diameter.

$\eta_2 = 1,0$ for $\phi \leq 32 \text{ mm}$

$\eta_2 = \frac{132 - \phi}{100}$ for $\phi \leq 32 \text{ mm}$

Material parameters are given in the tabular form:

Parameter	f_{yk}	f_{ctk}	f_{cd}	f_{ctd}	f_{bd}	
Value	500	45	30	1.5	3.38	MPa

Table 4-5: Material Properties

Basic anchorage length

The calculation of the required anchorage length considers the type of steel and bond properties of the bars.

The basic required anchorage length, $l_{b,rqd}$ for anchoring the force $A_s \cdot \sigma_{sd}$ is computed as follows:

$$l_{b,rqd} = \frac{\phi}{4} \cdot \frac{\sigma_{sd}}{f_{bd}}$$

Where:

σ_{sd} is the design stress of the bar.

Design anchorage length

The design anchorage length, l_{bd} , is defined as:

$$l_{bd} = \alpha_1 \cdot \alpha_2 \cdot \alpha_3 \cdot \alpha_4 \cdot \alpha_5 \cdot l_{b,rqd} \geq l_{b,min}$$

Where:

α_1 is for the effect of the form of the bars assuming adequate cover (EN1992.1.1.2004 Figure 8.1)

α_2 is for the effect of concrete minimum cover (EN1992.1.1.2004 Figure 8.3)

α_3 is for the effect of confinement by transverse reinforcement.

α_4 is for the influence of one or more welded transverse bars along the design anchorage length.

α_5 is for the effect of the pressure transverse to the plane of splitting along the design anchorage length.

Minimum anchorage length is defined as:

- For anchorages in tension: $l_{b,min} \geq \max \{0,3 \cdot l_{bd,rqd}; 10 \cdot \emptyset; 100 \text{ mm}\}$
- For anchorages in compression: $l_{b,min} \geq \max \{0,6 \cdot l_{bd,rqd}; 10 \cdot \emptyset; 100 \text{ mm}\}$

Lap length is defined as:

$$l_0 = \alpha_1 \cdot \alpha_2 \cdot \alpha_3 \cdot \alpha_5 \cdot \alpha_6 \cdot l_{b,rqd} \geq l_{0,min}$$

Where:

$$l_{0,min} \geq \max \{0,3 \cdot \alpha_6 \cdot l_{b,req}; 15 \cdot \emptyset; 200 \text{ mm}\}$$

α_6 is defined EN1992.1.1.2004 Table 8.3

These coefficients are derived from EN1992.1 Table 8.2 and put in a tabular form as follows.

- a) Straight bars
- b) Others

Design Anchorage Length		
$\emptyset$	$l_{b,req}$	$l_{b,min}$
[mm]	[mm]	[mm]
8	258	100
10	322	100
12	386	120
16	515	160
20	644	200
22	709	220
25	805	250
32	1031	320

Table 4-6: Design anchorage length.

Case a: Straight bars									
$\emptyset$	a_1	a_2	α_3	a_4	a_5	α_6	l_{bd}	l_0	$l_{0,min}$
[mm]	[-]	[-]	[-]	[-]	[-]	[-]	[mm]	[mm]	[mm]
8	1	0.70	1	0.7	1	1.5	126	271	200
10	1	0.70	1	0.7	1	1.5	158	338	200
12	1	0.70	1	0.7	1	1.5	189	406	200
16	1	0.73	1	0.7	1	1.5	263	563	240
20	1	0.81	1	0.7	1	1.5	366	785	300
22	1	0.84	1	0.7	1	1.5	418	896	330
25	1	0.88	1	0.7	1	1.5	496	1063	375
32	1	0.94	1	0.7	1	1.5	677	1452	480
40	1	0.98	1	0.7	1	1.5	885	1896	600

Table 4-7: Design parameters and anchorage length for straight bars.

Case b: Straight bars								
$\varnothing$ [mm]	a_1 [-]	a_2 [-]	α_3 [-]	a_4 [-]	a_5 [-]	α_6 [-]	l_{bd} [mm]	l_0 [mm]
8	1	0.70	1	0.7	1	1.5	126	271
10	1	0.78	1	0.7	1	1.5	175	374
12	1	0.89	1	0.7	1	1.5	240	514
16	0.7	1	1	0.7	1	1.5	252	541
20	0.7	1	1	0.7	1	1.5	316	676
22	0.7	1	1	0.7	1	1.5	347	744
25	0.7	1	1	0.7	1	1.5	395	845
32	0.7	1	1	0.7	1	1.5	505	1082
40	0.7	1	1	0.7	1	1.5	631	1353

Table 4-8: Design parameters and anchorage length for others.

5. Loads and load combinations.

The relevant traffic actions and other specific actions on bridges should be classified in accordance with EN 1990, section 4. For this project, the structure is analysed considering the following loads:

- Dead Loads (Structural & non-structural)
- Traffic Loads (Live loads)

5.1 Dead loads

5.1.1 Structural dead load

The structural dead load is computed as follows:

$$g_{k,1} = \gamma_c \cdot \frac{V}{b \cdot l} = 25 \frac{kN}{m^3} \cdot \frac{279 \text{ m}^3}{40 \text{ m} \cdot 12 \text{ m}} = 14,53 \frac{kN}{m^2}$$

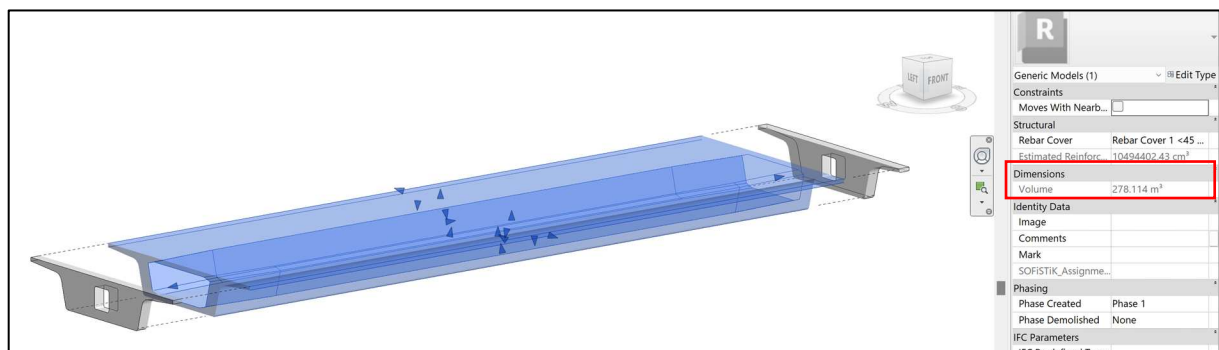


Figure 5-1: Volume of bridge girder.

5.1.2 Non-structural dead load

The non-structural dead load of the structure is computed as follows by considering the materials concrete, bitumen, and curbs.

$$\begin{aligned}
 g_{k,2} &= \gamma_{bitumen} \cdot t_{bitumen} + \gamma_c \cdot t_c + \gamma_c \cdot \frac{V_{kerbs}}{b \cdot l} \\
 &= 11 \frac{kN}{m^3} \cdot 0,06 \text{ m} + 25 \frac{kN}{m^3} \cdot 0,05 \text{ m} + 25 \frac{kN}{m^3} \cdot \frac{7,54 \text{ m}^3}{12 \text{ m} \cdot 40 \text{ m}} \\
 &= 2,3 \frac{kN}{m^2}
 \end{aligned}$$

Where:

$\gamma_{bitumen} = 11 \frac{kN}{m^3}$	Specific weight of bitumen
$t_{bitumen} = 0.06 \text{ m}$	Thickness of bitumen layer
$t_{concrete} = 0.05 \text{ m}$	Thickness of concrete layer
$V_{kerbs} = 7.543 \text{ m}^3$	Volume of kerbs

5.2 Live loads

The bridge is subjected to a uniformly distributed load (q) and a tandem system (Q) and is defined by Load Model 1 as suggested by EN1991.2.2003. The definition of the traffic actions on the bridge is determined by conventional lane model. The width of the conventional lane on a carriageway is defined as the maximum number of lanes that can fit in the carriageway. The distribution of conventional lanes is illustrated in the following figure.

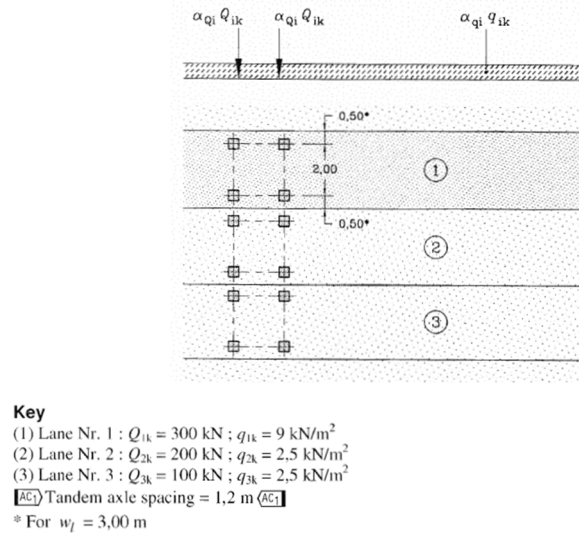


Figure 4.2a - Application of load Model 1

Figure 5-2: Notional lanes.

$w = 10.5 \text{ m}$	Total width of the carriageway
$L_{lane} = 3 \text{ m}$	Conventional lanes width
$L_r = 1,5 \text{ m}$	Remaining area

Load model 1 consists of uniformly distributed loads and concentrated loads (TS) in each lane. The system is given as follows:

- a) Double-axle concentrated loads (tandem system), each axle having the following weight:

$$Q = \alpha_q \cdot Q_k$$

Where:

α_q are adjustment factors.

- b) Uniformly distributed loads (UDL), having the following weight per square meter of notional lane:

$$q = \alpha_Q \cdot q_k$$

Where:

α_q are adjustment factors.

- c) On the remaining areas, the load magnitude is referred to as $\alpha_{qr} \cdot q_{rk}$.
- d) The characteristic values of Q_{ik} and q_{ik} should be taken from EN1991.2.2003 Table 4.2, shown below.

Location	Tandem system <i>TS</i>	<i>UDL</i> system
	Axle loads Q_{ik} (kN)	$\overline{AC_1} q_{ik}$ (or q_{ik}) (kN/m ²) $\overline{AC_1}$
Lane Number 1	300	9
Lane Number 2	200	2,5
Lane Number 3	100	2,5
Other lanes	0	2,5
Remaining area (q_{ik})	0	2,5

Table 5-1: EN1991.2.2003 Table 4.2

Hence, the axle loads (Q_{ik}) and the distributed loads q_{ik} are selected as follows:

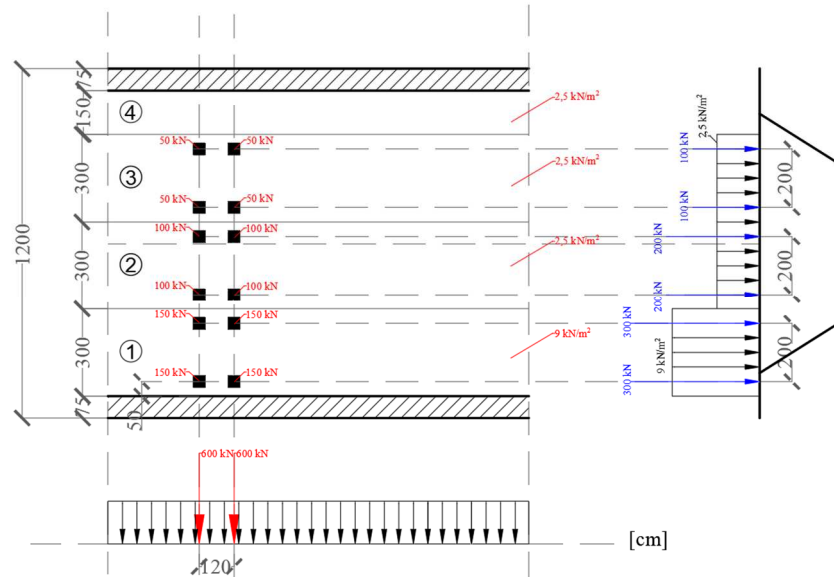


Figure 5-3: Scheme for the axle loads and distributed loads.

5.3 Load combinations

5.3.1 Combinations of actions: Ultimate Limit State

After defining the actions on the structure, load combination definitions are necessary to obtain the design loads to be used in the bridge design. The fundamental ULS combination is given as:

$$E_d = \gamma_{Sd} \cdot E \cdot \left\{ \gamma_{g,j} \cdot G_{k,j} ; \gamma_P \cdot P ; \gamma_{q,1} \cdot Q_{k,1} ; \gamma_{q,i} \cdot \psi_{0,i} \cdot Q_{k,i} \right\} \quad j \geq 1 ; i \geq 1$$

$$\sum_{j \geq 1} \gamma_{G,j} \cdot G_{k,j} + \gamma_P \cdot P + \gamma_{Q1} \cdot \psi_{01} \cdot Q_{k1} + \sum_{i > 1} \gamma_{Qi} \cdot \psi_{0i} \cdot Q_{ki}$$

$$E_d = \gamma_{G1} \cdot G_1 + \gamma_{G2} \cdot G_2 + \gamma_P \cdot P + \gamma_{Q1} \cdot Q_{k1} + \gamma_{Q2} \cdot \psi_{02} \cdot Q_{k2} + \gamma_{Q3} \cdot \psi_{03} \cdot Q_{k3} + \dots$$

The partial safety factors are provided in EN1990-Annex A Table A1.2. For this project, it has been selected as following:

$$\gamma_{G,j,sup} = 1.35 \text{ (unfavourable)}$$

$$\gamma_{Q,i} = 1.5 \text{ (unfavourable)}$$

5.3.2 Combinations of actions: Serviceability Limit State

The combinations of actions to be considered in the relevant design situations should be appropriate for the serviceability requirements and performance criteria being verified. The combinations of actions for SLS are defined symbolically in EN1990. The following three combinations are defined to check the structure during its service life.

a) *Characteristic combination:*

It is related to irreversible limit states.

$$E_d = E \cdot \left\{ G_{k,j} ; P ; Q_{k,1} ; \psi_{0,i} \cdot Q_{k,i} \right\}$$

$$\sum_{j \geq 1} G_{k,j} + P + Q_{k,1} + \sum_{i > 1} \psi_{0i} \cdot Q_{k,i}$$

$$E_d = G_1 + G_2 + P + Q_{k,1} + \psi_{02} \cdot Q_{k2} + \psi_{03} \cdot Q_{k3} + \dots$$

b) *Frequent combination:*

It is related to reversible limit states.

$$E_d = E \cdot \left\{ G_{k,j} ; P ; \psi_{1,1} \cdot Q_{k,1} ; \psi_{2,i} \cdot Q_{k,i} \right\}$$

$$\sum_{j \geq 1} G_{k,j} + P + \psi_{1,1} \cdot Q_{k,1} + \sum_{i > 1} \psi_{2i} \cdot Q_{k,i}$$

$$E_d = G_1 + G_2 + P + \psi_{11} \cdot Q_{k1} + \psi_{22} \cdot Q_{k2} + \psi_{23} \cdot Q_{k3} + \dots$$

c) *Quasi-permanent combination*

It is used for long-term effects and the appearance of the structure.

$$E_d = E \cdot \left\{ G_{k,j} ; P ; \psi_{2,i} \cdot Q_{k,i} \right\}$$

$$\sum_{j \geq 1} G_{k,j} + P + \sum_{i > 1} \psi_{2i} \cdot Q_{k,i}$$

$$E_d = G_1 + G_2 + P + \psi_{21} \cdot Q_{k1} + \psi_{22} \cdot Q_{k2} + \psi_{23} \cdot Q_{k3} + \dots$$

Recommended values of ψ factors for road bridges are taken from EN1990 Table A2.1

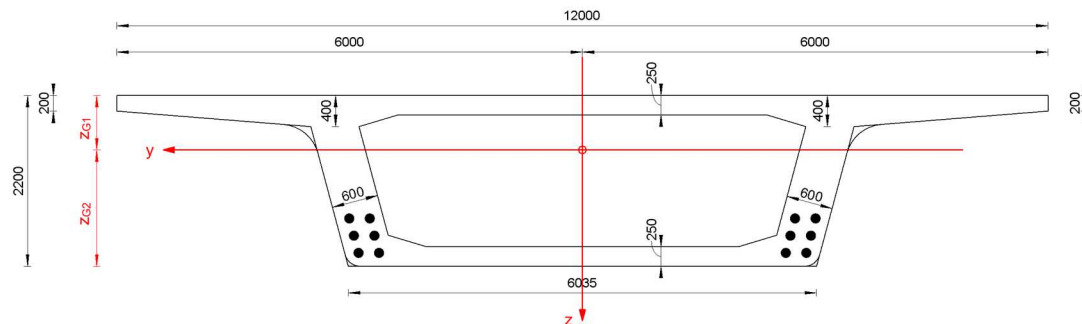
Part B: Girder design (Longitudinal behaviour)

6. First draft

The following sections deal with the overall scheme and preliminary computations for the longitudinal analysis of the bridge girder. It includes description of the scheme and loads, computation of the internal forces and an estimation of the required pre-stress. Lastly, the girder will be fully verified in ULS and SLS in both flexure and shear.

6.1 Cross-section

The cross-section of the bridge girder is shown in Figure 6-1. The cross-sectional parameters have been computed by Autodesk Robot, for which a report is seen in Appendix C. In SLS, homogenized parameters are computed, see section 9.2.1. Below are the values reported for the base concrete section:



Section	Parameters			
	z_{G1} [mm]	z_{G2} [mm]	A [mm ²]	I_y [mm ⁴]
Concrete section	866,6	1333,4	$7,03 \cdot 10^6$	$4,7 \cdot 10^{12}$

Figure 6-1: Cross-section properties

6.2 Structural scheme

For the longitudinal analysis of the bridge girder, we are referring to the simply supported scheme shown in Figure 6-2.

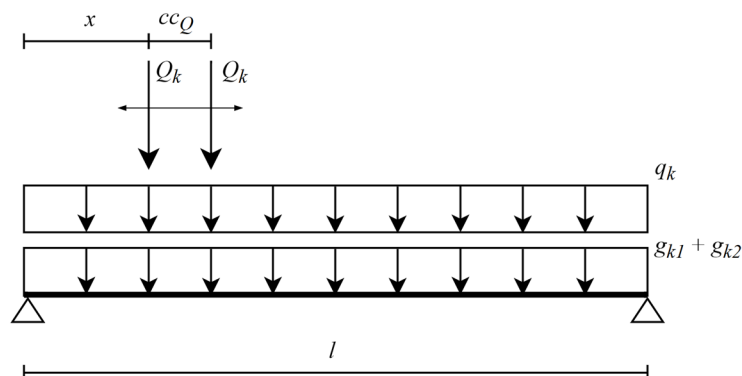


Figure 6-2: Longitudinal scheme of bridge

The geometric properties are reported below:

$$l = 40 \text{ m}, \quad cc_Q = 1,2 \text{ m}, \quad x \in [0, l]$$

The set of concentrated axle loads are moving loads and are spaced according to EN1991-2. For the verification of a specific section at a section at distance x , the moving loads are assumed to enforce the most critical effect when placed with the leftmost load at $x = x_Q$.

6.2.1 Loads and notional lanes.

The scheme refers to a distribution of notional lanes according to **Figure 6-3**. The surface loads and concentrated loads from section 5 are for the longitudinal scheme given in **Table 6-1**.

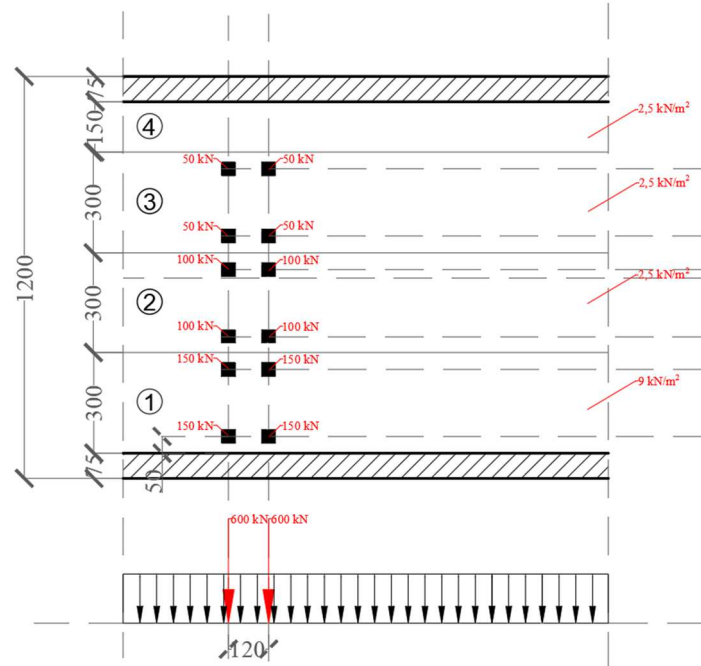


Figure 6-3: Notional lanes in longitudinal scheme

Line loads				
Type	Characteristic value [kN/m²]	Tributary width [m]	UDL, g_k, q_k [kN/m]	Safety factor, γ
Structural DL	14,53	12	174,4	1,35
Non-structural DL	2,45	12	29,5	1,35
LL, notional lane 1	9,5	3	28,5	1,5
LL, notional lane 2-4	2,5	7,5	18,75	1,5
Concentrated loads				
Set of axles	Characteristic value, Q_k [kN]		Safety factor, γ	
Left set	600 (= 2 · 150 + 2 · 100 + 2 · 50)		1,5	
Right set	600 (= 2 · 150 + 2 · 100 + 2 · 50)		1,5	

Table 6-1: Loads in longitudinal scheme.

6.3 Internal forces

6.3.1 Shear force

To compute the shear force diagrams, we computed the shear diagrams for each load case according to section 5 combined them according to the load combination given in section 5. For the uniformly distributed loads, the shear diagrams are computed simply by the formulas:

$$V_{g1}(x) = \frac{1}{2}g_{k1}l - g_{k1}x, \quad x \in [0, l]$$

$$V_{g2}(x) = \frac{1}{2}g_{k2}l - g_{k2}x, \quad x \in [0, l]$$

$$V_q(x) = \frac{1}{2}q_k l - q_k x, \quad x \in [0, l]$$

For the concentrated the loads, in each section $x \in [0, l]$, the maximum shear force is computed referring to the following scheme:

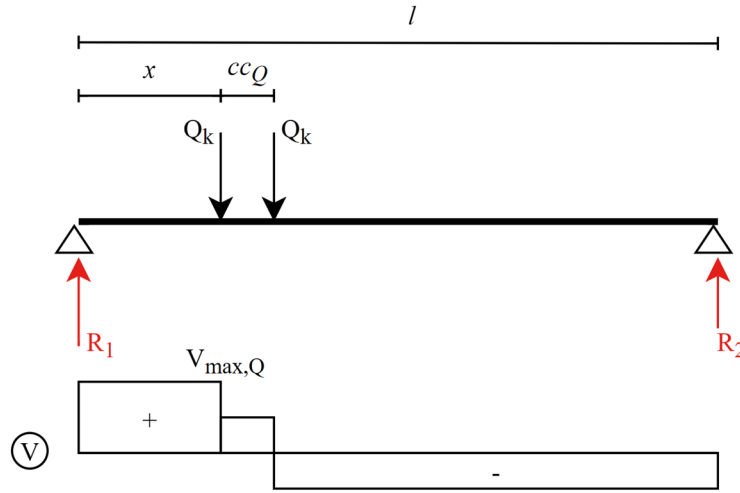


Figure 6-4: Scheme for computation of max. shear in each section due to concentrated forces

Applying rotational equilibrium at the right end point:

$$R_1 \cdot l - Q_k(l - x) - Q_k(l - x - cc_Q) = 0 \Rightarrow$$

$$R_1 = \frac{Q_k((l - x) + (l - x - cc_Q))}{l} = \frac{Q_k}{l} \cdot (2l - 2x - cc_Q)$$

$$V_{Q,max} = R_1 = \frac{Q_k}{l} \cdot (2l - 2x - cc_Q)$$

Finally, the design shear force diagram in ULS is given by:

$$V_{Ed}(x) = 1,35 \cdot (V_{g1} + V_{g2}) + 1,5 \cdot (V_q + V_{max,Q})$$

The shear diagrams are shown below for which all the values can be found in Appendix D.

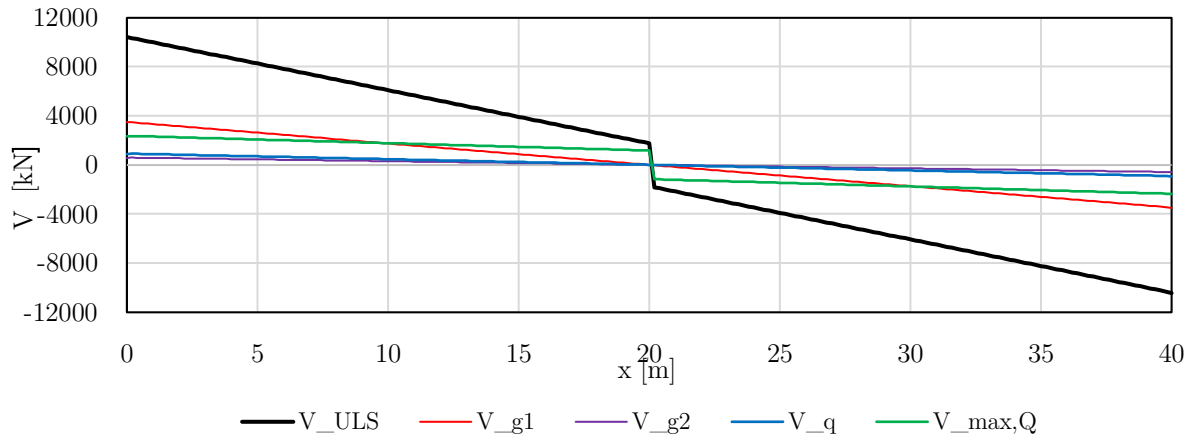


Figure 6-5: Shear diagrams, longitudinal scheme.

6.3.2 Bending moment

The bending moment diagrams are computed for the same load cases as for the shear, and have been performed according to the following formulas:

$$\begin{aligned}
 M_{g1}(x) &= -\frac{1}{2}g_{k1}x^2 + \frac{1}{2}g_{k1}lx \\
 M_{g2}(x) &= -\frac{1}{2}g_{k2}x^2 + \frac{1}{2}g_{k2}lx \\
 M_q(x) &= -\frac{1}{2}q_kx^2 + \frac{1}{2}q_klx \\
 M_{max,Q}(x) &= V_{Q,max} \cdot x
 \end{aligned}$$

The design bending moment diagram is given similarly as the shear diagram:

$$M_{Ed}(x) = 1,35 \cdot (M_{g1} + M_{g2}) + 1,5 \cdot (M_q + M_{max,Q})$$

The moment diagrams are shown below for which all the values can be found in Appendix D.

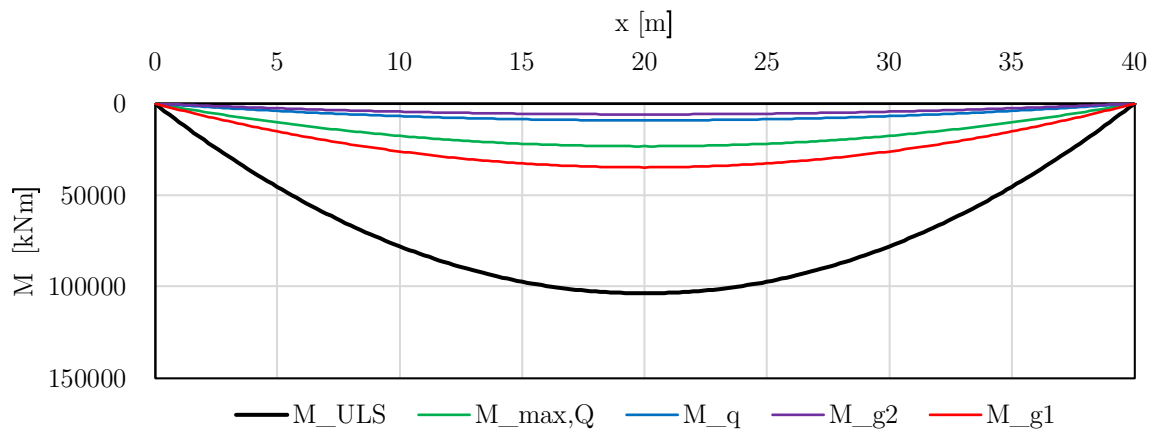


Figure 6-6: Moment diagrams, longitudinal scheme.

6.4 Estimation of required post-tension force.

To estimate the required post-tension force, we consider the situation in which the leftmost axle load is placed at $x_Q = \frac{l}{2} = 20 \text{ m}$. The maximum design bending moment, M_{Ed} , in the mid-span of the girder can be evaluated according to the formulas given in **section 6.3**. The maximum bending moment in the mid-section is evaluated to:

$$M_{Ed}(x = 20 \text{ m}) = 103686 \text{ kNm} = 103,7 \text{ MNm}$$

The internal lever arm can be estimated according to the lecture notes by the following formula:

$$z = h - \frac{3}{4}(t_3 + t_5) = 2,2 \text{ m} - \frac{3}{4}(0,25 \text{ m} + 0,25 \text{ m}) = 1.825 \text{ m}$$

Finally, the estimated total required post-tensioning force for the section is given by:

$$F_{sp,req} = \frac{M_{Ed}(x = 20 \text{ m})}{z} = \frac{103,7 \text{ MNm}}{1,825 \text{ m}} = \mathbf{56,8 \text{ MN}}$$

6.5 Preliminary verification of post-tension

The cable force of $F_{sp} = 57 \text{ MN}$ was chosen and verified through a similar procedure described in **section 9.2**. The only difference is that at this stage, we didn't have the actual cable force diagrams, so we assumed an instantaneous loss of 10% and a long-term loss of 15%. As seen in **section 8.3**, the actual losses for cable 1A are approx. 5% instantaneous and 19% long-term loss.

We needed to increase the pre-tension up to **59 MN**, to fulfil the preliminary verifications in SLS.

7. Cables layout

In this section, the definition of the cable paths will be described. The general principle governing the paths is that they are consisting of straight and parabolic segments. On **Figure 7-1**, the system of reference employed, (y^*, z^*) , and the naming of the cables are shown.

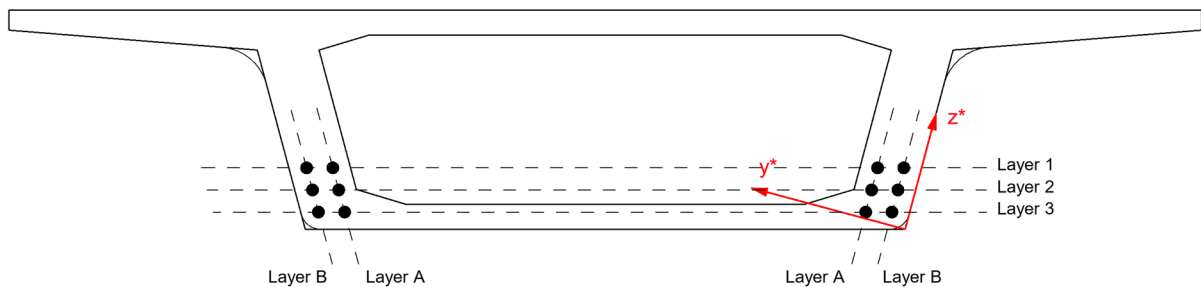


Figure 7-1: Reference system and naming for cable layout.

7.1 Cable properties

7.1.1 Prestressing force

As discussed in **sections 6.4** and **6.5**, in order to satisfy the preliminary check of the pre-tension force, we increased the force from 56,8 MN to 59 MN, satisfying still the following conditions:

$$\begin{aligned}
 F_{sp} = 59 \text{ MN} &\leq P_{max} = \min\{0,8 \cdot k_1 \cdot f_{pk} \cdot A ; 0,9 \cdot k_2 \cdot f_{p,0.1,k} \cdot A\} \\
 &= \min\{0,8 \cdot 12 \cdot 4050 \text{ mm}^2 \cdot 1860 \text{ MPa} ; 0,9 \cdot 1640 \text{ MPa} \cdot 12 \cdot 4050 \text{ mm}^2\} \\
 &= 71,73 \text{ MN} \\
 F_{sp} = 59 \text{ MN} &\geq F_{sp,req} = 56,8 \text{ MN}
 \end{aligned}$$

The pretension for each cable is hence:

$$F_p = \frac{F_{sp}}{n} = \frac{59 \text{ MN}}{12} = 4,92 \text{ MN}$$

7.1.2 Minimum distances

We consider first the limit distances for the anchorage devices in the end sections:

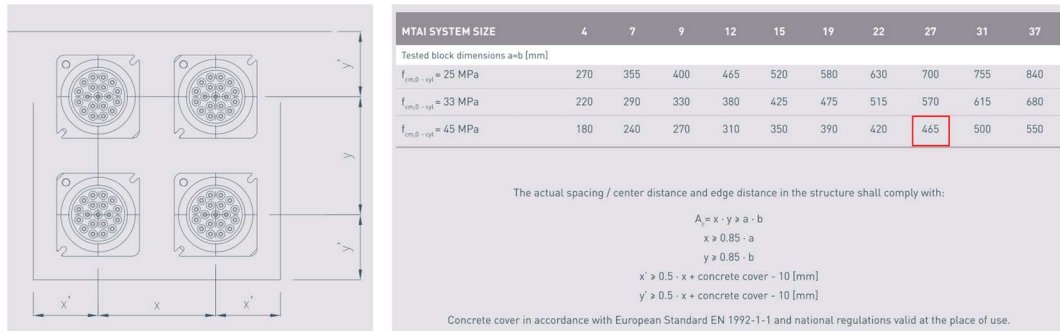


Figure 7-2: Minimum edge & inter-distances for anchorage devices, Tensa.

The illustrations are taken from Tensa's documentation for their pre-stress systems, in which the minimum inter-distance between anchorage devices is given by:

$$\begin{aligned}
 x &\geq 0,85 \cdot a = 0,85 \cdot 465 \text{ mm} = 395,25 \text{ mm} \Rightarrow x = 400 \text{ mm} \\
 y &\geq 0,85 \cdot b = 0,85 \cdot 465 \text{ mm} = 395,25 \text{ mm} \Rightarrow y = 400 \text{ mm}
 \end{aligned}$$

The minimum distance between each tendon and to the edge is given by EN1992-1-1, 8.10.1.3:

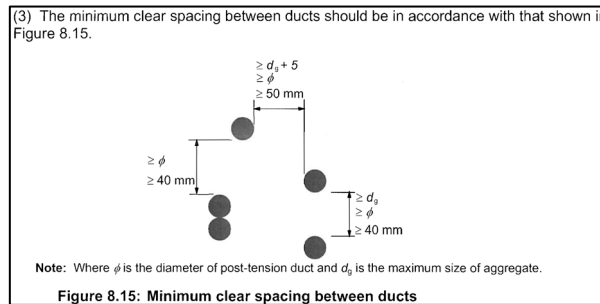


Figure 7-3: Minimum distances between post-tension ducts.

$$\text{Min. spacing} = \max\{32\text{mm} + 5\text{mm}; 115 \text{ mm}; 50 \text{ mm}\} = 115 \text{ mm}$$

7.2 Cable paths

The cable paths in (x, y^*) & (x, z^*) -planes consist of a straight, parabolic, and constant part:

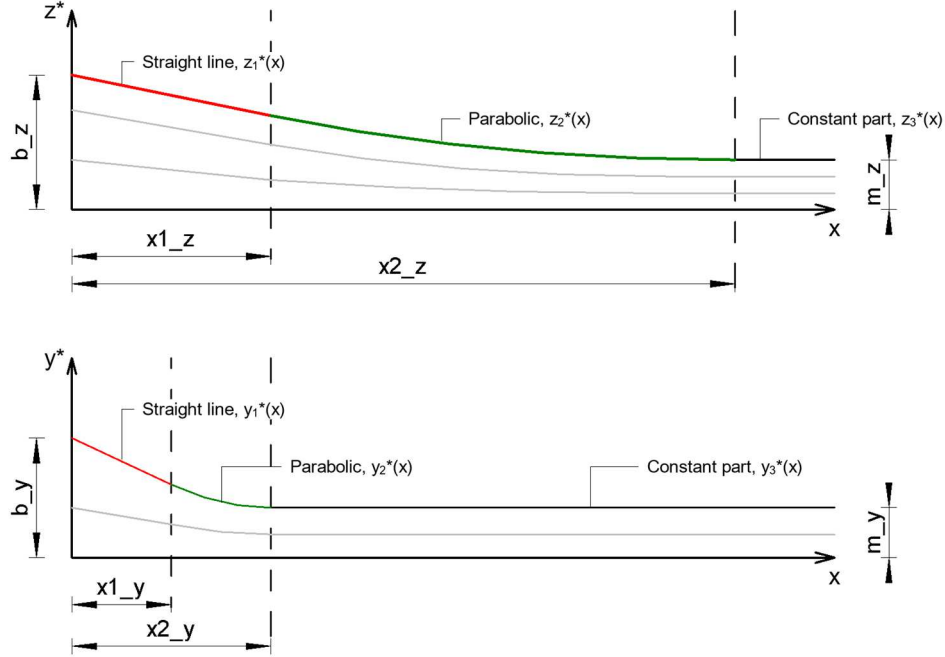


Figure 7-4: Schematisation of cable path.

Looking at the cable layers in e.g. (x, z^*) -plane, they can be described by:

$$z_1^*(x) = ax + b$$

$$z_2^*(x) = cx^2 + dx + e$$

$$z_3^*(x) = m$$

The boundary conditions to determine the 6 unknown coefficients are given by:

$$z_1^*(0) = b$$

$$z_1^*(x_1) = z_2^*(x_1) \Rightarrow ax_1 + b = cx_1^2 + dx_1 + e$$

$$z_1'^*(x_1) = z_2'^*(x_1) \Rightarrow a = 2cx_1 + d$$

$$z_2^*(x_2) = z_3^*(x_2) \Rightarrow cx_2^2 + dx_2 + e = m$$

$$z_2'^*(x_2) = z_3'^*(x_2) \Rightarrow 2cx_2 + d = 0$$

Defining the positions of the cables in both end section ($x = 0 \text{ m}$) and mid-section ($x = 20 \text{ m}$) effectively defines the values of b and m for each layer. The positions x_1 and x_2 , which marks the transitions between the curve parts, will be defined as well. This reduces the system to consist of 4 equations in 4 unknowns a , c , d and e , for which the solution is shown below:

$$c = \frac{m - b}{x_1^2 - x_2^2}$$

$$a = 2c(x_1 - x_2) = \frac{2(m - b)}{x_1 + x_2}$$

$$d = -2cx_2 = -\frac{2x_2(m - b)}{x_1^2 - x_2^2}$$

$$e = cx_2^2 + m = \frac{mx_1^2 - bx_2^2}{x_1^2 - x_2^2}$$

The paths in (x, y^*) -plane are defined in the same manner.

7.2.1 Cable path in (x, z^*) -plane

The choice of parameters for each cable layer is shown below along with the cable paths in defined in Excel. For all values, see Appendix E.

Input Parameter	Layer 1	Layer 2	Layer 3
b [m]	1,797	1,434	0,399
m [m]	0,592	0,362	0,132
x_1 [m]	5	5	5
x_2 [m]	20	18	16
Output parameter			
a [—]	-0,096	-0,093	-0,025
c [m ⁻¹]	0,003	0,004	0,001
d [—]	-0,128	-0,129	-0,037
e [m]	1,877	1,524	0,428

Table 7-1: Parameters defining cable path in (x, z^*) -plane.

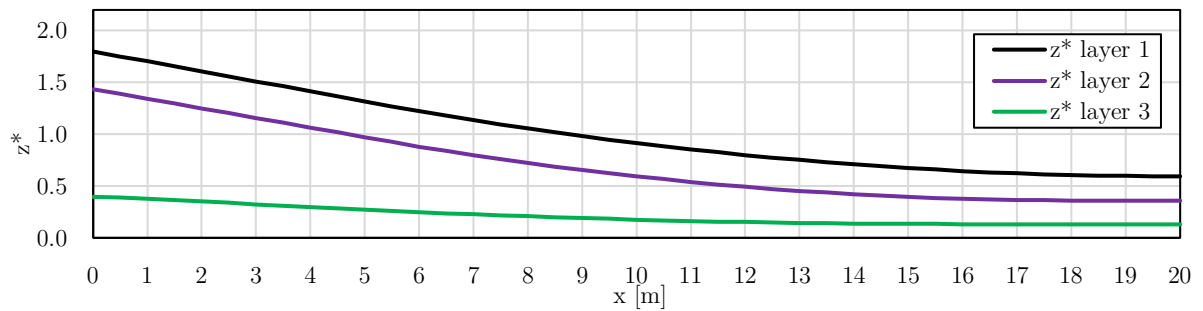


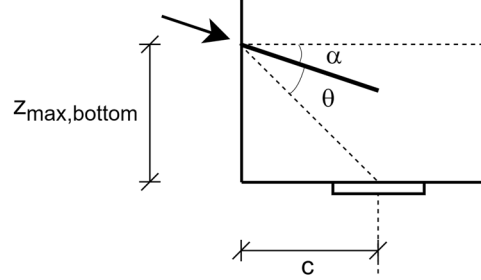
Figure 7-5: Cable paths in (x, z^*) -plane

Position of cables in $x = 0$ m:

The definition of the positions in end section, i.e. the value of b , is determined based on a goal of minimising the bending moment due to the pre-tension only. This is done by considering

rotational equilibrium around the centroidal axis in the end-section and finding a configuration of cables that results in an acceptable small bending moment. The procedure is as follows:

- Set approximate distance to centroid of bearing from edge and compute the critical height of the bottom cable such that its cone of diffusion stays in the concrete left of the support:

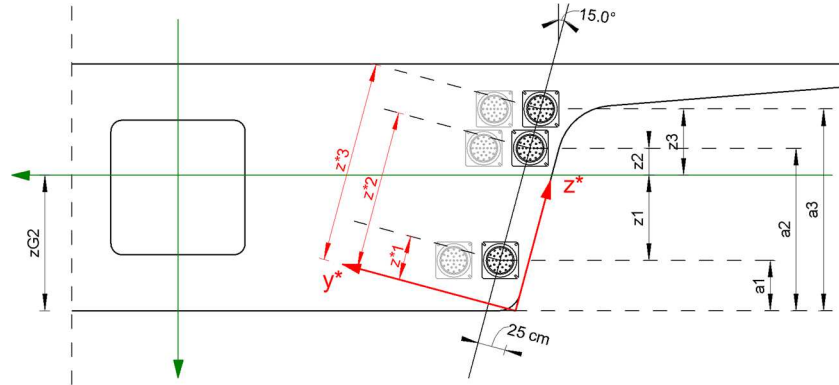


$$c \cong 0,7 \text{ m}, \quad \alpha = 1,45^\circ \text{ (from cable paths)}, \quad \theta = 33,7^\circ \text{ (Eurocode)}$$

$$z_{max,bottom} = c \cdot \tan(\alpha + \theta) = 0,493 \text{ m}$$

- Choose actual values of positions of cable layer 1 and 3, and the position of layer 2 is governed by the minimum distance allowed (by Tensa) to layer 3:

$$a_1 = 0,45 \text{ m} < 0,493 \text{ m}, \quad a_3 = 1,8 \text{ m}, \quad a_2 = a_3 - 0,35 \text{ m} = 1,45 \text{ m}$$



- Bending moments due to the pre-stress:

$$z_1 = z_{G2} - a_1 = 1,33 \text{ m} - 0,45 \text{ m} = 0,88 \text{ m}$$

$$z_3 = a_3 - z_{G2} = 1,8 \text{ m} - 1,33 \text{ m} = 0,47 \text{ m}$$

$$z_2 = a_3 - z_2 = 1,8 \text{ m} - 0,35 \text{ m} = 0,12 \text{ m}$$

$$\Sigma M = \frac{59 \text{ MN}}{3} \cdot 0,88 \text{ m} - \frac{F_{sp}}{3} \cdot 0,12 \text{ m} - \frac{F_{sp}}{3} \cdot 0,47 \text{ m} = 5,9 \text{ MNm OK!}$$

- Once an acceptable moment is achieved (by iteration), we can compute the z^* -values of each cable layer:

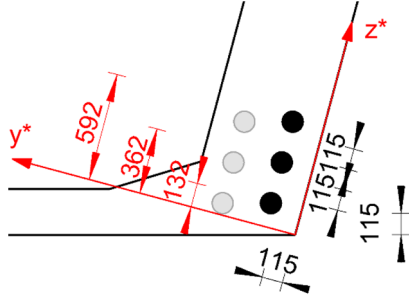
$$z_1^* = a_1 \cdot \cos(15^\circ) - 0,25 \text{ m} \cdot \tan(15^\circ) = 0,399 \text{ m} = b_{layer,1}$$

$$z_2^* = a_1 \cdot \cos(15^\circ) - 0,25 \text{ m} \cdot \tan(15^\circ) = 1,434 \text{ m} = b_{layer,2}$$

$$z_3^* = a_1 \cdot \cos(15^\circ) - 0,25 \text{ m} \cdot \tan(15^\circ) = 1,797 \text{ m} = b_{layer,3}$$

Position of cables in $x = 20$ m:

The position of cables in mid-section, i.e. value of m for each layer, is governed by limit edge distances presented in **section 7.1.2**. It is preferred to have the cables as close to the bottom edge as possible to maximise the flexural capacity. We have:



$$\begin{aligned} m_{\text{layer},1} &= 0,592 \text{ m} \\ m_{\text{layer},2} &= 0,362 \text{ m} \\ m_{\text{layer},3} &= 0,132 \text{ m} \end{aligned}$$

7.2.2 Cable path in (x, y^*) -plane

In order to avoid double-curvature, x_1 and x_2 are chosen differently compared with those for the (x, z^*) -plane. Results from the path definition is seen below, and for all values, see Appendix E.

Input Parameter	Layer A	Layer B
b [m]	0,65	0,25
m [m]	0,428	0,173
x_1 [m]	2	2
x_2 [m]	5	5
Output parameter		
a [—]	-0,064	-0,022
c [m ⁻¹]	0,011	0,004
d [—]	-0,106	-0,037
e [m]	0,692	0,265

Figure 7-6: Parameters defining cable path in (x, y^*) -plane.

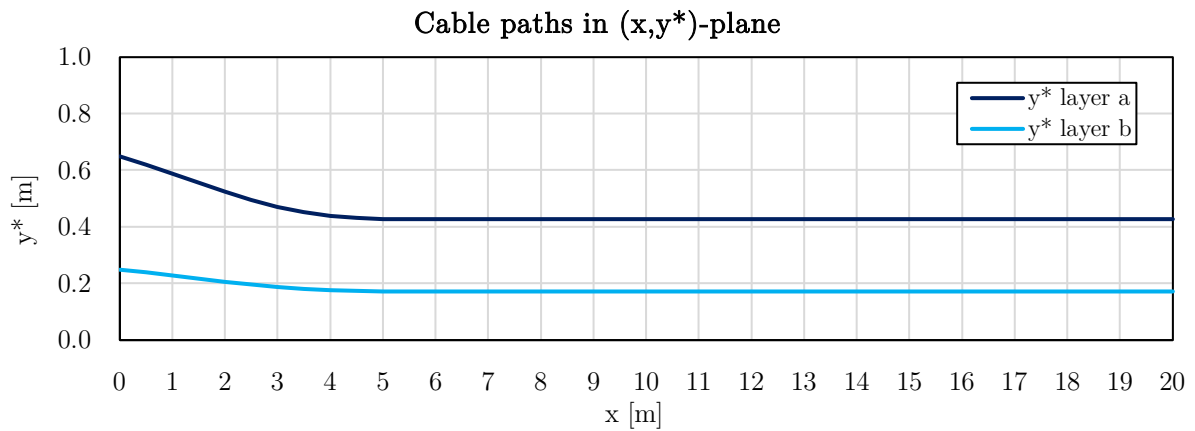
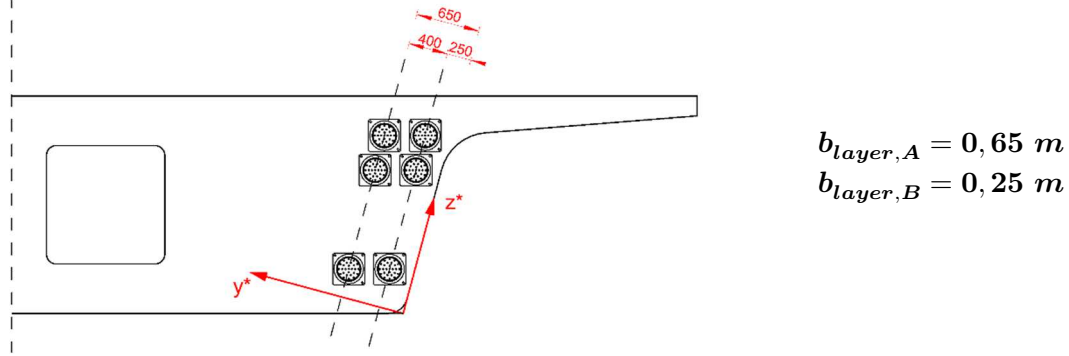


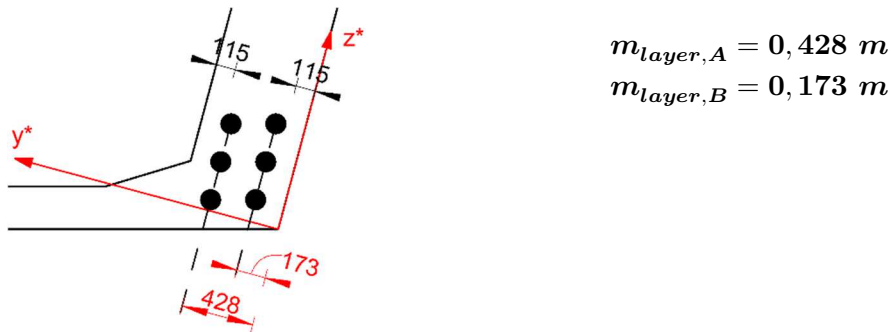
Figure 7-7: Cable paths in (x, y^*) -plane.

Position of cables in $x = 0$ m:

The position of cables (anchorage devices) in y^* -direction in the end-section is governed by minimum edge and inter-distances given in section 7.1.2. We have:

Position of cables in $x = 20$ m:

The position of the cables in y^* -direction in the mid-section is governed by having symmetry w.r.t the thickness of the web. We fix the edge dimension as the minimum distance according to section 7.1.2, so we have:



8. Prestressing losses

In **section 6.5**, in the preliminary assessment of cable pre-tension, we considered as a first estimate the total loss to be 25%, comprising 10% from instantaneous losses and 15% from long-term losses. Upon the definition of the cable paths in **section 7**, the exact losses throughout the girder can be computed. The losses have been divided into two parts:

- Instantaneous losses: Losses due to friction, wobble effect and draw-in of the wedges.
- Time-dependent losses: due to the shrinkage and creep of the concrete and the relaxation of steel.

8.1 Instantaneous Losses

8.1.1 Friction loss

Friction losses are generated at the interface between the cables and the ducts, as well as the friction between the tendons. This effect is highly influenced by curvature. However, in the straight segments, the ‘Wobble effect’ must be considered as the combination of losses due to the stretching and shortening of the cables and the friction between cables and ducts.

The losses due to friction in post-tensioned tendons is estimated according to section 5.10.5.2 of EN 1992-1-1.

$$\Delta P_\mu(x) = P_0 \cdot (1 - e^{-\mu(\theta + Kx)})$$

Where:

- θ = The sum of the angular deviations w.r.t horizontal over a distance x .
- $\mu = 0.19 \left[\frac{1}{rad} \right]$ = The coefficient for friction between the tendon and its duct, according to Tensa.
- $k = 0.005 \left[\frac{rad}{m} \right]$ = The coefficient for unintentional angular deviation (wobble factor), according to Tensa.
- x = The distance along the tendon from the point where the prestressing force is equal to P_0 force at the active end during tensioning.

8.1.2 Wedge loss

The draw-in of the wedges is a phenomenon connected to the release of the cables which are fixed in the pull jacks during the prestressing phase. This effect is related to a certain re-entry of the cables and the development of friction in the opposite direction with respect to the pull phase.

According to section 5.10.5.3 of EN 1992- 1.1: 2004 (E), in the post-tensioning systems, when the strands are locked off in the anchorage, they move through a fixed distance of 5 mm. Loss of prestressing due to wedge draw-in can be computed as:

$$l_w = \sqrt{\frac{\delta_w \cdot E_{sp} \cdot A_p}{p}}$$

$$\Delta P_w = 2 \cdot \Delta p \cdot l_w$$

where:

- δ_w draw-in of the wedges (5mm)
- E_{sp} modulus of elasticity of post tensioning steel [195000 (N/mm²)]
- A_{sp} steel cross section [N/mm²]
- $p = \frac{(P_0 - P_L)}{L}$ loss of force per unit of length [N/mm²]

In general, these losses do not have a significant influence, since the force required in the tendons is determined by the governing moment that is usually located at a point unaffected by distance δ_w . For short tendons with small curvature or where high normal stresses are required near a support or anchorage it may be advisable to compensate these losses.

In this section, we used a procedure that is explained in detail in the **section 8.3.1** of this chapter.

8.2 Time dependent losses

As per the section 5.10.5.3 of EN 1992-1.1:2004(E), the time dependent losses may be calculated by considering the following two reductions of the stress:

- a) Due to the reduction of strain, caused by the deformation of concrete due to creep and shrinkage, under the permanent loads.
- b) The reduction of stress in the steel caused by due to the relaxation under tension. The relaxation of steel depends on the concrete deformation due to the creep and shrinkage. This interaction can generally and approximately be considered by a reduction factor of 0.8.

A simplified method to evaluate time dependent losses at location x under the permanent loads is given by the equation:

$$\Delta P_{c+s+r} = A_p \cdot \Delta \sigma_{c+s+r} = A_p \cdot \frac{\varepsilon_{cs} \cdot E_p + 0.8 \cdot \Delta \sigma_{pr} + \frac{E_p}{E_{cm}} \varphi(t, t_0) \sigma_{c, Qp}}{1 + \frac{E_p}{E_{cm}} \cdot \frac{A_p}{A_c} \cdot \left(1 + \frac{A_c}{I_c} \cdot z_{cp}^2\right) \cdot [1 + 0.8 \cdot \varphi(t, t_0)]}$$

where:

- ΔP_{c+s+r} is the absolute value of the variation of stress in the tendons due to creep, shrinkage, and relaxation at location x at time t ;
- ε_{cs} is the estimated shrinkage strain according to eq. 3.1.4(6) in absolute value. Hence, the values of the total shrinkage strain ε_{cs} follows from

$$\varepsilon_{cs} = \varepsilon_{cd} + \varepsilon_{ca}$$

where:

- ε_{cs} is the total shrinkage strain.
- ε_{cd} is the drying shrinkage strain.
- ε_{ca} is the autogenous shrinkage strain.

- E_{sp} is the modulus of elasticity of the prestressing steel.
 - E_{cm} is the modulus of elasticity of concrete.
 - $\Delta\sigma_{pr}$ is the absolute value of the variation of stress in the tendons at location x at time t due to the relaxation of the prestressing steel. It is determined for a stress of $\sigma_p = \sigma_p(G + P_{m0} + \psi_2 Q)$, where σ_p is the initial stress in the tendons due to initial prestress and quasi-permanent actions.
 - $\varphi(t, t_0)$ is the creep co-efficient at a time t and load application at time t_0 ;
 - $\sigma_{c,Qp}$ is the stress in the concrete adjacent to the tendons, due to self-weight and initial prestress and other quasi-permanent actions. The value of $\sigma_{c,Qp}$ maybe the effect of part of self-weight and initial prestress or the effect of a full quasi-permanent combination of action $\sigma_c = \sigma_c(G + P_{m0} + \psi_2 Q)$, depending on the stage of construction considered.
 - A_c is the area of concrete section considered.
 - I_c is the second moment of inertia of the concrete section.
 - z_{cp} is the distance between the centre of gravity of the concrete section and the tendons.
- Compressive stresses and the corresponding strains should be used with a positive sign.

For our computations the input data used is mentioned in **section 4**.

Flowchart:

1) Calculate the parameters used for the calculation of drying shrinkage:

- Notional size, $h_0 = \left[2 \cdot \frac{A_c}{u}\right] = 517,39 \text{ mm}$
- Time for drying of concrete, $t_s = 28 \text{ days}$
- Age of concrete at the time of evaluation, $t = 365 \text{ days}$
- $\beta \cdot ds = \frac{(t-t_s)}{(t-t_s)+0,04\sqrt{h_0^3}} = 0,417$
- Relative Humidity = 75%
- $\varepsilon_{cdo} = 0,000261$ (Interpolation in table 3.2, EC2-1-1, 3.1.4)
- $k_h = 0,7$ (Table 3.3, EC2-1-1, 3.1.4)
- $\varepsilon_{cd}(t) = 0,000076$

2) Calculate the autogenous strain:

- $\varepsilon_{ca(t)} = \beta_{as}(t) \varepsilon_{ca(\infty)}$ (Autogenous strain)
- $\varepsilon_{ca(\infty)} = 2,5(f_{ck} - 10) \cdot 10^{-6} = 0,0000875$
- $\beta_{as}(t) = 1 - \exp(-0,2t^{0,5}) = 0,978$

3) Calculate the Total shrinkage strain:

- $\varepsilon_{cs} = \varepsilon_{cd} + \varepsilon_{ca} = 0,0001617$

4) Calculate the steel relaxation:

- $\sigma_p = \frac{\text{Total Post-Tension}}{\text{No of tendons}} \cdot \frac{1}{A_{sp}} = 1213.99 \frac{\text{N}}{\text{mm}^2}$
- $t = 500000 \text{ hours}$
- $\mu = \frac{\sigma_p}{f_{pk}} = 0,653$
- $\rho_{1000} = 2,5 \text{ (Tensa Acai p, 10)}$
- $\Delta\sigma_{pr} = 5,39 * \rho_{1000} * e^{6,7\mu} * \left(\frac{t}{1000}\right)^{(0.75(1-\mu))} \cdot 10^{-5} * \sigma_p = 65,46 \text{ (N/mm}^2\text{)}$
- $0,8\Delta\sigma_{pr} = 52,37 \text{ (N/mm}^2\text{)}$

5) Calculating the creep:

- $z_{cp} = 973,4 \text{ mm}$ (Centroid of tendons)
- $\sigma_{c,QP} = 20,29 \frac{\text{N}}{\text{mm}^2}$ (Stress in concrete at the position of centroid of tendons due to prestress and structural dead load)
- Total creep = $\varphi(t, t_0) \cdot \sigma_{c,QP} = 36,52 \frac{\text{N}}{\text{mm}^2}$

Time-dependent losses at mid-span			
Parameter	Symbol	Value	Unit
Total long-term loss as stress	$\Delta\sigma_{c+s+r}$	226,98	MPa
Total long-term as force	ΔP_{c+s+r}	11031,1	kN
Percentage of total pre-stress force		18,7	%
Long-term loss in each cable	ΔP_{∞}	919,25	kN

Table 8-1: Time-dependent losses at mid-span.

The long-term loss computed for the mid-span section is conservatively applied in every section along the girder, i.e. neglecting the fact that $\Delta\sigma_{pr}$ depends on the position along the girder.

8.3 Cable force diagrams

In the following section, the procedure and results of the computations of the cable forces considering the losses are shown below and visualised graphically with cable force diagrams for each layer. **Figure 8-1** shows again the naming of each cable, defined in **section 7**.

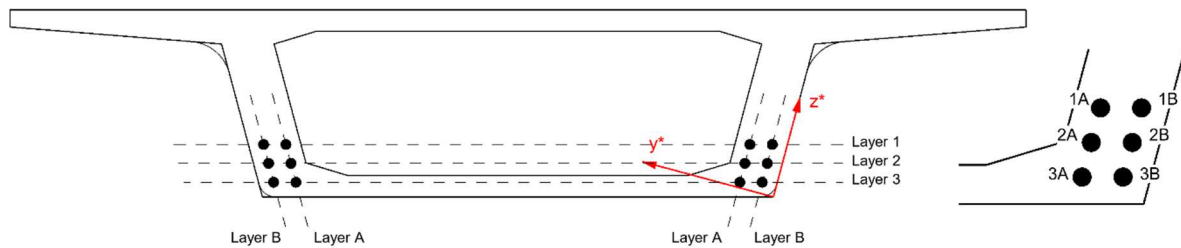


Figure 8-1: Cable layers.

8.3.1 Schematic procedure of computation

The following describes the procedure of determining the cable force diagrams, i.e. the cable force in each cable as a function of longitudinal position, x . Following the description, the results for cable 1A is shown. For the remaining cables, cable force diagrams and values can be found in Appendix F.

1) Obtain the sum of angular deviations:

The sum of angular deviations can be determined as the sum from the contributions in y^* - and z^* -directions:

$$\vartheta = \vartheta_y(x) + \vartheta_z(x) = \arctan\left(\frac{dy^*(x)}{dx}\right) + \arctan\left(\frac{dz^*(x)}{dx}\right)$$

This formula is applied in each sub path for each cable, respecting the bounds for linear, parabolic, and constant parts.

2) Compute the friction losses and the cable force after the application of friction loss:

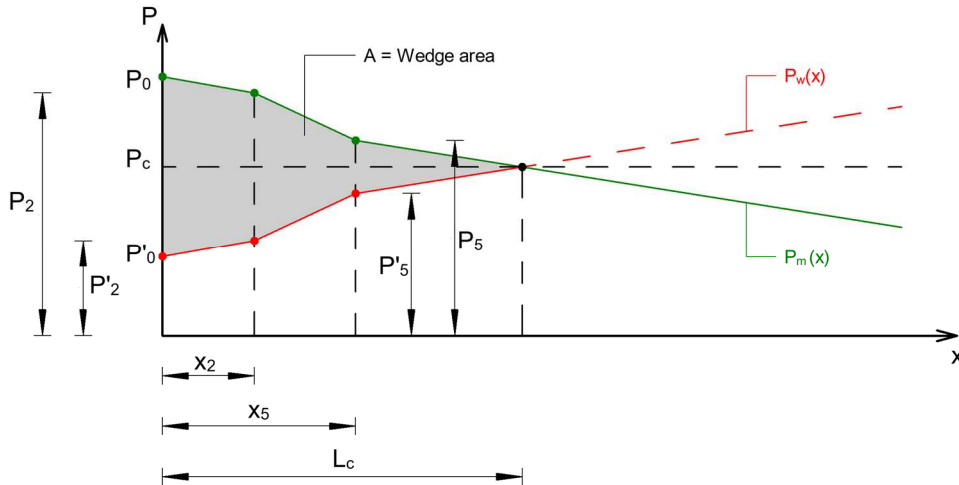
$$\Delta P_\mu(x) = P_0 \cdot (1 - e^{-\mu(\theta+Kx)}) \Rightarrow P_\mu(x) = P_0 - \Delta P_\mu(x) = P_0 \cdot e^{-\mu(\theta+Kx)}$$

3) Determine the critical length for wedge loss:

We set the wedge draw-in displacement and compute the wedge loss area:

$$\delta = 5 \text{ mm}, \quad W_{int} = E_p \cdot A_p \cdot \delta = 195 \cdot 10^3 \frac{\text{N}}{\text{mm}^2} \cdot 4050 \text{ mm}^2 \cdot 5 \text{ mm} \cdot 10^{-6} = 3948,7 \text{ kNm}$$

We consider the following sketch:



The force built up in the cable due to the wedge loss is assumed to be the reflected curve of $P_\mu(x)$:

$$P_w(x) = P_0' \cdot e^{+\mu(\theta+Kx)}$$

The procedure is to iterate a value of L_c , such that $A \approx W_{wedge}$. In that case, we will have the correct representation of the wedge draw-in loss function, $P_w(x)$. For each cable, the following iteration is done:

a) Choose L_c . We have that $x_2 = 2 \text{ m}$ and $x_5 = 5 \text{ m}$.

b) Compute:

$$P_c = P_\mu(x = L_c) = P_0 \cdot e^{-\mu(\theta + K \cdot L_c)}, \quad P_2 = P_0 \cdot e^{-\mu(\theta + Kx_2)}, \quad P_5 = P_0 \cdot e^{-\mu(\theta + Kx_5)}$$

c) Compute by exploiting the properties of two mirrored exponentials:

$$P'_0 = \frac{P_c^2}{P_0}, \quad P'_2 = \frac{P_c^2}{P_2}, \quad P'_5 = \frac{P_c^2}{P_5}$$

d) Compute the area:

$$\begin{aligned} A &= A_{trapezoid,1} + A_{trapezoid,2} + A_{triangle} \\ &= \frac{1}{2} \cdot ((P_0 - P'_0) + (P_2 - P'_2)) \cdot x_2 + \frac{1}{2} \cdot ((P_2 - P'_2) + (P_5 - P'_5)) \cdot (x_5 - x_2) + \frac{1}{2} \\ &\quad \cdot (P_5 - P'_5) \cdot (L_c - x_5) \end{aligned}$$

e) Check if $A \approx W_{int}$. Redo a)-d) until $\text{Error} = A - W_{int} < 0,02 \cdot W_{int}$

4) Determine the final cable force diagram in t_0 and t_∞ :

$$P_0(x) = \min\{P_w(x); P_\mu(x)\}, \quad \forall x \in [0; 20] \text{ m}$$

$$P_\infty(x) = P_0(x) - \Delta P_\infty, \quad \text{where } \Delta P_\infty \text{ is computed in section 8.2.}$$

Below is shown the results for cable 1A. See Appendix F for all cables.

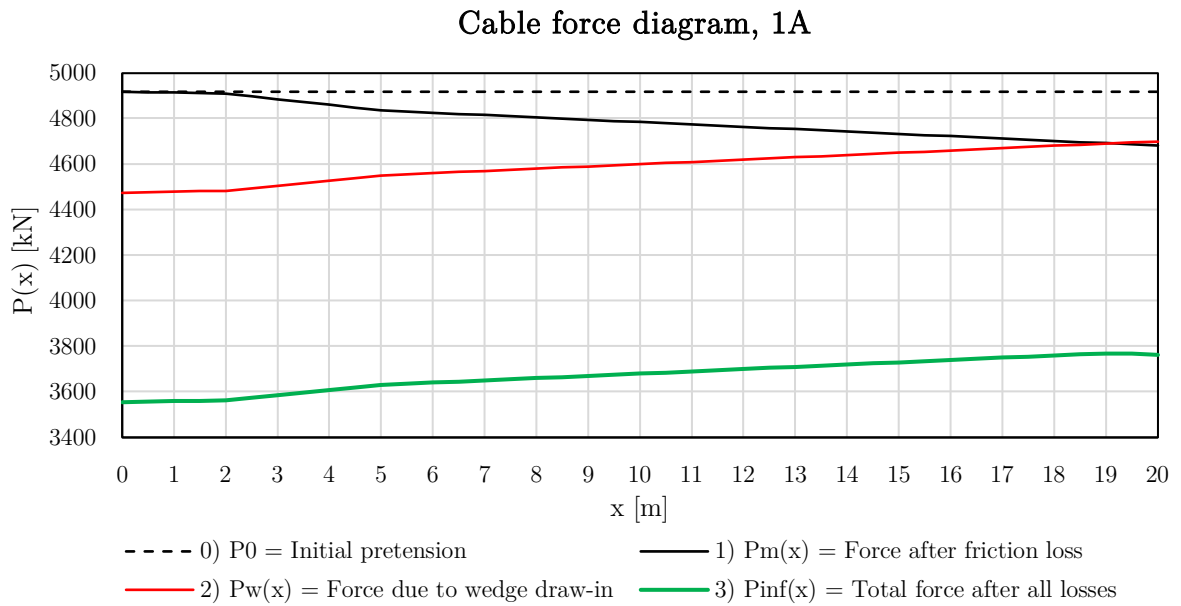


Table 1A														
x [m]	y^{**} [-]	z^{**} [-]	$atan(y^{**})$ [rad]	$atan(z^{**})$ [rad]	θ_y [rad]	θ_z [rad]	ΔP_μ [kN]	$P_\mu(x)$ [kN]	$Pw(x)$ [kN]	$P_0(x)$ [kN]	Loss ratio, t0 [%]	ΔP_∞ [kN]	$P_\infty(x)$ [kN]	Loss ratio, t _∞ [%]
0	-0.0636	-0.0963	0.0635	0.0960	0	0	0.00	4916.7	4472.9	4472.9		919.25	3553.7	
0.5	-0.0636	-0.0963	0.0635	0.0960	0	0	2.33	4914.3	4475.1	4475.1		919.25	3555.8	
1	-0.0636	-0.0963	0.0635	0.0960	0	0	4.67	4912.0	4477.2	4477.2		919.25	3557.9	
1.5	-0.0636	-0.0963	0.0635	0.0960	0	0	7.00	4909.7	4479.3	4479.3		919.25	3560.1	
2	-0.0636	-0.0963	0.0635	0.0960	0	0	9.33	4907.3	4481.4	4481.4		919.25	3562.2	
2.5	-0.0530	-0.0963	0.0529	0.0960	0.0106	0	21.49	4895.2	4492.6	4492.6		919.25	3573.3	
3	-0.0424	-0.0963	0.0424	0.0960	0.0211	0	33.64	4883.0	4503.7	4503.7		919.25	3584.5	
3.5	-0.0318	-0.0963	0.0318	0.0960	0.0317	0	45.76	4870.9	4515.0	4515.0		919.25	3595.7	
4	-0.0212	-0.0963	0.0212	0.0960	0.0423	0	57.85	4858.8	4526.2	4526.2		919.25	3606.9	
4.5	-0.0106	-0.0963	0.0106	0.0960	0.0529	0	69.93	4846.7	4537.5	4537.5		919.25	3618.2	
5	0	-0.0963	0.0000	0.0960	0.0635	0	81.97	4834.7	4548.8	4548.8		919.25	3629.5	
5.5	0	-0.0931	0.0000	0.0929	0.0635	0.00318	87.19	4829.5	4553.7	4553.7		919.25	3634.4	
6	0	-0.0899	0.0000	0.0897	0.0635	0.00637	92.40	4824.3	4558.6	4558.6		919.25	3639.4	
6.5	0	-0.0867	0.0000	0.0865	0.0635	0.00955	97.61	4819.1	4563.5	4563.5		919.25	3644.3	
7	0	-0.0835	0.0000	0.0833	0.0635	0.01274	102.81	4813.9	4568.5	4568.5		919.25	3649.2	
7.5	0	-0.0803	0.0000	0.0801	0.0635	0.01593	108.02	4808.7	4573.4	4573.4		919.25	3654.2	
8	0	-0.0771	0.0000	0.0769	0.0635	0.01912	113.21	4803.5	4578.4	4578.4		919.25	3659.1	
8.5	0	-0.0739	0.0000	0.0737	0.0635	0.02232	118.41	4798.3	4583.3	4583.3		919.25	3664.1	
9	0	-0.0706	0.0000	0.0705	0.0635	0.02551	123.60	4793.1	4588.3	4588.3		919.25	3669.0	
9.5	0	-0.0674	0.0000	0.0673	0.0635	0.02871	128.78	4787.9	4593.2	4593.2		919.25	3674.0	
10	0	-0.0642	0.0000	0.0641	0.0635	0.03190	133.96	4782.7	4598.2	4598.2		919.25	3679.0	
10.5	0	-0.0610	0.0000	0.0609	0.0635	0.03510	139.14	4777.5	4603.2	4603.2		919.25	3683.9	
11	0	-0.0578	0.0000	0.0577	0.0635	0.03830	144.31	4772.4	4608.2	4608.2		919.25	3688.9	
11.5	0	-0.0546	0.0000	0.0545	0.0635	0.04150	149.47	4767.2	4613.2	4613.2		919.25	3693.9	
12	0	-0.0514	0.0000	0.0513	0.0635	0.04471	154.64	4762.0	4618.2	4618.2		919.25	3698.9	
12.5	0	-0.0482	0.0000	0.0481	0.0635	0.04791	159.79	4756.9	4623.2	4623.2		919.25	3703.9	
13	0	-0.0450	0.0000	0.0449	0.0635	0.05111	164.95	4751.7	4628.2	4628.2		919.25	3709.0	
13.5	0	-0.0417	0.0000	0.0417	0.0635	0.05432	170.09	4746.6	4633.2	4633.2		919.25	3714.0	
14	0	-0.0385	0.0000	0.0385	0.0635	0.05752	175.24	4741.4	4638.3	4638.3		919.25	3719.0	
14.5	0	-0.0353	0.0000	0.0353	0.0635	0.06073	180.38	4736.3	4643.3	4643.3		919.25	3724.0	
15	0	-0.0321	0.0000	0.0321	0.0635	0.06394	185.51	4731.2	4648.3	4648.3		919.25	3729.1	
15.5	0	-0.0289	0.0000	0.0289	0.0635	0.06715	190.64	4726.0	4653.4	4653.4		919.25	3734.1	
16	0	-0.0257	0.0000	0.0257	0.0635	0.07036	195.76	4720.9	4658.4	4658.4		919.25	3739.2	
16.5	0	-0.0225	0.0000	0.0225	0.0635	0.07357	200.88	4715.8	4663.5	4663.5		919.25	3744.2	
17	0	-0.0193	0.0000	0.0193	0.0635	0.07678	205.99	4710.7	4668.5	4668.5		919.25	3749.3	
17.5	0	-0.0161	0.0000	0.0161	0.0635	0.07999	211.10	4705.6	4673.6	4673.6		919.25	3754.3	
18	0	-0.0128	0.0000	0.0128	0.0635	0.08320	216.20	4700.5	4678.7	4678.7		919.25	3759.4	
18.5	0	-0.0096	0.0000	0.0096	0.0635	0.08641	221.30	4695.4	4683.8	4683.8		919.25	3764.5	
19	0	-0.0064	0.0000	0.0064	0.0635	0.08962	226.39	4690.3	4688.8	4688.8		919.25	3769.6	
19.5	0	-0.0032	0.0000	0.0032	0.0635	0.09283	231.48	4685.2	4693.9	4685.2		919.25	3765.9	
20	0	0.0000	0.0000	0.0000	0.0635	0.09604	236.56	4680.1	4699.0	4680.1	4.81	919.25	3760.9	23.51

WEDGE LOSSES			
Ep	195000.00	MPa	Young's modulus
Ap	4050.00	mm ²	Area per tendon
delta	5.00	mm	Wedge take-in displacement
W_int	3948.75	kNm	Wedge loss area
Computing actual W_int for Lc			
Lc	19.07	m	Corrected such that the wedge loss area is sufficiently close to the exact value W_int (error < 2%).
Pc	4689.56	kN	
P'_0	4472.94	kN	
P'_2	4481.45	kN	
P'_5	4548.77	kN	
P_5 - P'_5	285.92	kN	
P_2 - P'_2	425.89	kN	
P_0 - P'_0	443.73	kN	
W_int approx	3948.80	kNm	Area of 2 trapezoids and a triangle
Error	0.05	kNm	< 0.02*W_int 79.0 OK

Figure 8-2: Cable force computations and diagrams for cable 1A.

9. Flexural verifications

9.1 ULS

The ULS flexural verification of the bridge girder is done in 4 different characteristic sections, as the position of the cables vary along the length, hence producing different resisting moments in each section. The verifications are done by means of an equivalent section, consisting of 5 rectangular lamellas, and the total equivalent section is matched with the real section in area, A , moment of inertia, I_y , and position of centroid, z_G .

9.1.1 Equivalent sections

Two equivalent sections are computed, due to sections of the girder with varying web thickness for $x \in [0; 5] \text{ m}$. The cross-sectional parameters to be matched are shown in **Table 9-1**.

Real cross-section data		
Parameters	Section I	Section II
Domain	$x \in [0,5] \text{ m}$	$x \in [5,20] \text{ m}$
Reference web thickness	775 mm	600 mm
Area, A	$7,457 \cdot 10^6 \text{ mm}^2$	$7,031 \cdot 10^6 \text{ mm}^2$
Moment of inertia, I_y	$4,886 \cdot 10^{12} \text{ mm}^4$	$4,699 \cdot 10^{12} \text{ mm}^4$
Centroid, z_{G1}	897,4 mm	866,6 mm

Table 9-1: Real cross-sectional data for equivalent sections

The equivalent section and its geometric parameters are shown in **Figure 9-1**.

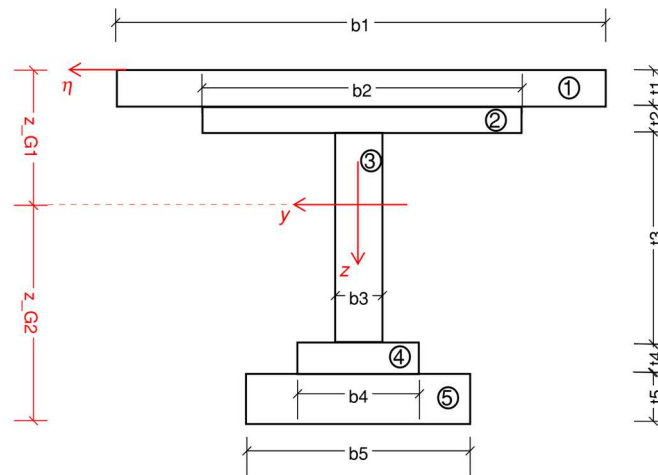


Figure 9-1: Scheme for equivalent section

Procedure of determining the equivalent cross-sections:

- 1) Set geometric parameters:

$$b_1 = 1200 \text{ mm}, \quad b_5 = 6000 \text{ mm}$$

- 2) Choose parameters:

$$b_2, b_3, b_4 \text{ and } t_1, t_2, t_3, t_4, t_5 \text{ such that } \sum_{i=1}^5 t_i = h = 2200 \text{ mm}$$

- 3) Compute area, moment of inertia and centroid:

$$\begin{aligned} A_{eq} &= \sum_{i=1}^5 b_i t_i \\ S_{\eta,eq} &= b_1 t_1 \frac{t_1}{2} + b_2 t_2 \left(t_1 + \frac{t_2}{2} \right) + b_3 t_3 \left(t_1 + t_2 + \frac{t_3}{2} \right) + b_4 t_4 \left(t_1 + t_2 + t_3 + \frac{t_4}{2} \right) \\ &\quad + b_5 t_5 \left(h - \frac{t_5}{2} \right) \\ z_{G1,eq} &= \frac{S_{\eta}}{A}, \quad z_{G2,eq} = h - z_{G1} \\ I_{y,i} &= \frac{1}{12} b_i t_i^3 + A_i z_i^2, \quad z_1 = z_{G1} - \frac{t_1}{2}, \quad z_2 = z_{G1} - \left(t_1 + \frac{t_2}{2} \right), \\ z_3 &= z_{G1} - \left(t_1 + t_2 + \frac{t_3}{2} \right), \quad z_4 = z_{G1} - \left(t_1 + t_2 + t_3 + \frac{t_4}{2} \right), \\ z_5 &= z_{G1} - \left(z_{G2} - \frac{t_5}{2} \right) \\ I_{eq,y} &= \sum_{i=1}^5 I_{y,i} \end{aligned}$$

- 4) Compute errors:

$$\varepsilon_A = \frac{A_{real} - A_{eq}}{A_{real}}, \quad \varepsilon_{zG} = \frac{z_{G1,real} - z_{G1,eq}}{z_{G1,real}}, \quad \varepsilon_I = \frac{I_{y,real} - I_{y,eq}}{I_{y,real}}$$

- 5) If $\varepsilon_A, \varepsilon_{zG}, \varepsilon_I \leq 0,1$, end.

If $\varepsilon_A, \varepsilon_{zG}, \varepsilon_I > 0,1$, redo 2)-4).

The final values of equivalent cross-section parameters are shown below:

Equivalent cross-section data		
Parameters	Section I	Section II
Domain	$x \in [0,5] \text{ m}$	$x \in [5,20] \text{ m}$
Area, $\mathbf{A}$	$7,456 \cdot 10^6 \text{ mm}^2$	$7,027 \cdot 10^6 \text{ mm}^2$
Moment of inertia, $\mathbf{I_y}$	$4,864 \cdot 10^{12} \text{ mm}^4$	$4,703 \cdot 10^{12} \text{ mm}^4$
Centroid, $\mathbf{z_{G1}}$	897,3 mm	866 mm

Table 9-2: Equivalent cross-section parameters

9.1.2 Flexural resistance

The resisting moment of each section is computed based on the following scheme in which the plastic capacity of the section is computed:

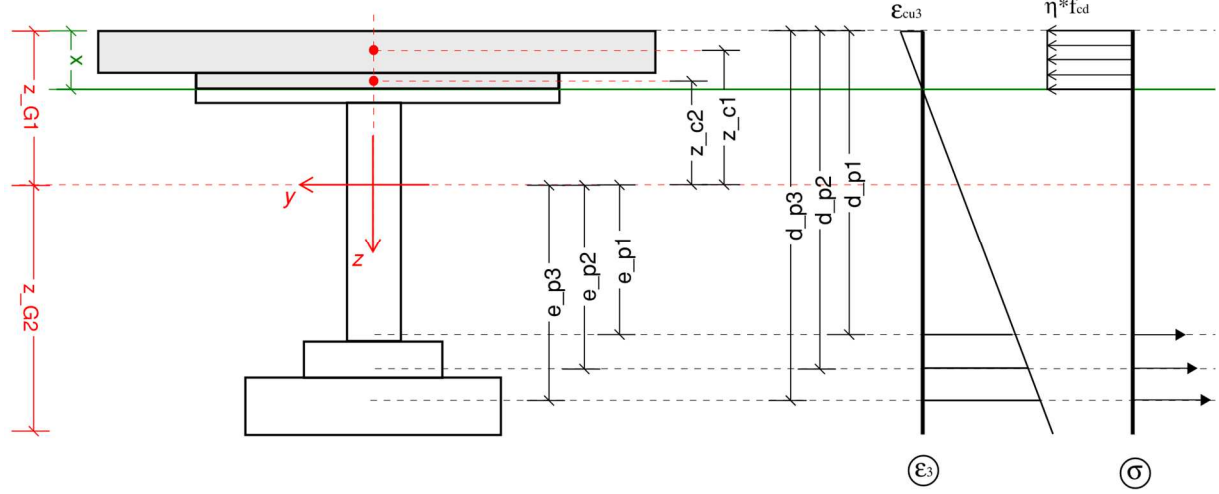


Figure 9-2: Scheme for flexural resistance in ULS

The procedure implemented is as follows:

- 1) Assume a compressive zone by setting the position of neutral axis x . Place in layer 1 for the first iteration.
- 2) Compute the compressive zone area and distance to overall centroid:

$$A_{c1} = b_1 t_1, \quad z_{c1} = z_{G1} - \frac{t_1}{2}$$

- 3) Compute geometric parameters, strains, and stresses for all tendons, $i = 1, 2, 3$ and $j = a, b$. Note, due to symmetry, we consider:

$$e_{p,ij} = z_{G2} - z_{ij}, \quad \text{where } z_{ij} \text{ are given by the cable curves}$$

$$d_{p,ij} = h - z_{ij}$$

$$\varepsilon_{p,ij} = \varepsilon_{1,ij} + \varepsilon_{2,ij} + \varepsilon_{3,ij} \leq \varepsilon_{p,ud} = 0,035$$

$$\varepsilon_{1,ij} = \frac{F_{p,\infty,ij}}{2E_p A_p}, \quad \text{with } F_{p,\infty,ij} = 2 \cdot \text{Force in tendon layer } ij, \quad A_p = \text{Area per tendon}$$

$$\varepsilon_{2,ij} = \frac{1}{E_c I_{cy}} \left(\frac{F_{p,\infty,ij}}{A_c} + \frac{F_{p,\infty,ij} \cdot e_{p,ij}}{I_{cy}} \cdot e_{p,ij} \right)$$

$$\varepsilon_{3,ij} = \varepsilon_{cu3} \cdot \frac{d_{p,ij} - x}{x}$$

$$\sigma_{p,ij} = \begin{cases} E_p \cdot \varepsilon_{p,ij}, & \text{if } \varepsilon_{p,ij} \leq \varepsilon_{yd} \\ f_{yd}, & \text{if } \varepsilon_{p,ij} > \varepsilon_{yd} \end{cases}$$

The strain limit of the tendons has been confirmed by TENSA to be 0,035.

- 4) Check the compressive zone by enforcing horizontal equilibrium:

$$F_c = \eta \cdot f_{cd} \cdot A_{c1} = F_p = \sum_{i=1}^3 \sigma_{p,ia} \cdot A_{p,ia} + \sigma_{p,ib} \cdot A_{p,ib}$$

- a. If $F_s = F_c$ then:

Bending capacity is computed:

$$M_c = \eta \cdot f_{cd} \cdot (A_{c1} \cdot z_{c1})$$

$$M_p = \sum_{i=1}^3 \sigma_{p,ia} \cdot A_{p,ia} \cdot e_{p,ia} + \sigma_{p,ib} \cdot A_{p,ib} \cdot e_{p,ib},$$

$$M_{Rd} = M_c + M_p$$

b. If $F_s \neq F_c$:

Iterate through 1)-4). If the neutral axis is set in layer 2, step 2) gets altered:

$$A_{c1} = b_1 t_1, \quad z_{c1} = z_{G1} - \frac{t_1}{2}$$

$$A_{c2} = A_{c1} + t_2(x - t_1), \quad z_{c2} = z_{G1} - \left(t_1 + \frac{1}{2}(x - t_1) \right)$$

Bending capacity is computed:

$$M_c = \eta \cdot f_{cd} \cdot (A_{c1} \cdot z_{c1} + A_{c2} \cdot z_{c2}),$$

$$M_p = \sum_{i=1}^3 \sigma_{p,ia} \cdot A_{p,ia} \cdot e_{p,ia} + \sigma_{p,ib} \cdot A_{p,ib} \cdot e_{p,ib},$$

$$M_{Rd} = M_c + M_p$$

Below is seen the summary of ULS verifications in 4 different sections. For the full computations in Excel, see Appendix B.

Flexural verifications			
Section	x (m)	M_{Ed} (kNm)	M_{Rd} (kNm)
I	20	103686,9	112227,1
II	10	78035,1	99564,7
III	5	45565,5	80832,9
IV	3	28910,8	74474,0

Figure 9-3: Flexural verifications, ULS

9.2 SLS

The serviceability limit state is verified by checking that no tensile stresses above 0,5 MPa arise in the bridge girder in the 3 different SLS combinations. Hence, we are verifying the pre-stress force introduced in the system. The verifications are based on a purely elastic consideration of the stress distribution in the section. As reported in **section 5**, we consider the following situations:

Load situation	Symbol	Description
Instantaneous	t_0	Only post-tensioning applied on the section along with the structural dead load and instantaneous losses. Use of pure concrete cross-sectional parameters.
Quasi-permanent	t_∞	Non-structural dead load and long-term losses are added. Use of long-term Young's modulus of concrete, $E_{c,eff}$
Characteristic	t_m	Live loads are further added.

Table 9-3: Load situations in SLS.

9.2.1 Homogenized cross-sectional parameters

To compute the elastic stress distributions, we must consider a homogenized cross-section for which the real section consisting of concrete and steel is transformed into an equivalent section consisting of a homogenized material. This is done by multiplying all the steel terms by the homogenization factor, $\alpha = \frac{E_s}{E_c}$. This factor depends on which load situation is considered:

Quasi-permanent situation:

$$E_{c,eff} = \frac{E_{cm}}{1 + \varphi} = \frac{35 \text{ GPa}}{1 + 1,8} = 12,86 \text{ GPa}, \quad \alpha_\infty = \frac{E_s}{E_{c,eff}} = \frac{195 \text{ GPa}}{12,86 \text{ GPa}} = \mathbf{15,56}$$

where φ = creep coefficient = 1,8 according to EC2-1-1, 3.1.4.

Characteristic situation:

$$E_{cm} = 35 \text{ GPa}, \quad \alpha_m = \frac{E_s}{E_{cm}} = \frac{195 \text{ GPa}}{35 \text{ GPa}} = \mathbf{5,56}$$

All cross-sectional parameters in the instantaneous situation, referring to the pure concrete section, have been computed in Autodesk Robot and are shown in Appendix C. They are based on the section and cable configuration in $x = 20 \text{ m}$, i.e. the mid-span section. The procedure of determining the cross-sectional parameters in the remaining load situations is schematically shown below:

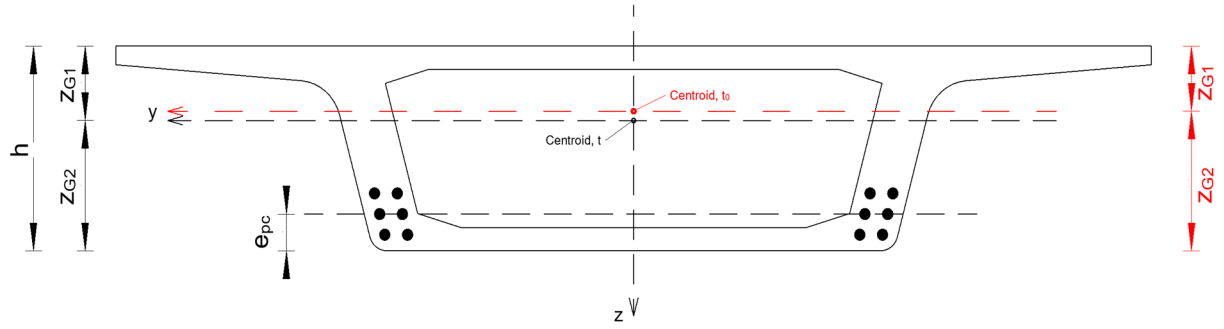


Figure 9-4: Scheme for homogenized cross-section parameters.

- 1) Computation of area:

$$A = A_{c0} + \alpha \cdot n \cdot A_p, \quad n = \text{no. of tendons}, A = \text{area per tendon}$$

- 2) Computation of centroid:

$$S_{top} = S_{top,0} + \alpha \cdot n \cdot A_p \cdot (h - e_{pc})$$

$$z_{G1} = \frac{S_{top}}{A}, \quad z_{G2} = h - z_{G1}$$

- 3) Computation of moment of inertia:

$$I_c = I_{c0} + A_{c0} \cdot (z_{G1} - z_{G1,0})^2, \quad \text{where } z_{G1,0} = \text{centroid in } t_0$$

$$I_s = n \cdot A_p \cdot (z_{G2} - e_{pc})^2$$

$$I_t = I_c + I_s$$

Cross-sectional parameters, $x = 20 \text{ m}$							
Time	α	A	z_{G1}	z_{G2}	I_c	I_s	I_t
	[-]	[mm ²]	[mm]	[mm]	[mm ⁴]	[mm ⁴]	[mm ⁴]
t_0		7.03E+06	866.60	1333.40	4.70E+12		4.70E+12
t_∞	15.56	7.79E+06	962.56	1237.44	4.76E+12	3.87E+10	5.37E+12
t_m	5.56	7.30E+06	903.15	1296.85	4.71E+12	4.40E+10	4.95E+12

Table 9-4: Homogenized cross-sectional parameters from Excel.

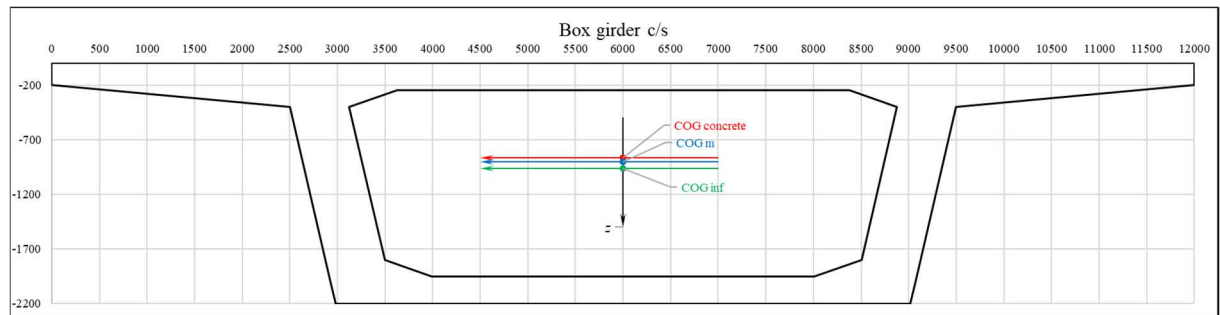


Figure 9-5: Centroids in each load situation, SLS

9.2.2 Elastic stress distributions

The stress distributions are verified in 3 different sections along the girder, in which the contributions from the cables change as the positions of the cables are different in each of these sections.

Section	Position, x (m)
I	20
II	10
III	5

Table 9-5: Sections to verify in SLS.

In the following, the schematic procedure of determining the stress distributions for each section is shown along with results from all 3 sections. All calculations refer to this schematic representation:

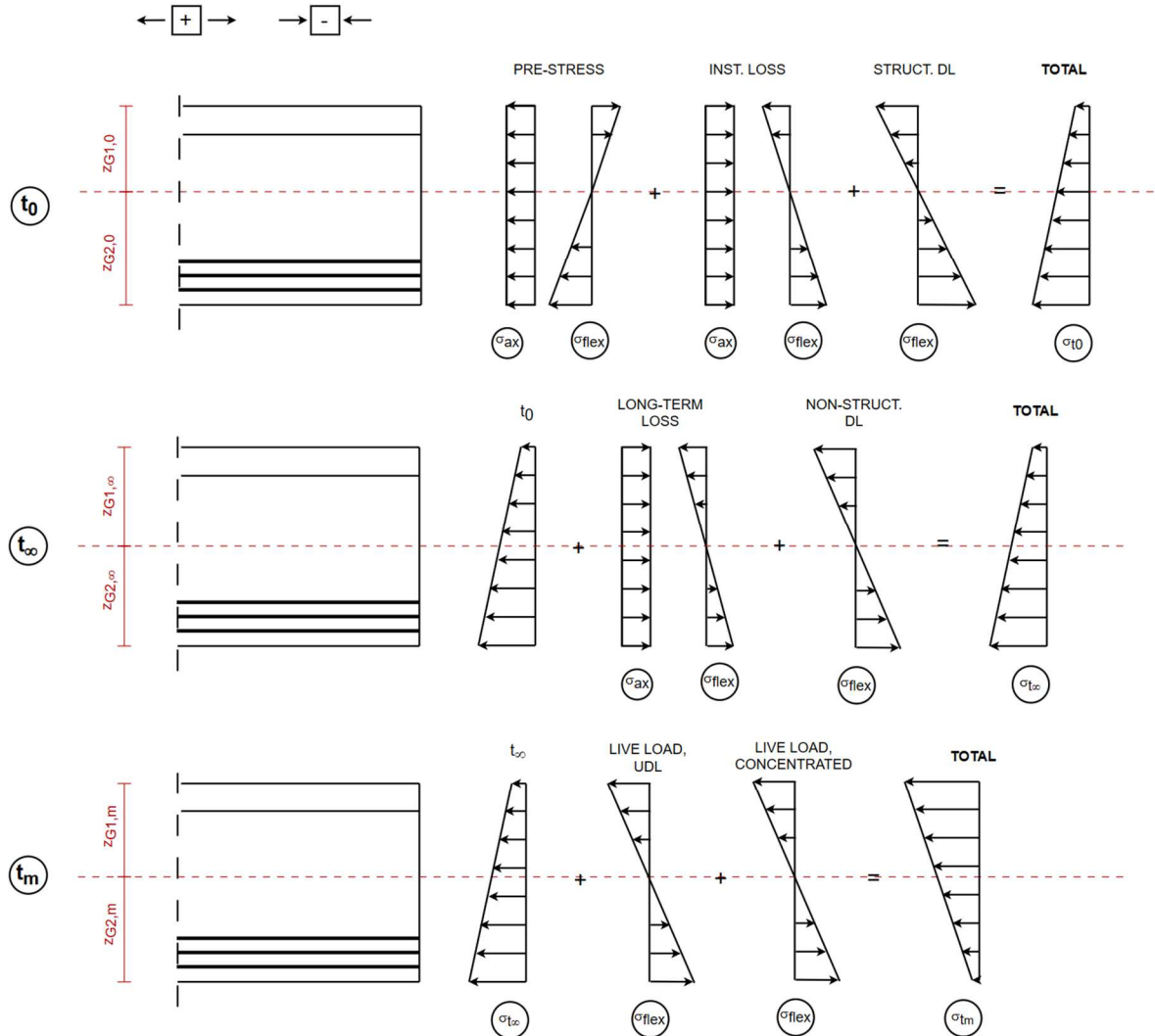


Figure 9-6: Schematic view of the elastic stresses in the 3 load situations in SLS.

For each section, the following formulas have been adopted:

1) Determine the contributions from cables:

Based on the cable forces from section 7, the force, eccentricity, and moment for each cable w.r.t the centroid is computed:

Eccentricity of cable w.r.t centroid, in t_0 :

$$e_{t0} = z_{G2,0} - z(x), \quad \text{with } z(x) = \text{Position of cable defined in Appendix E}$$

Eccentricity of cable w.r.t centroid, in t_∞ :

$$e_{t\infty} = z_{G2,\infty} - z(x)$$

Forces in cables are referred to section 8, here using:

P_0 = Force in each cable in t_0 after initial loss

ΔP_∞ = The longterm loss in each cable

ΣP_0 = Sum of forces in cables

$\Sigma \Delta P_\infty$ = Sum of longterm losses in cables

The moments due to the cables in each load situation t_0 and t_∞ can be computed:

$$M_0 = P_0 \cdot e_{t0}$$

$$\Delta M_\infty = \Delta P_\infty \cdot e_{t\infty}$$

ΣM_0 = Sum of moments due to cables

$\Sigma \Delta M_\infty$ = Sum of moments from longterm losses due to cables

2) Elastic stresses in t_0 :

Stresses due to pre-stress:

$$\sigma_{ax} = \frac{\Sigma P_0}{A_0}, \quad \sigma_{flex} = \begin{cases} \frac{\Sigma M_0}{I_{y0}} \cdot z_{G1,0} \\ \frac{\Sigma M_0}{I_{y0}} \cdot z_{G2,0} \end{cases}$$

Stresses due to structural dead load refer to section 6.3, in which all internal forces are

$$\text{computed: } \sigma_{flex} = \begin{cases} \frac{M_{g1}}{I_{y0}} \cdot z_{G1,0} \\ \frac{M_{g1}}{I_{y0}} \cdot z_{G2,0} \end{cases}$$

3) Elastic stresses in t_∞ :

Stresses due to non-structural dead load refer to section 6.3, in which all internal forces

$$\text{are computed: } \sigma_{flex} = \begin{cases} \frac{M_{g2}}{I_{t,\infty}} \cdot z_{G1,\infty} \\ \frac{M_{g2}}{I_{t,\infty}} \cdot z_{G2,\infty} \end{cases}$$

$$\text{Stresses due to long-term loss: } \sigma_{ax} = \frac{\Sigma \Delta P_\infty}{A_\infty}, \quad \sigma_{flex} = \begin{cases} \frac{\Sigma \Delta M_\infty}{I_{t,\infty}} \cdot z_{G1,\infty} \\ \frac{\Sigma \Delta M_\infty}{I_{t,\infty}} \cdot z_{G2,\infty} \end{cases}$$

4) Elastic stresses in t_m :

Stresses due to live loads (UDL and concentrated) refer to section 6.3, in which all internal forces are computed, and the value for the relevant section is used, M_q and M_Q :

$$\sigma_{flex,q} = \begin{cases} \frac{M_q}{I_{t,m}} \cdot z_{G1,m} \\ \frac{M_q}{I_{t,m}} \cdot z_{G2,m} \end{cases}, \quad \sigma_{flex,Q} = \begin{cases} \frac{M_Q}{I_{t,m}} \cdot z_{G1,m} \\ \frac{M_Q}{I_{t,m}} \cdot z_{G2,m} \end{cases}$$

The results are shown below:

SECTION I, x = 20 m							
POST-TENSION CABLES							
	e_t0	e_t∞	P_t0	ΔP_t∞	M_t0	ΔM_t∞	
	[m]	[m]	[kN]	[kN]	[kNm]	[kNm]	
1a	0.72	0.62	9360.21	1838.51	-6707.86	1141.12	
1b	0.72	0.62	9434.03	1838.51	-6760.77	1141.12	
2a	0.94	0.84	9365.73	1838.51	-8792.54	1549.57	
2b	0.94	0.84	9439.60	1838.51	-8861.88	1549.57	
3a	1.16	1.07	9384.95	1838.51	-10895.57	1958.02	
3b	1.16	1.07	9442.53	1838.51	-10962.41	1958.02	
Total			56427.05	11031.06	-52981.03	9297.43	
STRESSES, INSTANTANEOUS LOAD SITUATION t0							
		Post-tension		Struct. DL	Total	Limit	
	z [mm]	σ_ax [MPa]	σ_flex [MPa]	σ_flex [MPa]	σ_t0 [MPa]		
Top fiber	-866.60	-8.17	9.77	-6.43	-4.83	< fck = 45 MPa	
Bottom fiber	1333.40	-8.17	-15.03	9.90	-13.31	< fck = 45 MPa	
STRESSES, QUASI-PERMANENT LOAD SITUATION, t_∞							
		t_0	Non-struct. DL	Long-term loss		Total	Limit
	z [mm]	σ_t0 [MPa]	σ_flex [MPa]	σ_ax [MPa]	σ_flex [MPa]	σ_t∞ [MPa]	
Top fiber	-962.6	-4.83	-1.06	1.42	-1.67	-6.14	< 0,45*fck=20,25
Bottom fiber	1237.4	-13.31	1.36	1.42	2.14	-8.39	< 0,45*fck=20,25
STRESSES, CHARACTERISTIC LOAD SITUATION, t_m							
		t_∞	Distr. traffic load	Conc. traffic load	Total	Limit	
	z [mm]	σ_t∞ [MPa]	σ_flex [MPa]	σ_flex [MPa]	σ_tm [MPa]		
Top fiber	-903.2	-6.14	-1.67	-4.24	-12.05	< 0,6*fck = 27 MPa	
Bottom fiber	1296.8	-8.39	2.40	6.10	0.10	< 0,5 MPa	

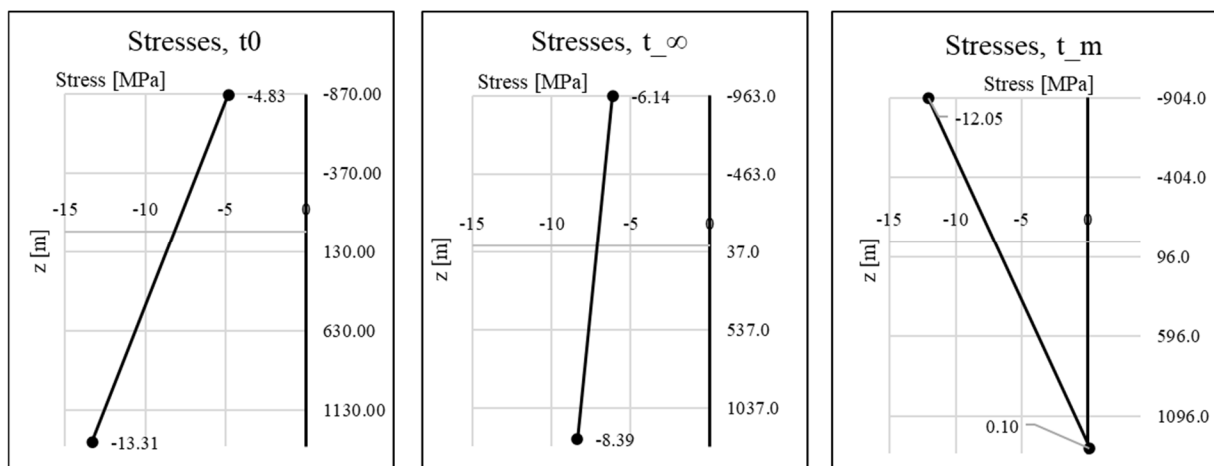


Figure 9-7: Elastic stresses, SLS, section I

SECTION I, x = 10 m							
POST-TENSION CABLES							
	e_t0	e_t∞	P_t0	ΔP_t∞	M_t0	ΔM_t∞	
	[m]	[m]	[kN]	[kN]	[kNm]	[kNm]	
1a	0.41	0.31	9196.44	1838.51	-3737.94	570.85	
1b	0.41	0.31	9236.06	1838.51	-3754.05	570.85	
2a	0.72	0.62	9190.70	1838.51	-6591.45	1142.14	
2b	0.72	0.62	9249.11	1838.51	-6633.35	1142.14	
3a	1.12	1.02	9271.79	1838.51	-10392.11	1884.24	
3b	1.12	1.02	9328.67	1838.51	-10455.87	1884.24	
Total			55472.77	11031.06	-41564.76	7194.46	
STRESSES, INSTANTANEOUS LOAD SITUATION t0							
		Post-tension		Struct. DL	Total	Limit	
	z [mm]	σ_ax [MPa]	σ_flex [MPa]	σ_flex [MPa]	σ_t0 [MPa]		
Top fiber	-866.60	-8.03	7.67	-4.82	-5.19	< fck = 45 MPa	
Bottom fiber	1333.40	-8.03	-11.79	7.42	-12.40	< fck = 45 MPa	
STRESSES, QUASI-PERMANENT LOAD SITUATION, t_∞							
		t_0	Non-struct. DL	Long-term loss		Total	Limit
	z [mm]	σ t0 [MPa]	σ flex [MPa]	σ_ax [MPa]	σ flex [MPa]	σ_∞ [MPa]	
Top fiber	-962.6	-5.19	-1.06	1.42	-1.29	-6.12	< 0,45*fck=20,25
Bottom fiber	1237.4	-12.40	1.36	1.42	1.66	-7.97	< 0,45*fck=20,25
STRESSES, CHARACTERISTIC LOAD SITUATION, t_m							
		t_∞	Distr. traffic load	Conc. traffic load	Total	Limit	
	z [mm]	σ_∞ [MPa]	σ_flex [MPa]	σ_flex [MPa]	σ_tm [MPa]		
Top fiber	-903.2	-6.12	-1.25	-3.22	-10.59	< 0,6*fck = 27 MPa	
Bottom fiber	1296.8	-7.97	1.80	4.62	-1.55	< 0,5 MPa	

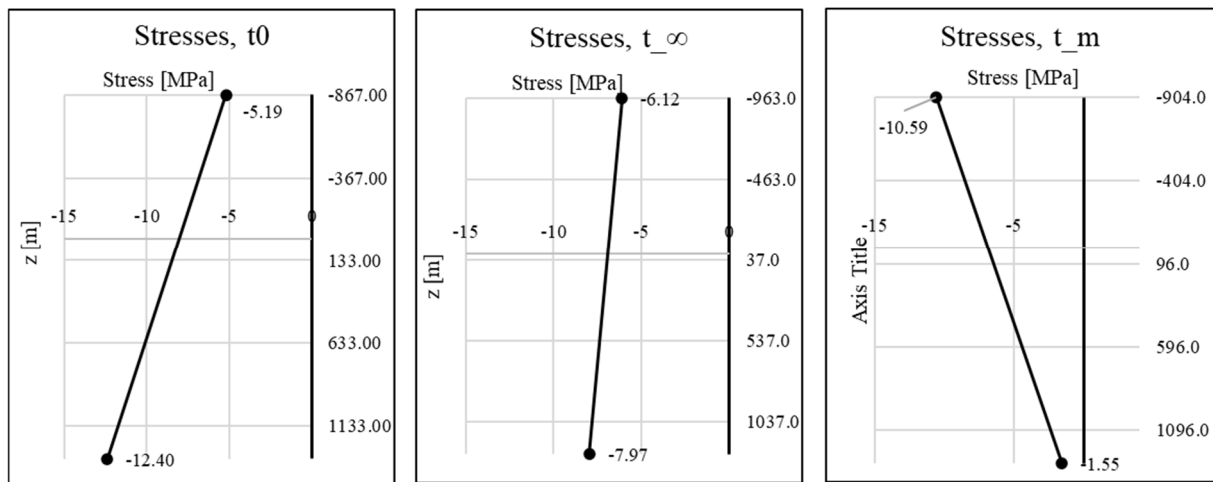


Figure 9-8: Elastic stresses, SLS, section II

SECTION I, x = 5 m							
POST-TENSION CABLES							
	e_t0	e_∞	P_t0	ΔP_∞	M_t0	ΔM_∞	
	[m]	[m]	[kN]	[kN]	[kNm]	[kNm]	
1a	0.02	-0.08	9097.55	1838.51	-170.39	-141.98	
1b	0.02	-0.08	9136.74	1838.51	-171.13	-141.98	
2a	0.35	0.26	9085.41	1838.51	-3212.66	473.69	
2b	0.35	0.26	9143.15	1838.51	-3233.08	473.69	
3a	1.03	0.93	9207.65	1838.51	-9447.77	1710.04	
3b	1.03	0.93	9264.14	1838.51	-9505.74	1710.04	
Total			54934.64	11031.06	-25740.76	4083.48	
STRESSES, INSTANTANEOUS LOAD SITUATION t0							
		Post-tension		Struct. DL	Total	Limit	
	z [mm]	σ_ax [MPa]	σ_flex [MPa]	σ_flex [MPa]	σ_t0 [MPa]		
Top fiber	-866.60	-7.95	4.75	-2.81	-6.02	< fck = 45 MPa	
Bottom fiber	1333.40	-7.95	-7.30	4.33	-10.93	< fck = 45 MPa	
STRESSES, QUASI-PERMANENT LOAD SITUATION, t_∞							
		t_0	Non-struct. DL	Long-term loss		Total	Limit
	z [mm]	σ_t0 [MPa]	σ_flex [MPa]	σ_ax [MPa]	σ_flex [MPa]	σ_∞ [MPa]	
Top fiber	-962.6	-6.02	-1.06	1.42	-0.73	-6.39	< 0,45*fck=20,25
Bottom fiber	1237.4	-10.93	1.36	1.42	0.94	-7.21	< 0,45*fck=20,25
STRESSES, CHARACTERISTIC LOAD SITUATION, t_m							
		t_∞	Distr. traffic load	Conc. traffic load	Total	Limit	
	z [mm]	σ_∞ [MPa]	σ_flex [MPa]	σ_flex [MPa]	σ_tm [MPa]		
Top fiber	-903.2	-6.39	-0.73	-1.88	-9.01	< 0,6*fck = 27 MPa	
Bottom fiber	1296.8	-7.21	1.05	2.70	-3.46	< 0,5 MPa	

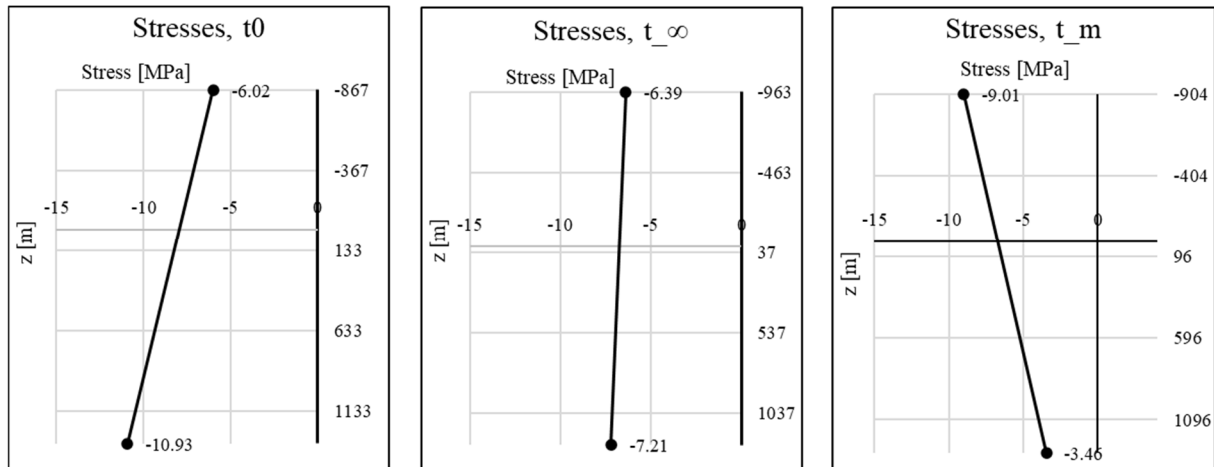


Figure 9-9: Elastic stresses, SLS, section III.

10. Shear and torsion verifications

10.1 Introduction

In order to obtain the total acting shear action in the structure, it is needed to obtain an equivalent total force, as the prestressing cables make a vertical upwards force in the inclined parts. Then, the total acting shear in the bridge is.

$$V = V_q + V_p$$

Where:

V_q is the shear induced by the acting forces,

V_p is the shear induced by the tension in cables.

The term V_q can be divided into two contributions;

- V_s , that is the contribution related to the shear force computed with longitudinal analysis; that value has been then divided by two, in order to obtain the contribution in every web, as it is a reasonable hypothesis of the fact that the vertical parts are the only ones that contribute to counteract the shear force.
- V_t , the value of resultant shear force coming from the value of shear flow (obtained from Bredt's formula) multiplied to the distance of the element, in order to obtain the resultant force, as

$$V_t = d \cdot \frac{M_t}{\Omega}$$

where,

Ω is the double of the area enclosed by the webs and slabs.

The shear force due to the prestressing cables is computed as

$$V_p = -N \cdot \sin \alpha$$

where,

α is the angle of the cable, N is the prestressing force in every cable; clearly, in the central part of the cables, the trajectory is horizontal, so no upward forces arise.

As both the shear force and the torsional moment come from a longitudinal analysis, it is position of the loads in the section that will influence the results; for example, in order to maximize the shear force, all the lanes should be loaded with the maximum possible force indicated by the codes, but for maximizing the distributed torsional moments, the loads should be calibrated with the eccentricities in order to obtain the worst case scenario.

The longitudinal load configuration that maximizes shear and torsion at this section is given by:

- (1) Permanent loads due to self-weight of the beam, which is distributed along the length and its linear variation that has been considered for the widening zone.
- (2) Permanent non-structural loads such as pavement which is distributed equally along the length of the bridge.

10.1.1 Load case A: Maximise shear

The following load case maximises the total vertical load:

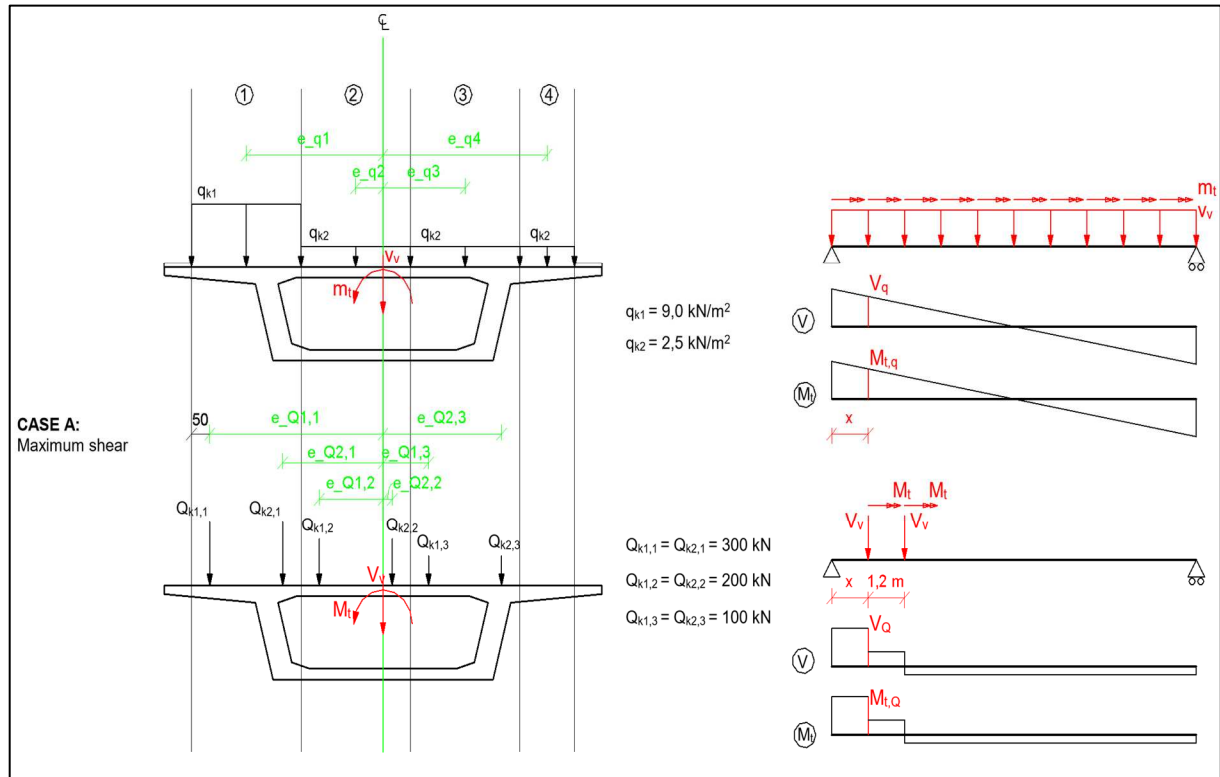


Figure 10-1: Load case configuration for maximum shear

The above figure shows the load configuration that has been assessed to obtain the values for the shear analysis. The sections for evaluation that are considered are 0m, 2m, 5m, 15m, 20m. The calculations are based on the following data:

Input Data		
Parameter	Value	Unit
Width of Box Girder, b	12	[m]
Length of span, l	40	[m]
Longitudinal distance between axle loads, cc	1,2	[m]
Width of the parapet	0,75	[m]

Table 10-1: Input parameters data

The following procedure has been adopted to compute the results:

- 1) **Calculating the contribution from cables**(Refer Appendix G), **uniformly distributed loads.** (Refer, Appendix D)

The load for every notional lane is given in detail in the following table.

Parameter	Notional lane no. i			
	1	2	3	4
W_i	3	3	3	1,5
q_{ki}	9	2,5	2,5	2,5
Q_{ki}	300	200	100	0
e_{qi}	3,75	0,75	-2,25	-4,5
e_{Q1i}	4,75	1,75	-1,25	
e_{Q2i}	3,75	-0,25	-3,25	

Table 10-2: Loads on the notional lane

Here, after the loads have been placed, we will compute the Total Shear and Torsional moments for the Case A for all the sections mentioned above.

Case A: Maximise shear							
	x	0	2	5	15	20	[m]
Effect from q_k	v_v	45,75	45,75	45,75	45,75	45,75	[kN/m]
	m_t	73,13	73,13	73,13	73,13	73,13	[kNm/m]
	V_q	915	823,50	686,25	228,75	0,00	[kN]
	$M_{t,q}$	1462,5	1316,25	1096,88	365,63	0,00	[kNm]
Effect from Q_k	v_v	1200	1200	1200	1200	1200	[kN/m]
	m_t	2100	2100	2100	2100	2100	[kNm/m]
	V_Q	2364	2244	2064	1464	1164	[kN]
	$M_{t,Q}$	4137	3927	3612	2562	2037	[kNm]
Total	V_a	3279	3067,5	2750,3	1692,8	1164,0	[kN]
	$M_{t,a}$	5599,5	5243,3	4708,9	2927,6	2037,0	[kNm]

Table 10-3: Contribution of the variable loads for Case (a)

- 2) **Design Internal Forces, ULS without the effect of pretension:**

Here, we do not consider the effect of pretension first, but later we divide the forces into both the sides of webs to incorporate the effect of pretension.

Partial safety factor for calculations : $\gamma_g = 1,35$ and $\gamma_q = 1,5$

So, the design of internal forces will be calculated in the table below:

Case A		
x [m]	V_{Ed} [kN]	M_{tEd} [kNm]
0	10422,7	8399,3
2	9555,0	7864,9
5	8253,5	7063,3
15	3915,2	4391,4
20	1746,0	3055,5

Table 10-4: Shear and Torsional Moments for Case a (Maximum Shear)

10.1.2 Load case B: Maximise torsion.

The transversal load distribution that maximizes the torsion is obtained by increasing the eccentricity of the loads. Therefore, only lanes 1 and 2 are loaded and remaining area is neglected. The following shows this distribution.

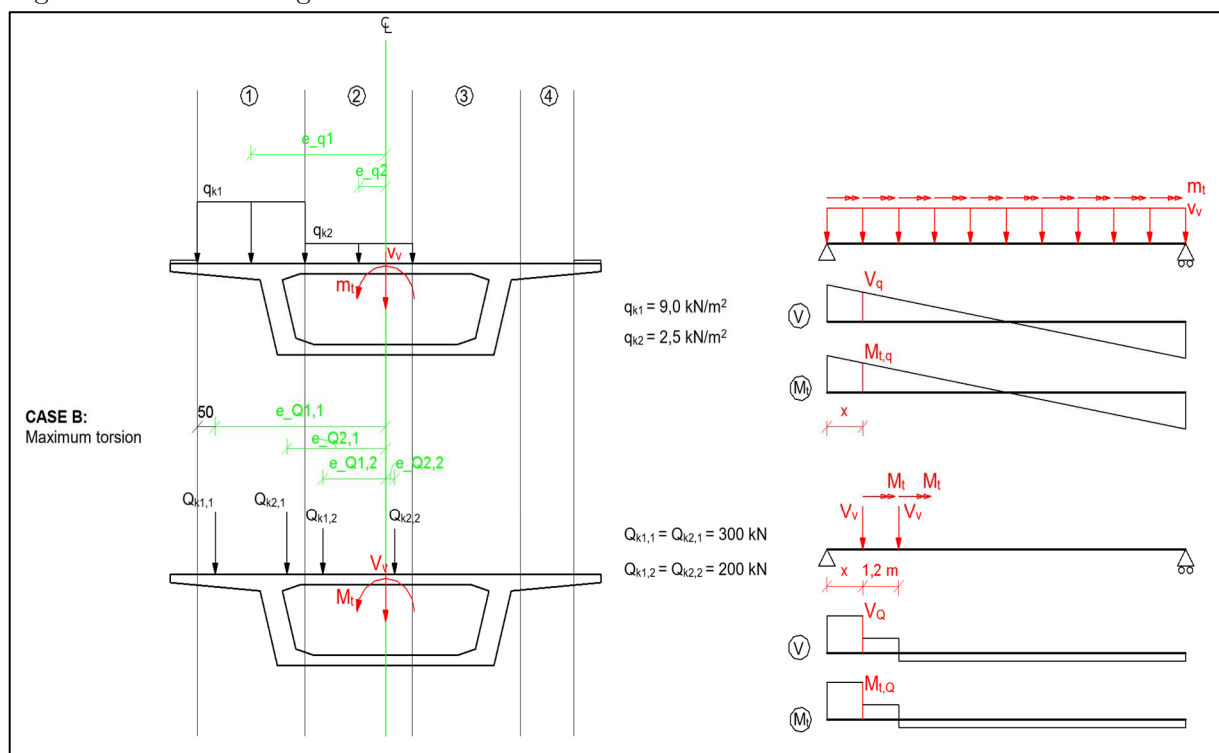


Figure 10-2: Load Case configuration for maximum torsion

The above figure shows the load configuration that has been assessed to obtain the values for the shear analysis. The sections for evaluation that are considered are 0m, 2m, 5m, 15m, 20m. The calculations are based on the following data:

The following procedure has been adopted to compute the results:

- 1) **Calculating the contribution from cables** (Refer, Appendix G), **uniformly distributed loads.** (Refer, Appendix D)

The loads on the notional lanes are mentioned in the section 10.1.1.

Here, after the loads have been placed, we will compute the Total Shear and Torsional moments for the Case (b) for all the sections mentioned above.

Case (b) (Maximum Torsion)							
	x	0	2	5	15	20	[m]
qk	v_v	34.5	34.5	34.5	34.5	34.5	[kN/m]
	m_t	106.88	106.88	106.88	106.88	106.88	[kNm/m]
	V_q	690	621	517.5	172.5	0	[kN]
	$M_{t,q}$	2137.5	1923.75	1603.125	534.375	0	[kNm]
Qk	v_v	1000	1000	1000	1000	1000	[kN/m]
	m_t	2550	2550	2550	2550	2550	[kNm/m]
	V_Q	1970	1870	1720	1220	970	[kN]
	$M_{t,Q}$	5023.5	4768.5	4386	3111	2473.5	[kNm]
Total	v_a	2660	2491	2237.5	1392.5	970	[kN]
	$M_{t,a}$	7161	6692.25	5989.125	3645.375	2473.5	[kNm]

Table 10-5: Contribution of the variable loads for case (b)

2) Design Internal Forces, ULS without the effect of pretension

Here, we do not consider the effect of pretension first, but later we divide the forces into both the sides of webs to incorporate the effect of pretension.

Partial safety factor for calculations: $\gamma_g = 1,35$ and $\gamma_q = 1,5$.

So, the design of internal forces will be calculated in the table below:

Case B		
x [m]	V_{Ed} [kN]	M_{tEd} [kNm]
0	9494,2	10741,5
2	8690,3	10038,4
5	7484,4	8983,7
15	3464,8	5468,1
20	1455,0	3710,3

Table 10-6: Shear and Torsional Moments for Case b (Maximum Torsion)

10.2 Verifications of shear and torsion of sections

The verification of shear and torsion, i.e. design of stirrups, follows the schematisation shown below:

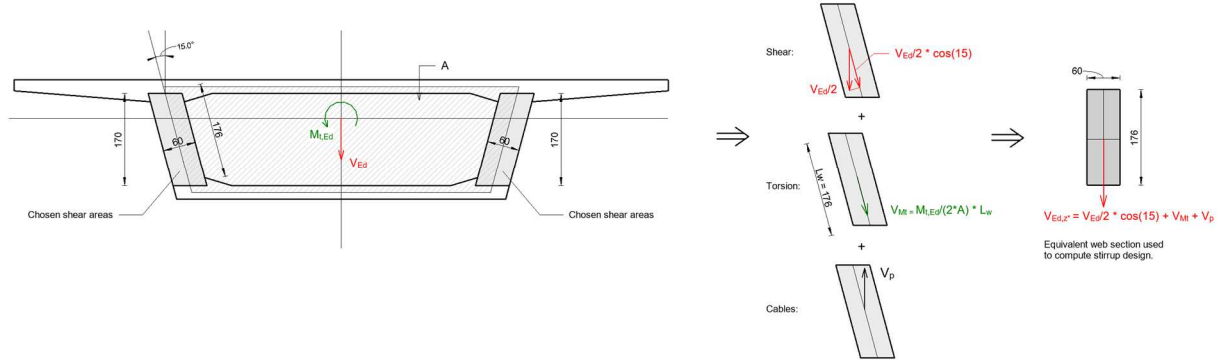


Figure 10-3: Scheme for the design of stirrups.

The properties of the materials and geometry considered is given here:

Materials		
Concrete		
f_{ck}	45	MPa
γ_c	1,5	[-]
f_{cd}	30	MPa
v_1	0,492	[-]
α_{cw}	1	[-]
Rebar steel		
f_{yk}	500	MPa
γ_s	1,2	[-]
f_{yd}	416,7	MPa
Geometry		
Height of the c/s, h	2,2	m
Compression angle, θ	30	deg (EC2.1.1)
$\cot(\theta)$	1,73	[-]

Table 10-7: Material and Geometry for calculation

The required shear forces can now be calculated using these equations:

$$V_{rd,s} = \frac{A_{sw} \cdot f_{yd} \cdot z \cdot \cot \vartheta}{s}$$

$$V_{rd,max} = \frac{\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}}{\cot \vartheta + \tan \vartheta}$$

Below we have the calculations for both the maximum shear (case a) and torsion (case b) cases and their verifications.

SECTION I, x = 0 m				
	Case a	Case b		
Ω	11,2		mm^2	
α	15		deg	Inclination of web
h_w	1760,0		mm	Height of web
b_w	850		mm	Width of web
z	1584,0		mm	
$\emptyset$	16		mm	Stirrup diameter
s	100		mm	Stirrup spacing
A_{sw}, z^*	402,12		mm^2	Area of stirrups
q	749,93	959,06	kN/m	Shear flow due to torsion
V_{Mt}	1514,0	1936,1	kN	Shear force in web due to torsion
V_{Ed,z^*}	4323,4	4297,2	kN	Design shear force in web w. effect of pre-tension
V_{Rd,s,z^*}	4596,8		kN	$V_{Rd,s} = \frac{A_{sw} \cdot f_{yd} \cdot z \cdot \cot(\vartheta)}{s}$
V_{Rd,max,z^*}	8605,1		kN	$V_{Rd,max} = \frac{\alpha_{cw} \cdot b_w \cdot z \cdot \nu_1 \cdot f_{cd}}{\cot \vartheta + \tan \vartheta}$
V_{Rd,z^*}	4596,8		kN	

Table 10-8: Forces at section x= 0 [m]

SECTION II, x = 2 m				
	Case a	Case b		
Ω	11,4		mm^2	
α	15		deg	Inclination of web
h_w	1760,0		mm	Height of web
b_w	750		mm	Width of web
z	1584,0		mm	
$\emptyset$	16		mm	Stirrup diameter
s	100		mm	Stirrup spacing
A_{sw}, z^*	402,12		mm^2	Area of stirrups
q	689,90	880,56	kN/m	Shear flow due to torsion
V_{Mt}	1392,8	1777,7	kN	Shear force in web due to torsion
V_{Ed,z^*}	3777,9	3745,1	kN	Design shear force in web w. effect of pre-tension
V_{Rd,s,z^*}	4596,8		kN	$V_{Rd,s} = \frac{A_{sw} \cdot f_{yd} \cdot z \cdot \cot(\vartheta)}{s}$
V_{Rd,max,z^*}	7592,7		kN	$V_{Rd,max} = \frac{\alpha_{cw} \cdot b_w \cdot z \cdot \nu_1 \cdot f_{cd}}{\cot \vartheta + \tan \vartheta}$
V_{Rd,z^*}	4596,8		kN	

Table 10-9: Forces at section x= 2 [m]

SECTION III, x = 5 m				
	Case a	Case b		
Ω	11,7		mm^2	
α	15		deg	Inclination of web
h_w	1760,0		mm	Height of web
b_w	600		mm	Width of web
z	1584,0		mm	
$\emptyset$	16		mm	Stirrup diameter
s	150		mm	Stirrup spacing
A_{sw}, z^*	402,12		mm^2	Area of stirrups
q	603,70	767,84	kN/m	Shear flow due to torsion
V_{Mt}	1218,7	1550,1	kN	Shear force in web due to torsion
V_{Ed,z^*}	2944,4	2904,3	kN	Design shear force in web w. effect of pre-tension
V_{Rd,s,z^*}	3064,5		kN	$V_{Rd,s} = \frac{A_{sw} \cdot f_{yd} \cdot z \cdot \cot(\vartheta)}{s}$
V_{Rd,max,z^*}	6074,2		kN	$V_{Rd,max} = \frac{\alpha_{cw} \cdot b_w \cdot z \cdot \nu_1 \cdot f_{cd}}{\cot \vartheta + \tan \vartheta}$
V_{Rd,z^*}	3064,5		kN	

Table 10-10: Forces at section x= 5 [m]

SECTION IV, x = 15 m				
	Case a	Case b		
Ω	11,7		mm^2	
α	15		deg	Inclination of web
h_w	1760,0		mm	Height of web
b_w	600		mm	Width of web
z	1584,0		mm	
$\emptyset$	16		mm	Stirrup diameter
s	200		mm	Stirrup spacing
A_{sw}, z^*	402,12		mm^2	Area of stirrups
q	375,34	467,36	kN/m	Shear flow due to torsion
V_{Mt}	757,7	943,5	kN	Shear force in web due to torsion
V_{Ed,z^*}	1989,7	1958,0	kN	Design shear force in web w. effect of pre-tension
V_{Rd,s,z^*}	2298,4		kN	$V_{Rd,s} = \frac{A_{sw} \cdot f_{yd} \cdot z \cdot \cot(\vartheta)}{s}$
V_{Rd,max,z^*}	6074,2		kN	$V_{Rd,max} = \frac{\alpha_{cw} \cdot b_w \cdot z \cdot \nu_1 \cdot f_{cd}}{\cot \vartheta + \tan \vartheta}$
V_{Rd,z^*}	2298,4		kN	

Table 10-11: Forces at section x= 15 [m]

SECTION V, x = 20 m				
	Case a	Case b		
Ω	11,7		mm^2	
α	15		deg	Inclination of web
h_w	1760,0		mm	Height of web
b_w	600		mm	Width of web
z	1584,0		mm	
ϕ	16		mm	Stirrup diameter
s	200		mm	Stirrup spacing
A_{sw}, z^*	402,12		mm^2	Area of stirrups
q	261,15	317,12	kN/m	Shear flow due to torsion
V_{Mt}	527,2	640,2	kN	Shear force in web due to torsion
V_{Ed,z^*}	1370,5	1342,9	kN	Design shear force in web w. effect of pre-tension
V_{Rd,s,z^*}	2298,4		kN	$V_{Rd,s} = \frac{A_{sw} \cdot f_{yd} \cdot z \cdot \cot(\vartheta)}{s}$ $V_{Rd,max} = \frac{\alpha_{cw} \cdot b_w \cdot z \cdot \nu_1 \cdot f_{cd}}{\cot \vartheta + \tan \vartheta}$
V_{Rd,max,z^*}	6074,2		kN	
V_{Rd,z^*}	2298,4		kN	

Table 10-12: Forces at section x= 20 [m].

Part C: Box girder transversal analysis

11. Introduction

This chapter explains the transversal analysis of the box-girder cross section to design the transversal steel reinforcement of the bridge. The transversal analysis is based on a reduced box model of the girder, in which a segment of 1 m of slab is loaded and distributed along the height of the section. Two methods are applied in the analysis:

- The effect of concentrated loads on the top slab is analysed by implementing Homberg's Influence surfaces.
- The effect of the distributed loads is analysed by means of FEM in Autodesk Robot, in which the effect of the concentrated loads is subsequently added.

To conclude, the results of these two approaches are combined to obtain the moment envelope for top slab, webs, and bottom slab. The analyses are based on the following representation of the cross-section:

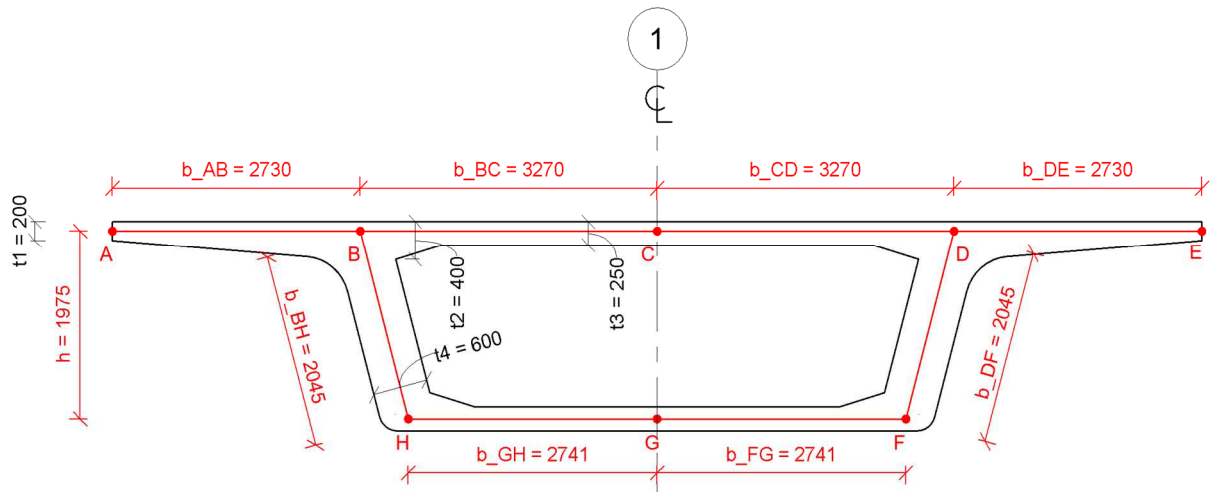


Figure 11-1: Bridge cross-section for transversal analysis.

The structure of the chapter is as follows:

- Computations of bending moments due to concentrated loads are explained in **Section 13** through Homberg's analysis,
- Combined loads are explained in **Section 14** through FE-model.
- Finally, safety verifications of transversal rebar are carried out in **Section 16**.

12. Load cases and combinations

12.1 Structural and non-structural permanent loads

The bending moment due to structural and non-structural dead loads are computed with considering following load distributions. The bending moment values are given in **Section 14.1**, FE-Model using Autodesk Robot.

$$g_{k,1} = 25 \frac{kN}{m^3} \cdot \frac{279 \text{ m}^3}{40 \text{ m} \cdot 12 \text{ m}} \cdot 1 \text{ m} = \frac{465 \cdot kN}{32 \cdot m} = 14,53 \frac{kN}{m}$$

$$g_{k,2} = 29,48 \frac{kN}{m} \cdot \frac{1}{12 \text{ m}} \cdot 1 \text{ m} = 2,46 \cdot \frac{kN}{m}$$

Load Case	Structural Dead Load	Non-structural Dead Load
Value	$14,53 \frac{kN}{m}$	$2,46 \text{ kN/m}$

Table 12-1 Summarized dead load values

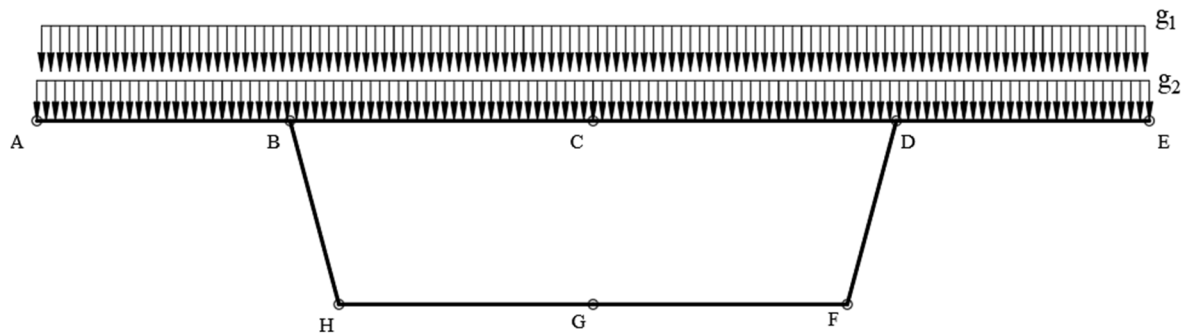


Figure 12-1: Structural and non-structural permanent load

12.2 Live loads

The bending moment due to distributed and concentrated live loads are computed with considering following load distributions. The bending moment values are given in **Section 1413**, FE-Model using AUTODESK ROBOT.

Live Load Case 1

Load Case	$Q_{1,i}$	$Q_{2,i}$	$Q_{3,i}$	q_1	q_2, q_3
Value	300 kN	200 kN	100 kN	9,5 kN/m	2,5 kN/m

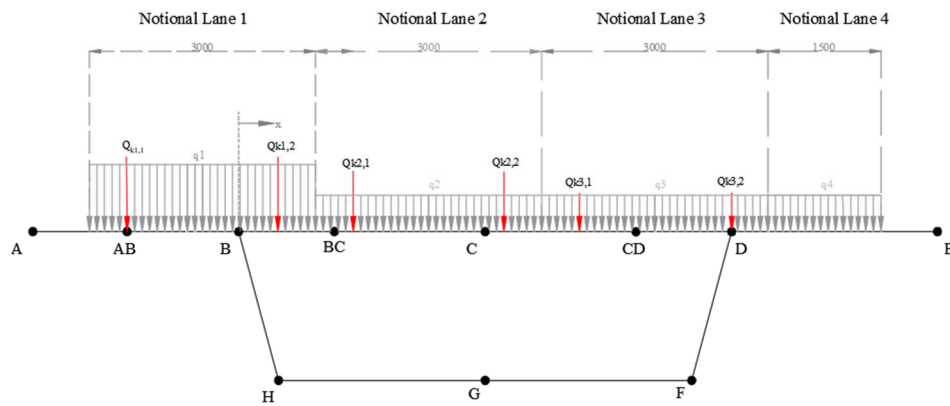


Figure 12-2 Static scheme for live load case 1

Live Load Case 2

Load Case	$Q_{1,i}$	$Q_{2,i}$	q_1	q_2
Value	300 kN	200 kN	9,5 kN/m	2,5 kN/m

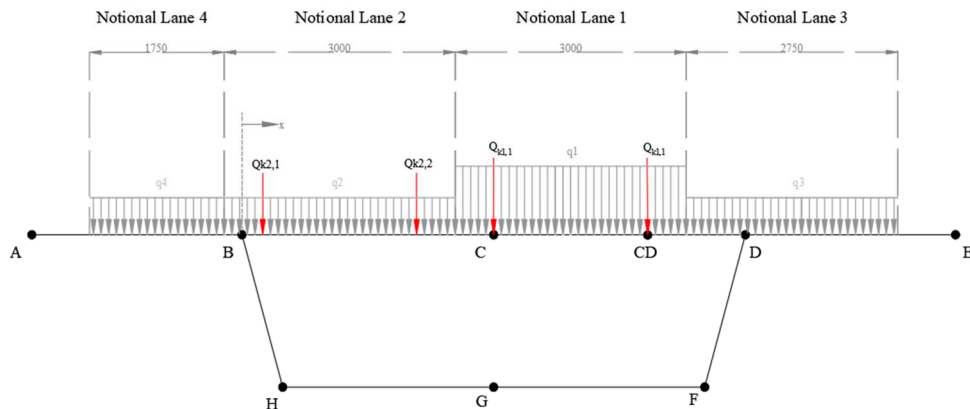


Figure 12-3 Static scheme for live load case 2

Live Load Case 3

Load Case	$Q_{1,i}$	$Q_{2,i}$	$Q_{3,i}$	q_1	q_2, q_3
Value	300 kN	200 kN	100 kN	9,5 kN/m	2,5 kN/m

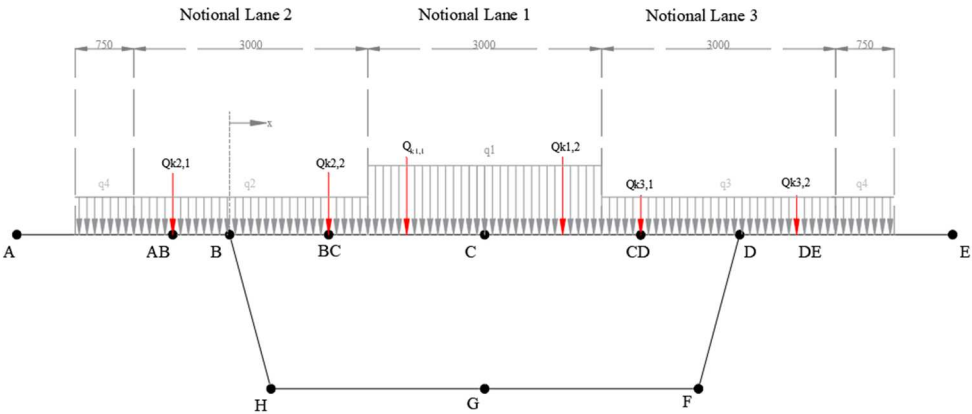


Figure 12-4 Static scheme for live load case 3

Live Load Case 4

Load Case	$Q_{1,i}$	$Q_{2,i}$	q_1	q_2
Value	300 kN	200 kN	9,5 kN/m	2,5 kN/m

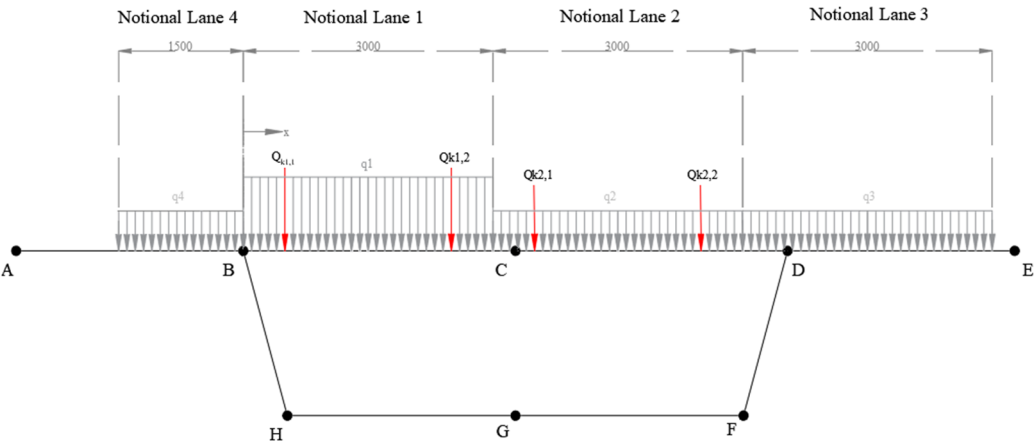


Figure 12-5 Static scheme for live load case 4

13. Analysis through influence surfaces

To investigate the effects of concentrated live loads such as traffic loads (vehicles) acting on the structure, Homberg's influence surface model is considered for computations of finding the maximizing bending moment, thus the designing the top and bottom slab reinforcement.

Homberg surfaces are a set of contour plots which essentially gives you the bending moment at specific points on a structural surface due to a moving, concentrated load Q . The Homberg surfaces used in this analysis are the ones referring to the following schemes:



Figure 13-1: Schemes for Homberg's Approach

For the analysis of concentrated loads, top slab is divided into 5 sections to observe the bending moments in each these 5 sections.

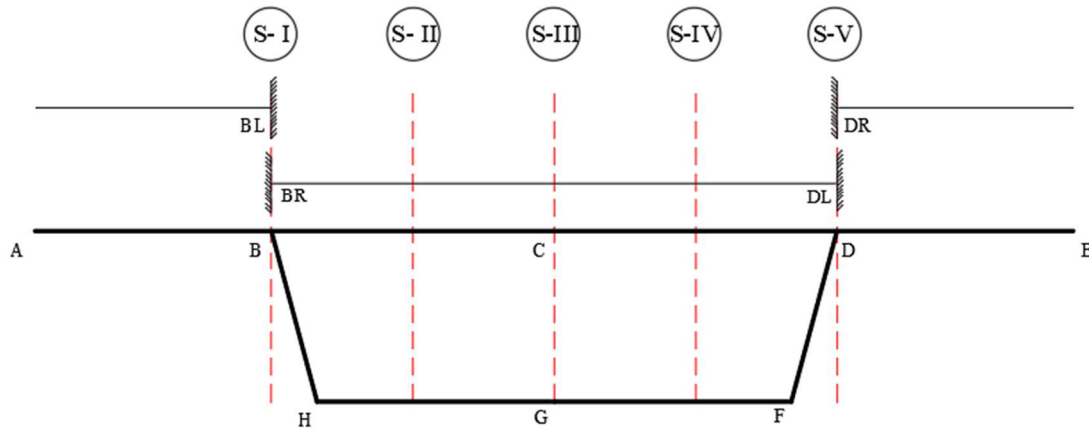


Figure 13-2: Cross-sections for top slab

Homberg charts are used to find bending moments. For each load, Q (vehicle wheel), applied on the Homberg's chart, first it is necessary to determine the dimensions of load footprint on the surface. It is defined in EN1991.2.2003 4.3.6 as dispersal of concentrated loads that needs to be considered for local verifications. The dispersal through the pavement and concrete slabs is defined as follows:

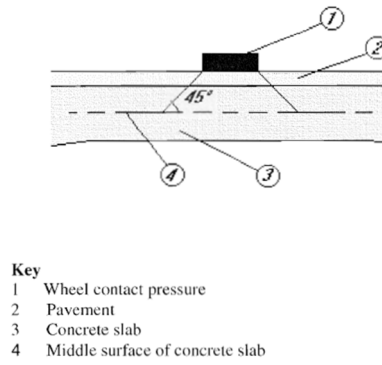


Figure 13-3: Dispersal of concentrated loads EN1991.2.2003

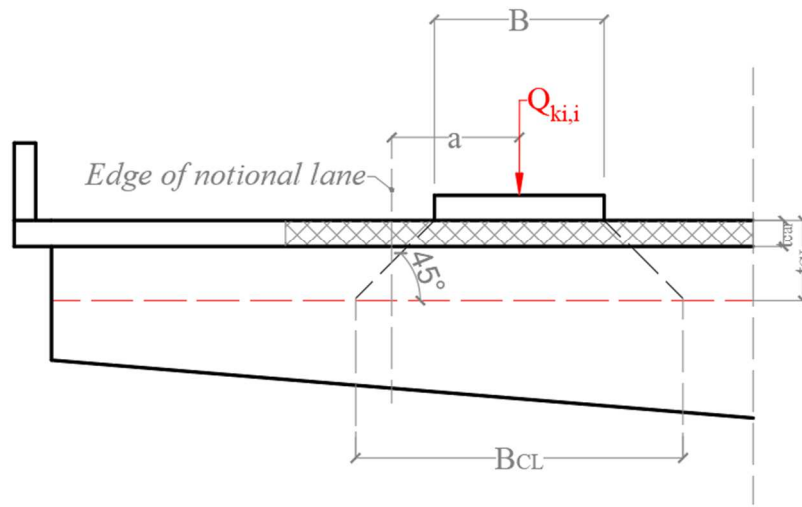


Figure 13-4: Scheme for concentrated loads

Where:

$$\begin{aligned}
 B_{CL} &= B + 2 \cdot t_{CL} \cdot \tan(45^\circ) = B + 2 \cdot \left(t_{car} + \frac{t_1}{2} \right) \cdot \tan(45^\circ) \\
 &= 0,4 \text{ m} + 2 \cdot \left(0,1 \text{ m} + \frac{0,2 \text{ m}}{2} \right) \cdot \tan(45^\circ) = 0,8 \text{ m}
 \end{aligned}$$

According to given figure in Eurocode, the width of load application is selected as 40x40 cm by considering the angle of diffusion of concentrated load as 45 degrees, hence the width of load application at centre line of top slab is 80x80 cm.

The moment m_x of Homborg surfaces is defined by following analytical equation, considering the load is constant gives:

$$m_x = \int_{x_1}^{x_2} \int_{y_1}^{y_2} q(x, y) \cdot \zeta(x, y) \cdot dx \cdot dy = q \cdot a \cdot b \cdot \left[\frac{1}{a \cdot b} \int_{x_1}^{x_2} \int_{y_1}^{y_2} \zeta(x, y) \cdot dx \cdot dy \right] = \zeta_{med} \cdot Q$$

Where:

Q : Applied concentrated load.

ζ_{med} : Mean value of approximate volume below the surface, computed through Simpson's rule.

Application of Simpson's rule will give a numerical solution for a grid net 3x3 that contains the points corresponding to value of the height of the influence surface. The grid net is illustrated as follows:

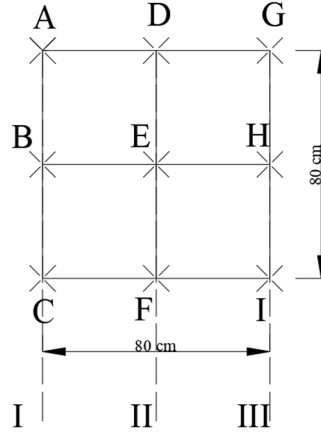


Figure 13-5: Loading Area

Finally, the equation becomes:

$$m_x = Q \cdot \zeta_{med}$$

ζ_{med} is evaluated for each line given in the figure above, ζ_I , ζ_{II} , ζ_{III} , and then the following rule is applied:

$$\zeta_{med} = \frac{\Delta x}{3} \cdot (\zeta_I + 4 \cdot \zeta_{II} + \zeta_{III}) \cdot \frac{1}{2 \cdot \Delta x}$$

Where:

$$\begin{aligned} \zeta_I &= \frac{\Delta y}{3} \cdot (z_A + 4 \cdot z_B + z_C) \cdot \frac{1}{2 \cdot \Delta y} \\ \zeta_{II} &= \frac{\Delta y}{3} \cdot (z_D + 4 \cdot z_E + z_F) \cdot \frac{1}{2 \cdot \Delta y} \\ \zeta_{III} &= \frac{\Delta y}{3} \cdot (z_G + 4 \cdot z_H + z_I) \cdot \frac{1}{2 \cdot \Delta y} \end{aligned}$$

Each load on the top slab is applied one by one and the bending moments are read accordingly. Following charts represents total moment curve for Load Case 1, Load Case 2, Load Case 3, Load Case 4. Computation of Live Load Case 1, $Q_{k1,i}$ (i : lane number) is given in Appendix.

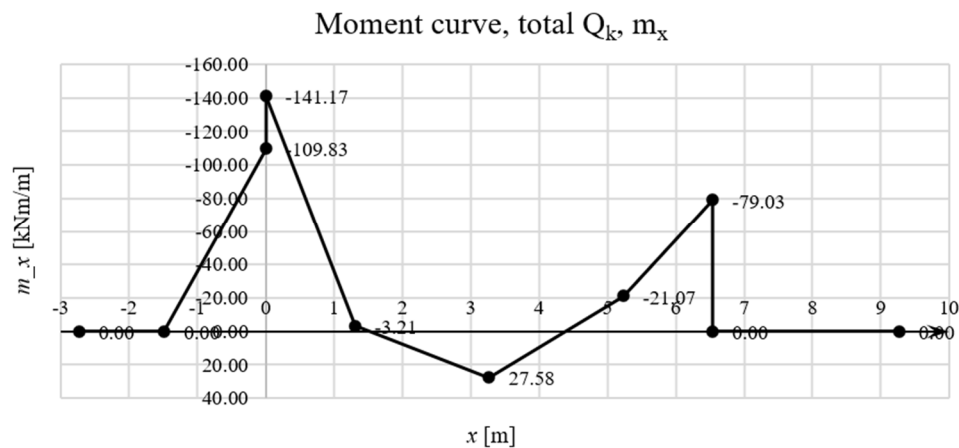


Figure 13-6: Moment curve for Concentrated Live Load Case 1

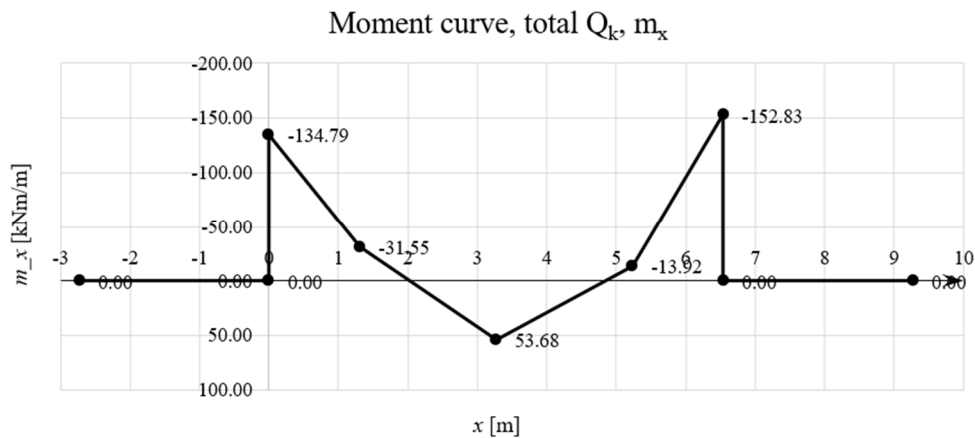


Figure 13-7: Moment curve for Concentrated Live Load Case 2

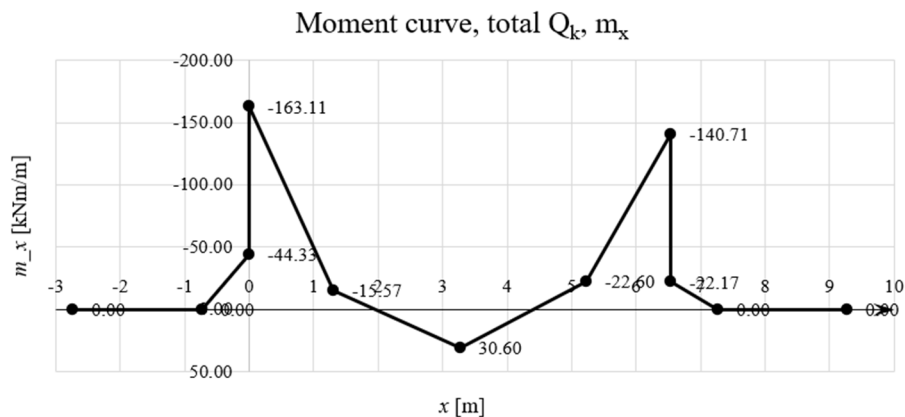


Figure 13-8: Moment curve for Concentrated Live Load Case 3

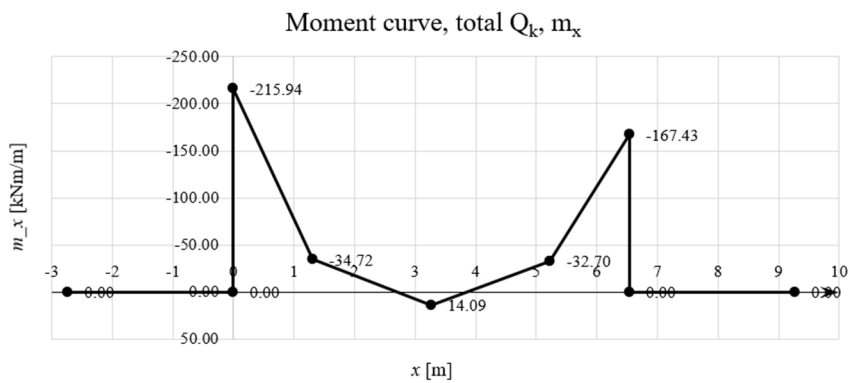


Figure 13-9: Moment curve for Concentrated Live Load Case 4

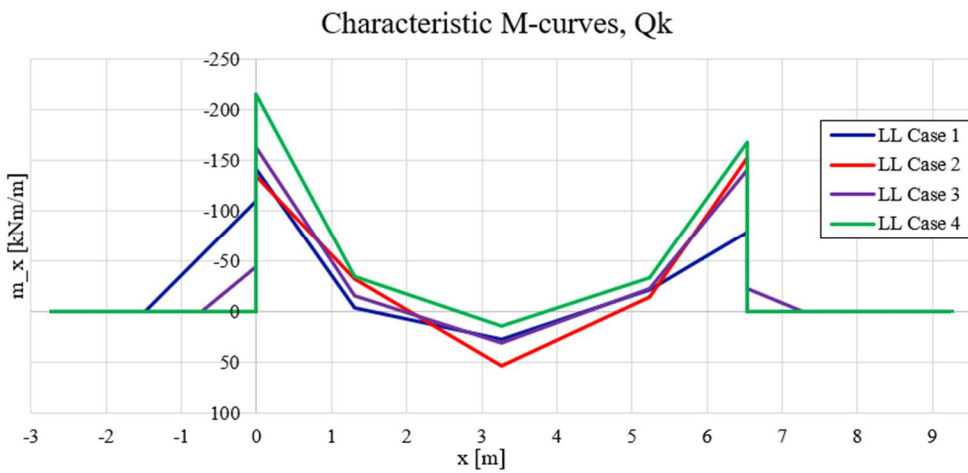


Figure 13-10: Characteristic M-curves

All computations are added to the following table that shows both characteristics and design bending moments for concentrated loads.

m_x : Characteristic moment values

$m_{x,Ed}$: Design moment values

These moments are later applied on the finite element model of the cross section as distributed moments at the nodes that is shown in the following sections.

CONCENTRATED FORCES										
		LL Case 1		LL Case 2		LL Case 3			LL Case 4	
Point	x [m]	m x [kNm/m]	m x,Ed [kNm/m]	m x [kNm/m]	m x,Ed [kNm/m]	x [m]	m x [kNm/m]	m x,Ed [kNm/m]	m x [kNm/m]	m x,Ed [kNm/m]
A	-2.73	0.00	0.00	0.00	0.00	-2.73	0.00	0.00	0.00	0.00
AB	-1.48	0.00	0.00	0.00	0.00	-0.73	0.00	0.00	0	0.00
BR	0.00	-109.83	-164.75	0	0.00	0	-44.33	-66.50	0	0.00
BL	0.00	-141.17	-211.75	-134.79	-202.18	0	-163.11	-244.66	-215.94	-323.91
BC	1.31	-3.21	-4.82	-31.55	-47.33	1.308	-15.57	-23.36	-34.72	-52.07
C	3.27	27.58	41.38	53.68	80.53	3.27	30.60	45.90	14.09	21.13
CD	5.23	-21.07	-31.60	-13.92	-20.88	5.232	-22.60	-33.90	-32.70	-49.05
DR	6.54	-79.03	-118.54	-152.83	-229.25	6.54	-140.71	-211.07	-167.43	-251.15
DL	6.54	0.00	0.00	0.00	0.00	6.54	-22.17	-33.25	0.00	0.00
	6.54	0	0	0.00	0.00	7.27	0.00	0.00	0	0.00
E	9.27	0.00	0.00	0.00	0.00	9.27	0.00	0.00	0.00	0.00

Table 13-1: Moment values for concentrated loads.

14. Reduced Box Model

The construction of a reduced box model consists in this case in attributing, to the top and bottom slabs and to the lateral walls, flexural stiffnesses suitable to represent the flexural effects in the box perimeter. Reduced frame model distributes the fixed-end moments at the side of the top slab in the other elements, webs, in this case lateral walls, and bottom slab according to structural response. As for the plates, the moments are given in per unit width.

The criterion for defining the stiffness of the box walls is described as follows:

- For the top slab, it is considered a strip of length, $l_0 = 1 \text{ m}$
- For the lateral walls, it is assumed a 45-degree diffusion. Therefore, the mean length of the diffusion results in the sum of l_0 and of the height of the lateral walls.
($l_1 = l_0 + h$)

These criteria are illustrated as follows:

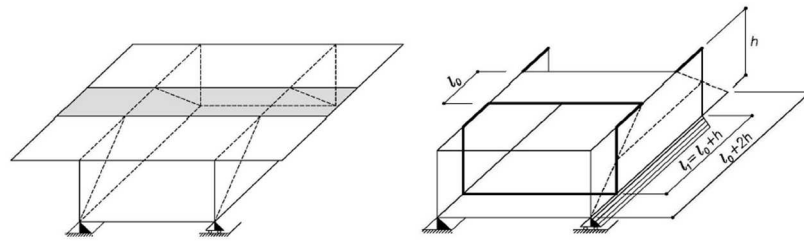


Figure 14-1: Schematics of a reduced box model.

14.1 FE-Model using Autodesk ROBOT

In this section, first application of load cases is shown, later the bending moment diagram for the top slab due to structural load, non-structural load, distributed load, and concentrated load is shown by exploiting the superposition principle. Combinations are summarized in the following table:

Load Combinations				
Load Cases	ULS Comb 1	ULS Comb 2	ULS Comb 3	ULS Comb 4
Structural DL	1.35	1.35	1.35	1.35
Non-Structural DL	1.35	1.35	1.35	1.35
LL LC 1 Concentrated	1.5	-	-	-
LL LC 2 Concentrated	-	1.5	-	-
LL LC 3 Concentrated	-	-	1.5	-
LL LC 4 Concentrated	-	-	-	1.5
LL LC1 Distributed	1.5	-	-	-
LL LC2 Distributed	-	1.5	-	-
LL LC3 Distributed	-	-	1.5	-
LL LC4 Distributed	-	-	-	1.5

Table 14-1: Load combinations applied on the FE-Model

Application of each load case is shown on the FE-Model in the next pages.

Top Slab: structural dead load:

$$g_{k,1} = 25 \frac{kN}{m^3} \cdot \frac{279 \text{ m}^3}{40 \text{ m} \cdot 12 \text{ m}} \cdot 1 \text{ m} = \frac{465 \cdot kN}{32 \cdot m} = 14,53 \text{ kN/m}$$

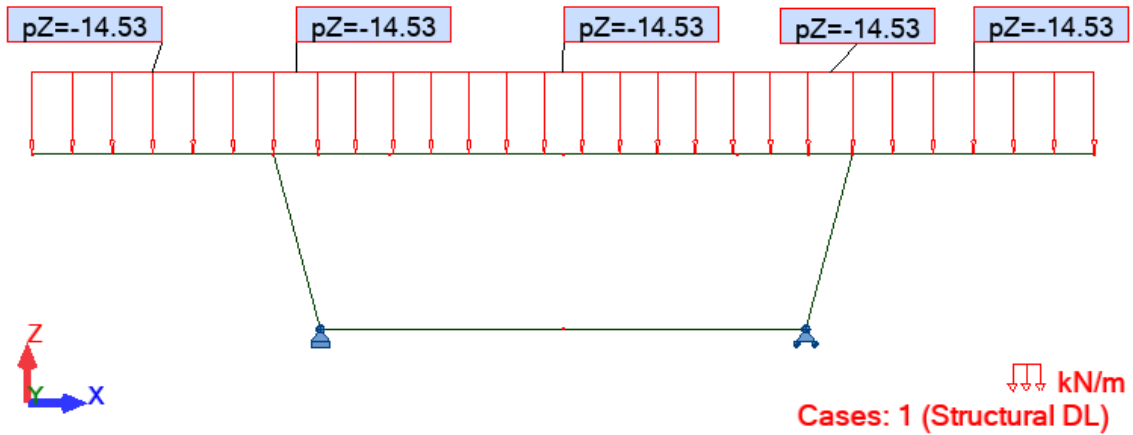


Figure 14-2: Structural dead load applied in FEM

Top Slab: non-structural dead load:

$$g_{k,2} = 29,48 \frac{kN}{m} \cdot \frac{1}{12 \text{ m}} \cdot 1 \text{ m} = 2,46 \cdot \frac{kN}{m}$$

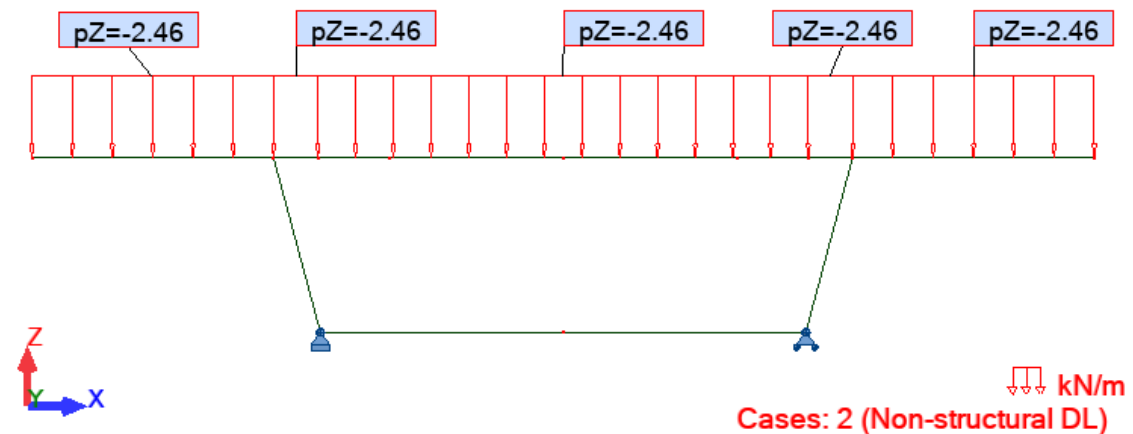


Figure 14-3: Non-structural dead load applied in FEM

Live Load Case 1: Concentrated

$$m_{x,BR} - m_{x,BL} = -141,17 \frac{kN}{m} - \left(-109,83 \frac{kN}{m} \right) = \frac{-31,34 \cdot kN}{m}$$

$$m_{x,DR} - m_{x,DL} = -79,03 \frac{kN}{m}$$

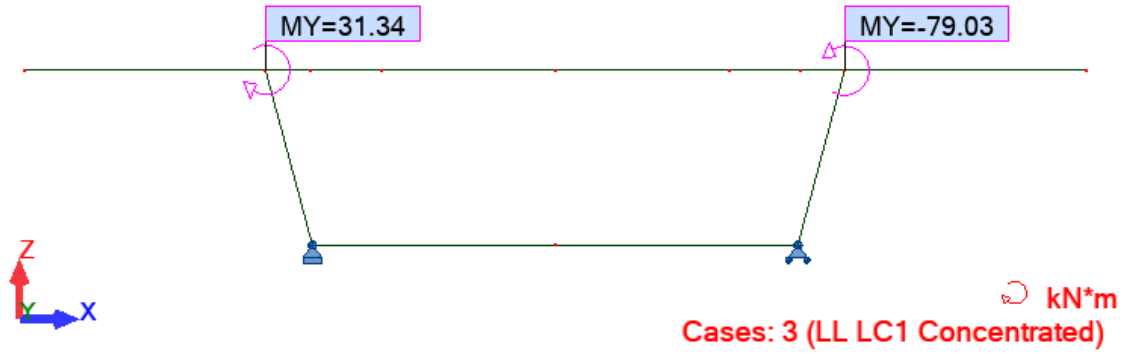


Figure 14-5: Live Load Case 1(Concentrated) in FEM

Live Load Case 2: Concentrated

$$m_{x,BR} - m_{x,BL} = -134,79 \text{ kNm}$$

$$m_{x,DR} - m_{x,DL} = -152,83 \text{ kNm}$$

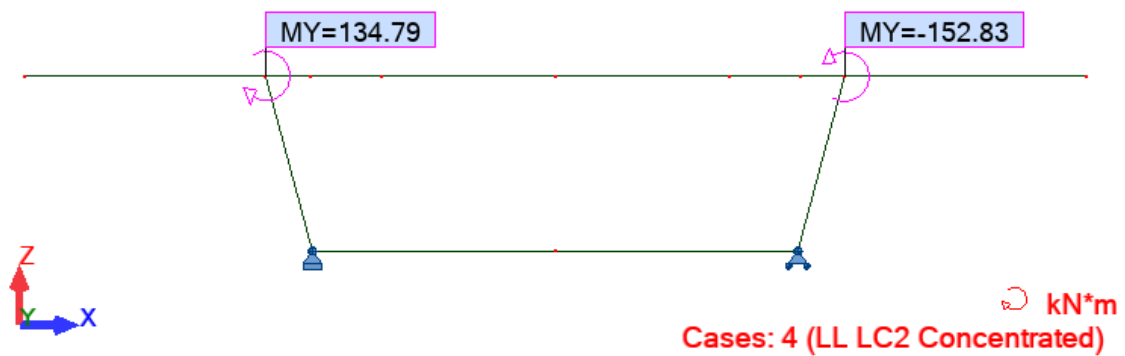


Figure 14-6 Live Load Case 2 (Concentrated) in FEM

Live Load Case 3: Concentrated

$$m_{x,BR} - m_{x,BL} = -163,11 \text{ kNm} - (-44,33 \text{ kNm}) = -118,78 \cdot \text{kNm}$$

$$m_{x,DR} - m_{x,DL} = -140,71 \text{ kNm} - (-22,17 \text{ kNm}) = -118,54 \cdot \text{kNm}$$

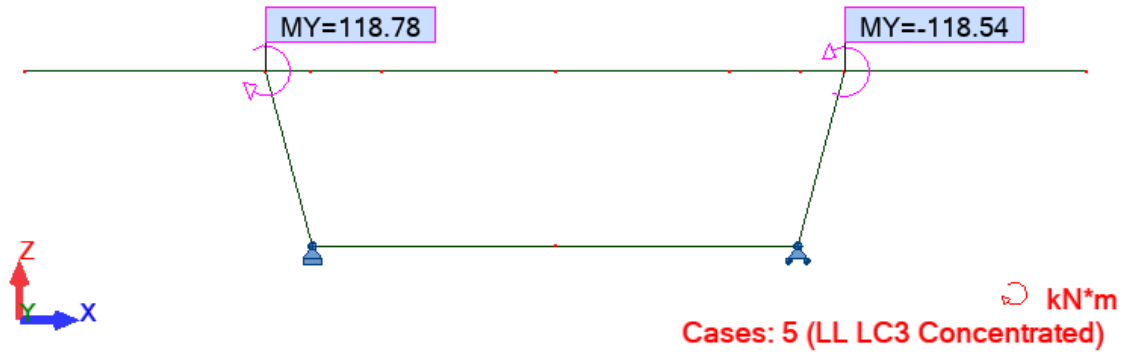


Figure 14-7 Live Load Case 3(Concentrated) in FEM

* $m_{x,jR}$: the bending moment at node j right $m_{x,jL}$: the bending moment at node j left

Live Load Case 4: Concentrated

$$m_{x,BR} - m_{x,BL} = -215.94 \text{ kNm}$$

$$m_{x,DR} - m_{x,DL} = -167.43 \text{ kNm}$$

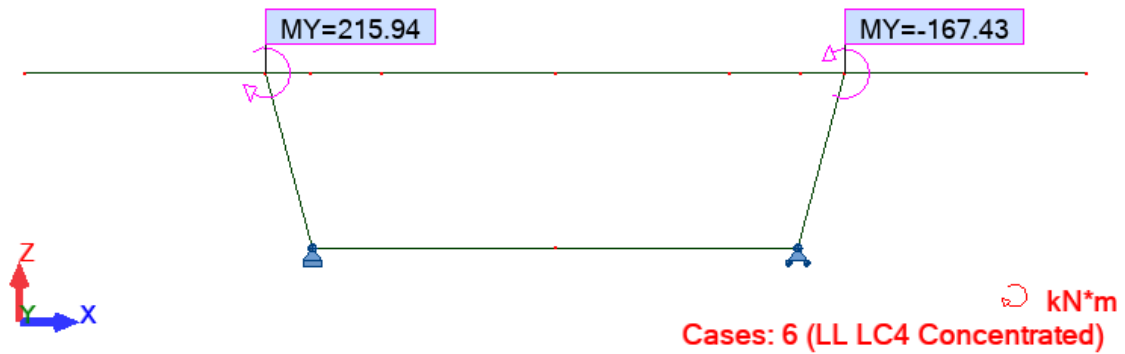


Figure 14-8 Live Load Case 4 (Concentrated) in FEM

* $m_{x,jR}$: the bending moment at node j right $m_{x,jL}$: the bending moment at node j left

Live Load Case 1: Distributed

$$q_1 = 9,5 \frac{kN}{m}, q_2 = q_3 = q_3 = 2,5 \frac{kN}{m}$$

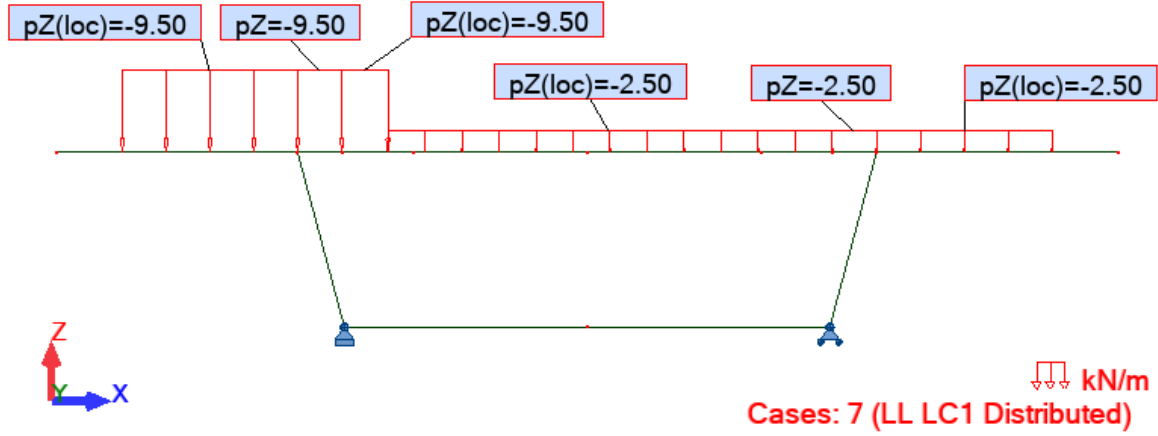


Figure 14-9 Live Load Case 1 (Distributed) in FEM

Live Load Case 2: Distributed

$$q_1 = 9,5 \frac{kN}{m}, q_2 = q_3 = q_3 = 2,5 \frac{kN}{m}$$

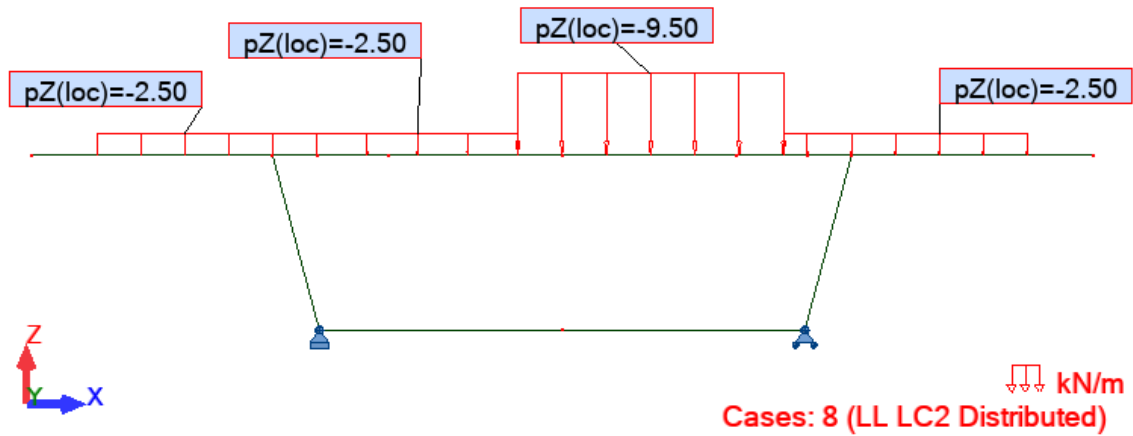


Figure 14-10 Live Load Case 2 (Distributed) in FEM

Live Load Case 3: Distributed

$$q_1 = 9,5 \frac{kN}{m}, q_2 = q_3 = q_3 = 2,5 \frac{kN}{m}$$

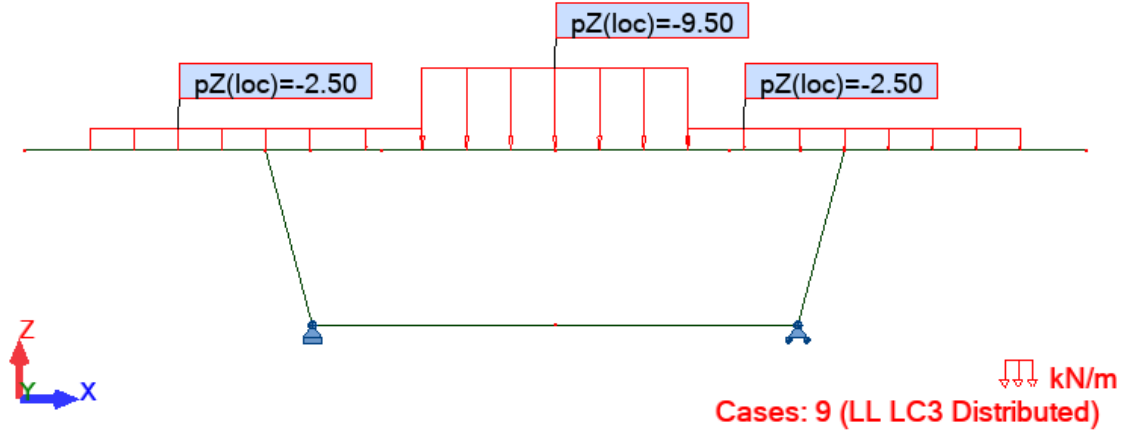


Figure 14-11: Live Load Case 3 (Distributed) in FEM

Live Load Case 4: Distributed

$$q_1 = 9,5 \frac{kN}{m}, q_2 = q_3 = q_3 = 2,5 \frac{kN}{m}$$

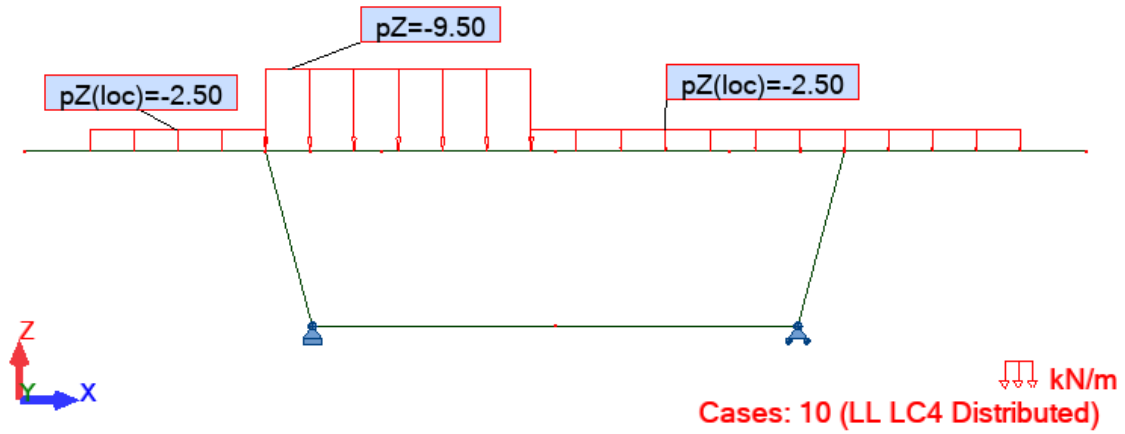


Figure 14-12: Live Load Case 4 (Distributed) in FEM

14.1.1 Bending moment curves: ULS Combinations

The bending moment curves for load combinations that are summarized in **Section 14.1** is shown in this section for top slab.

ULS Comb 1

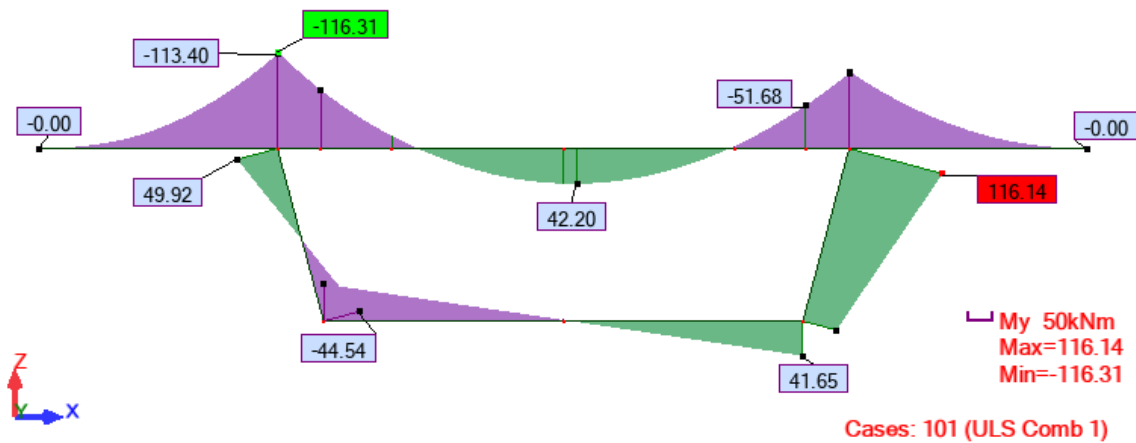


Figure 14-13: Top slab bending moment diagram for ULS Comb 1

ULS Comb 2

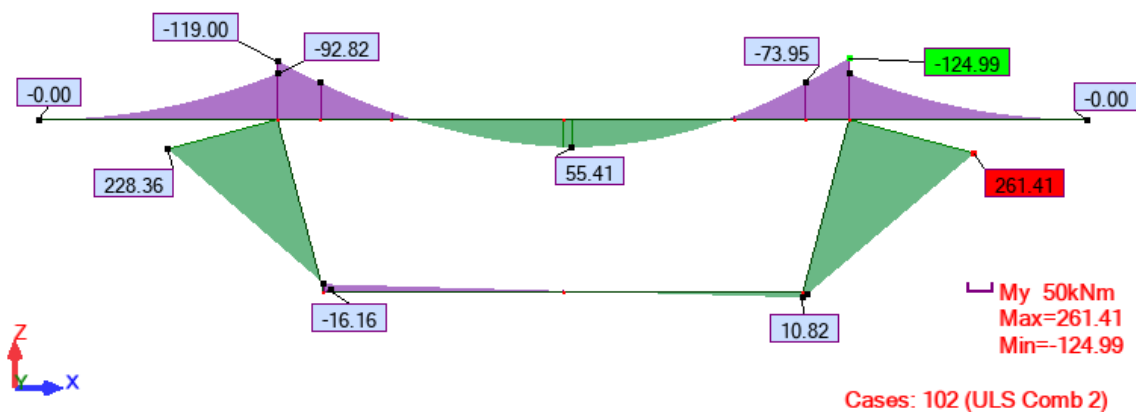


Figure 14-14: Top slab bending moment diagram for ULS Comb 2

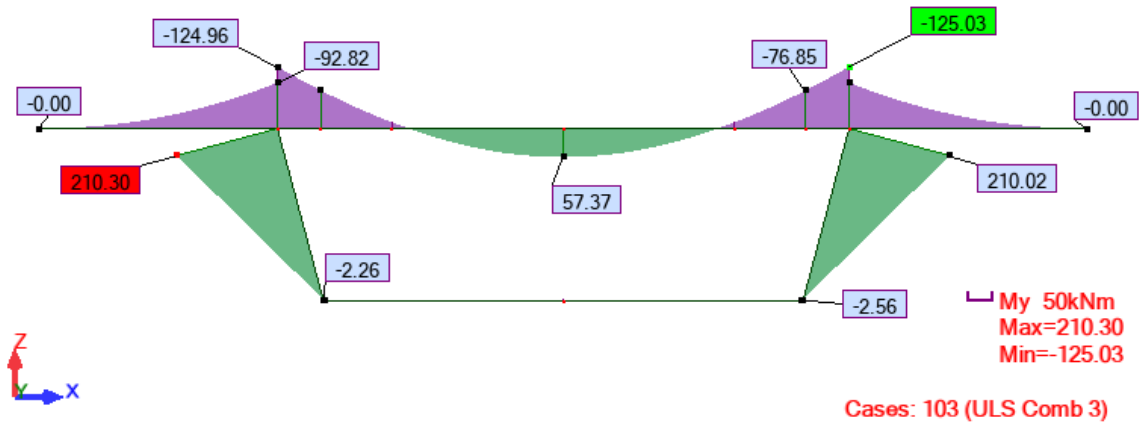
ULS Comb 3

Figure 14-15: Top slab bending moment diagram for ULS Comb 3

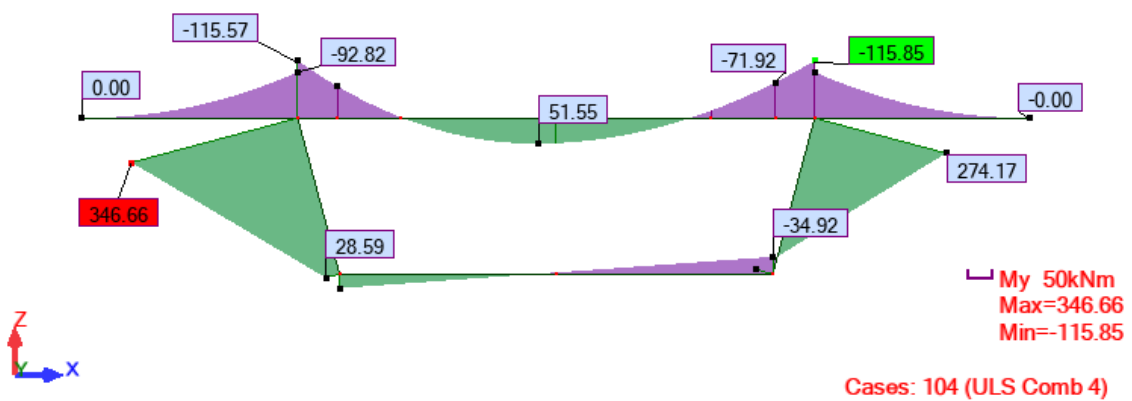
ULS Comb 4

Figure 14-16: Top slab bending moment diagram for ULS Comb 4

15. Bending moment envelope: ULS

Regarding to the bending moments, finally, the bending moment envelope is obtained by superposition of all load combinations that has been considered in **Section 14.1**. The procedure for obtaining the envelope for top slab is done by superimposing the design moment values (maximum positive and negative value) that is already obtained. A table summarizing all design moment values and final bending moment envelope for top slab is given below:

Gray columns show the bending moment values in each section that is derived from FE-Model, indicated as Robot*. Other columns show the design bending moments for concentrated loads that are obtained from Homberg's method.

TOP SLAB ENVELOPE															
		Load Case 1			Load Case 2			Load case 3			Load case 4			Envelope	
		Robot*	Conc. Loads	Total	Robot*	Conc. Loads	Total	Robot*	Conc. Loads	Total	Robot*	Conc. Loads	Total	Min	Max
Point	x	m _{x,Ed}	m _{x,Ed}	m _{x,Ed}	m _{x,Ed}	m _{x,Ed}	m _{x,Ed}	m _{x,Ed}	m _{x,Ed}	m _{x,Ed}	m _{x,Ed}	m _{x,Ed}	m _{x,Ed}	m _{x,min}	m _{x,max}
	[m]	[kNm/m]	[kNm/m]	[kNm/m]	[kNm/m]	[kNm/m]	[kNm/m]	[kNm/m]	[kNm/m]	[kNm/m]	[kNm/m]	[kNm/m]	[kNm/m]	[kNm/m]	[kNm/m]
A															
AB															
BR	0.00	-113.40	-164.75	-278.15	-92.82	0.00	-92.82	-92.82	-66.50	-159.32	-92.82	0.00	-92.82	-278.15	-92.82
BL	0.00	-116.31	-211.75	-328.06	-119.00	-202.18	-321.18	-124.96	-244.66	-369.62	-115.57	-323.91	-439.48	-439.48	-321.18
BC	1.31	-15.45	-4.82	-20.27	-14.58	-47.33	-61.91	-13.06	-23.36	-36.42	-1.54	-52.07	-53.61	-61.91	-20.27
C	3.27	42.05	41.38	83.43	55.12	80.53	135.65	57.37	45.90	103.27	51.00	21.13	72.13	72.13	135.65
CD	5.23	-3.19	-31.60	-34.79	-7.10	-20.88	-27.98	-13.10	-33.90	-47.00	-15.03	-49.05	-64.08	-64.08	-27.98
DR	6.54	-90.41	-118.54	-208.95	-124.99	-229.25	-354.24	-125.03	-211.07	-336.10	-115.85	-251.15	-367.00	-367.00	-208.95
DL	6.54	-92.82	0.00	-92.82	-92.82	0.00	-92.82	-92.82	-33.25	-126.07	-92.82	0.00	-92.82	-126.07	-92.82
E															

Figure 15-1: Top slab bending moment envelope.

The bending moment envelope for top slab is:

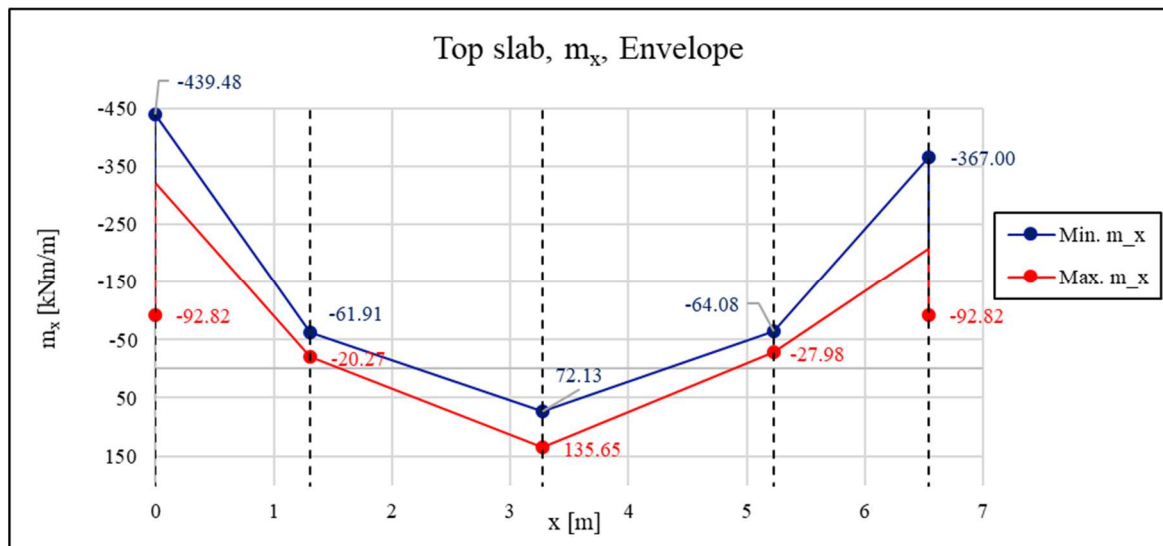


Figure 15-2: Top slab envelope

The bending moment envelope for webs and bottom slab is obtained through FE-Model.

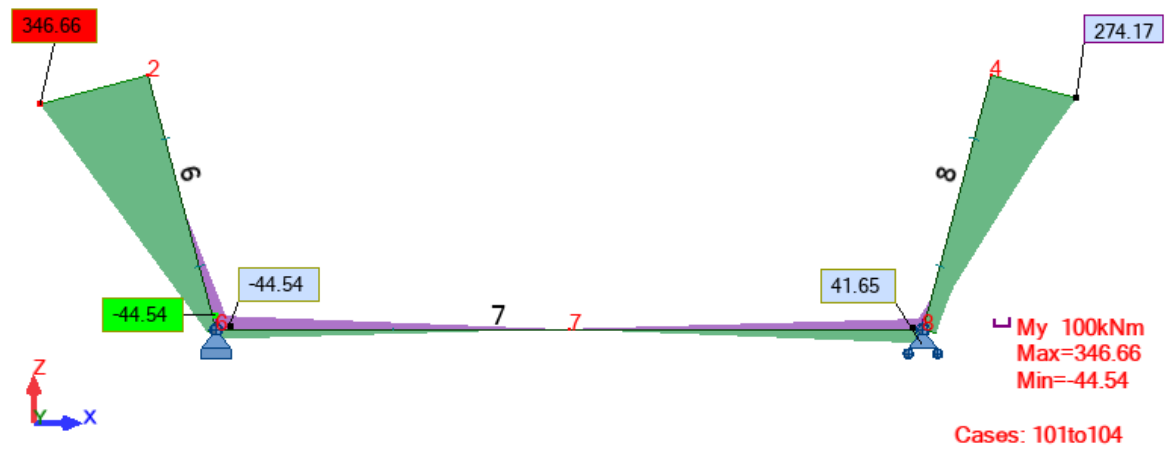


Figure 15-3 Bottom slab and webs

16. Safety verifications

16.1 Transversal reinforcement: ULS check

As a final computation, the amount of reinforcement required for top slab, webs and bottom slab is calculated by flexural verifications, obtaining acting bending moments envelope that is found in **Section 15** comparing with resistant bending moment of the section.

$$m_{Ed}^- \leq m_{Rd}^- \text{ (Top reinforcement)}$$

$$m_{Ed}^+ \leq m_{Rd}^+ \text{ (Bottom reinforcement)}$$

Flexural resistant verification is done by following the scheme:

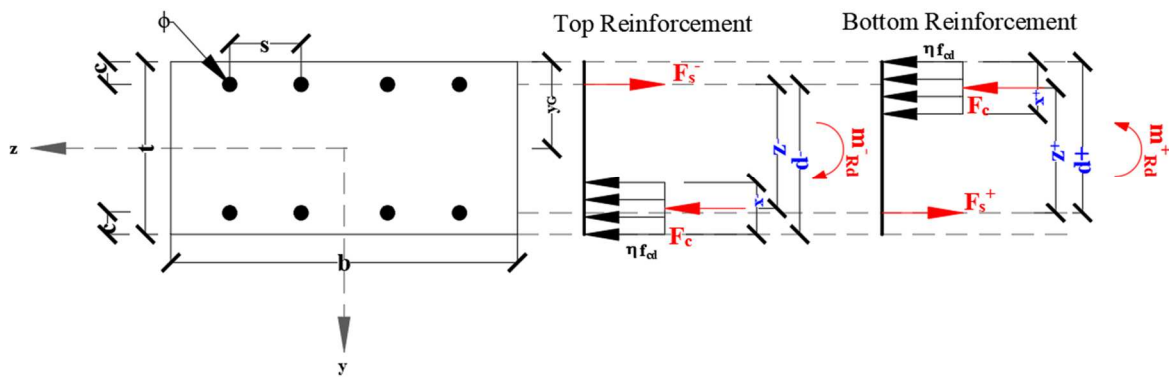


Figure 16-1: Single reinforced section for top and bottom reinforcement.

Where:

t is thickness of the cross section.

b is width of the cross section.

s is span of the bars.

c is concrete cover.

ϕ is concrete cover.

d is effective depth of the cross section.

x is depth of the neutral axis defined as $x = \lambda \cdot y_n$

z is the moment arm.

Material properties are defined as follows:

Material Properties		
f_{ck}	45	[MPa]
f_{ctm}	3.5	[MPa]
γ_c	1.5	[-]
f_{cd}	30	[MPa]
ϵ_{cu3}	0.0035	[-]
f_{yk}	500	[MPa]
E_s	200000	[MPa]
ϵ_{yk}	0.0025	[-]
γ_s	1.15	[-]
ϵ_{yd}	0.0022	[-]
f_{yd}	370	[MPa]
λ	0.8	[-]
η	0.85	[-]

Table 16-1: Material input

The computations are done by following procedure:

- Output parameters such as $A_s^-, A_s^+, A_s^{min}, A_s^{max}, d^-, d^+, x^-, x^+, z^-, z^+$ that are defined in **Section 16.1** is calculated by using following equations:

Minimum reinforcement area is defined in EN1992.9.2.1.1, should not be less than A_s^{min} ;

$$A_s^{min} = 0.26 \cdot \frac{f_{ctm}}{f_{yk}} \cdot b \cdot d \leq 0.0013 \cdot b \cdot d \left[\frac{mm^2}{m} \right]$$

Where:

f_{ctm} is determined with respect to relevant strength class in **Section 4.1**.

$$d^- = t - c^- [mm]$$

$$d^+ = t - c^+ [mm]$$

- Concrete cover is determined in **Section 4.1** for both reinforcements as:

$$c^- = c^+ = 45 \text{ mm}$$

Maximum reinforcement area is defined as:

$$A_s^{max} = 0.04 \cdot A_c = 0.04 \cdot t \cdot h [mm^2]$$

Where:

A_c is the area of concrete.

Area of chosen diameter:

$$A_s^- = \frac{\left(\pi \cdot \frac{\emptyset^{-2}}{4}\right)}{s^- \cdot b} [\frac{mm^2}{m}] \quad A_s^+ = \frac{\left(\pi \cdot \frac{\emptyset^{+2}}{4}\right)}{s^+ \cdot b} [\frac{mm^2}{m}]$$

Imposing equilibrium of stress resultants in the section.

$$F_C + F_s = 0$$

Where:

F_c is the compressive force in the concrete, defined as:

$$F_c = -\eta \cdot x \cdot f_{cd} \cdot b \quad [kN]$$

$$F_c = -\eta \cdot \lambda \cdot y_n \cdot f_{cd} \cdot b \quad [kN]$$

F_s is the tensile force in the bottom and top reinforcement, defined as:

$$F_s^- = A_s^- \cdot f_{yd} \quad [kN]$$

$$F_s^+ = A_s^+ \cdot f_{yd} \quad [kN]$$

The following formulation is obtained:

$$x^- = \frac{A_s^- \cdot f_{yd}}{\eta \cdot \lambda \cdot f_{cd} \cdot b} \quad [mm]$$

$$x^+ = \frac{A_s^+ \cdot f_{yd}}{\eta \cdot \lambda \cdot f_{cd} \cdot b} \quad [mm]$$

The resisting bending moment is obtained by imposing rotational equilibrium of the section about the point of action of the compressive force in concrete:

$$m_{Rd}^- = F_s^- \cdot z^- \quad [kNm]$$

$$m_{Rd}^+ = F_s^+ \cdot z^+ \quad [kNm]$$

Where:

z^-, z^+ is corresponding lever arm for the tensile force, defined as:

$$x^- = d^- - z^- \quad [mm]$$

$$x^+ = d^+ - z^+ \quad [mm]$$

16.1.1 Reinforcement for top slab

- Geometry input for each cross sections of the top slab is defined and illustrated as follows:

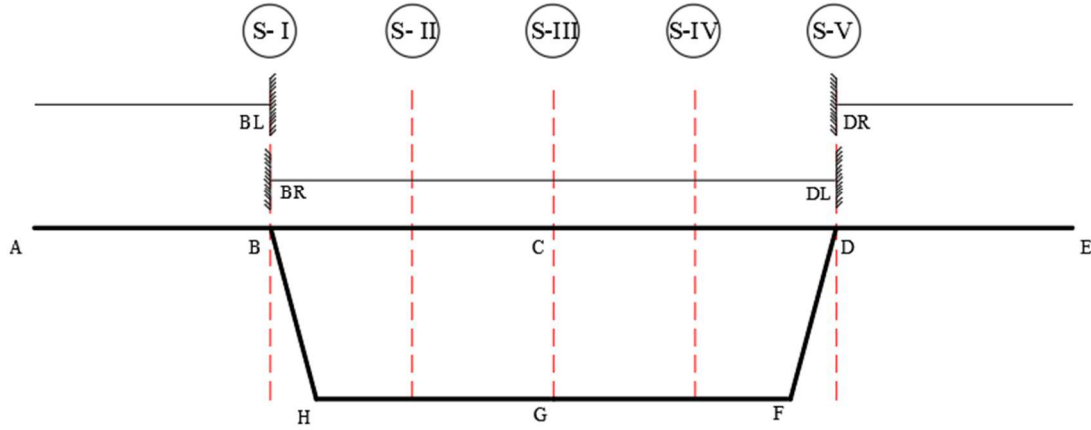


Figure 16-2: Cross-sections for top slab.

Input						
Parameter	Section I	Section II	Section III	Section IV	Section V	Units
t	400	250	250	250	400	[mm]
b	1000	1000	1000	1000	1000	[mm]

Table 16-2: Geometry input for top slab.

- Pre-dimensioning for bar diameter and bar spacing is defined. The bar diameters are kept uniform throughout each cross-section. For the top reinforcement, higher value of diameter is selected due to the higher negative moments in the cross section 1 and 5 which corresponds to the nodes.

$$\emptyset^- = 22 \text{ mm and } \emptyset^+ = 20 \text{ mm}$$

Spacing of bars is defined in EN1992.9.3.1.1, the spacing should not exceed $s_{max,slabs}$:

$$s_{max,slabs} = 2 \cdot h \leq 250 \text{ mm}$$

$$s^- = s^+ = 100 \text{ mm}$$

- Negative and positive acting bending moments are summarized in the following table:

Input						
Parameter	Section I	Section II	Section III	Section IV	Section V	Units
m_{Ed}^-	-439.48	-61.91	72.13	-64.08	-367.00	[kNm]
m_{Ed}^+	-321.18	-20.27	135.65	-27.98	-208.95	[kNm]

Table 16-3: Acting bending moments for top slab.

Output						
Parameter	Section I	Section II	Section III	Section IV	Section V	Units
A_s^-	3801.33	3801.33	3801.33	3801.33	3801.33	[mm ²]
A_s^+	3141.59	3141.59	3141.59	3141.59	3141.59	[mm ²]
d^+	355.00	205.00	205.00	205.00	355.00	[mm]
d^-	355.00	205.00	205.00	205.00	355.00	[mm]
x^-	68.86	68.86	68.86	68.86	68.86	[mm]
x^+	45.53	45.53	45.53	45.53	45.53	[mm]
z^-	327.45	177.45	177.45	177.45	327.45	[mm]
z^+	336.79	186.79	186.79	186.79	336.79	[mm]

Table 16-4: Output design parameters for top slab.

M

- Finally, the results for resisting design bending moment are given in a tabulated format:

Output						
Parameter	Section I	Section II	Section III	Section IV	Section V	Units
m_{Rd}^-	460.02	249.29	249.29	249.29	460.02	[kNm]
m_{Rd}^+	391.02	216.87	216.87	216.87	391.02	[kNm]

Table 16-5: Resisting bending moments for top slab.

The values for resisting design bending moments are found higher than the acting bending moment in each section on top slab reinforcement which are shown in green colour in the table above. Therefore, the design is verified in ultimate limit state.

Final selected reinforcement design for top slab is given below:

Input		
Parameter	Top Slab Reinforcement	Units
$\emptyset^-$	22	[mm]
$\emptyset^+$	20	[mm]
s^-	100	[mm]
s^+	100	[mm]

Table 16-6: Top slab reinforcement.

16.1.2 Reinforcement for webs

Same procedure is implemented for the reinforcement of webs. For webs, only 1 section is considered.

- Geometry of the section is considered as follows:

Input		
Parameter	Section I	Units
h	600	[mm]
b	1000	[mm]

Table 16-7: Geometry input for webs.

- Pre-dimensioning of rebars is done by selecting the diameter as 16 mm and spacing as 100 mm.

$$\emptyset^+ = \emptyset^- = 16 \text{ mm}$$

$$s^- = s^+ = 100 \text{ mm}$$

- Acting bending moments are taken as follows:

Input		
Parameter	Section I	Units
m_{Ed}^-	-44.54	[kNm]
m_{Ed}^+	346.66	[kNm]

Table 16-8: Acting bending moments for webs.

Output		
Parameter	Section I	Units
A_s^-	2010.62	[mm ²]
A_s^+	2010.62	[mm ²]
d^-	555.00	[mm]
d^+	555.00	[mm]
x^-	36.42	[mm]
x^+	36.42	[mm]
z^-	540.43	[mm]
z^+	540.43	[mm]

Table 16-9: Output design parameters for webs.

- Finally, the results for resisting design bending moment are given in a tabulated format:

Output		
Parameter	Section I	Units
m_{Rd}^-	401.57	[kNm]
m_{Rd}^+	401.57	[kNm]

Table 16-10: Resisting bending moments for webs.

Selected reinforcement design for webs is given below:

Input		
Parameter	Top Slab Reinforcement	Units
$\emptyset^-$	16	[mm]
$\emptyset^+$	16	[mm]
s^-	100	[mm]
s^+	100	[mm]

Table 16-11: Web reinforcement.

16.1.3 Reinforcement for bottom slab

- Geometry of the section is considered as follows:

Input		
Parameter	Section I	Units
h	250	[mm]
b	4950	[mm]

Table 16-12: Geometry input for bottom slab.

- Pre-dimensioning of rebars is done by selecting the diameter as 16 mm and spacing as 100 mm.

$$\emptyset^+ = \emptyset^- = 12 \text{ mm}$$

$$s^- = s^+ = 200 \text{ mm}$$

- Acting bending moments are taken as follows:

Input		
Parameter	Section I	Units
m_{Ed}^-	-44.54	[kNm]
m_{Ed}^+	41.65	[kNm]

Table 16-13: Acting bending moments for bottom slab.

Output		
Parameter	Section I	Units
A_s^-	2799.16	[mm ²]
A_s^+	2799.16	[mm ²]
d^-	205.00	[mm]
d^+	205.00	[mm]
x^-	8.20	[mm]
x^+	8.20	[mm]
z^-	200.90	[mm]
z^+	200.90	[mm]

Table 16-14: Output design parameters for bottom slab.

- Finally, the results for resisting design bending moment are given in a tabulated format:

Output		
Parameter	Section I	Units
m_{Rd}^-	207.83	[kNm]
m_{Rd}^+	207.83	[kNm]

Table 16-15: Resisting bending moments for bottom slab.

Selected reinforcement design for webs is given below:

Input		
Parameter	Bottom Slab Reinforcement	Units
$\emptyset^-$	12	[mm]
$\emptyset^+$	12	[mm]
s^-	200	[mm]
s^+	200	[mm]

Table 16-16: Bottom slab reinforcement.

Part D: Special regions

17. Rigid diaphragm

A diaphragm in bridge is used to strengthen the box-girder concerning torsion and transmit shear stresses and loads to the bearings. Behaviour of box-girder bridges consists of torsion, distortion and bending in longitudinal and transverse directions. The longitudinal bending stress distribution in wide flange girders is distributed non-uniformly throughout the width. It remains maximum at the edge and reduces towards the centre. The complications that may arise during design as follows:

- a) Wide-ranging span-to-depth ratio: The span-to-depth ratio of the diaphragm act as a deep beam.
- b) Presence of manhole: The presence of opening for bridge inspection in the deep beam induces geometric discontinuity which enhances the non-linear stress distribution along the depth.

Designing areas of structural components affected by a load and/or geometrical discontinuities is where the strut-and-tie model is most frequently employed. Within the vicinity, a quadratic distribution of stresses is brought about by loads and architectural imperfections. As a result, it is no longer possible to trust that plane portions will stay that way within of the discontinuity-affected area.

17.1 Strut and tie model

The strut-and-tie model was developed to design reinforcement for structural members with disturbed or D-regions where the linear strain distribution across depth is not possible. The successful application of the strut-and-tie model can be done by truss elements. A truss in a strut-and-tie model comprises tensile ties, compressive strut, and nodal zones where the struts and ties intersect.

EN1992.2005 Annex OO states that the strut-and-tie model is used to design and determine the internal force effects near supports and points of application of loads.

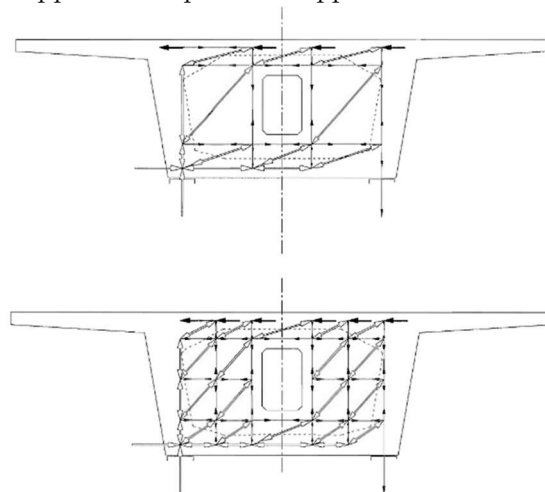


Figure 17-1: Strut and tie model for a solid-type diaphragm with a manhole.

Following initial model is made, but some inclined members are seen in tension, and it is difficult to provide the reinforcement inclined direction, hence a simplified model of strut-tie is considered along with some modification with respect to our model.

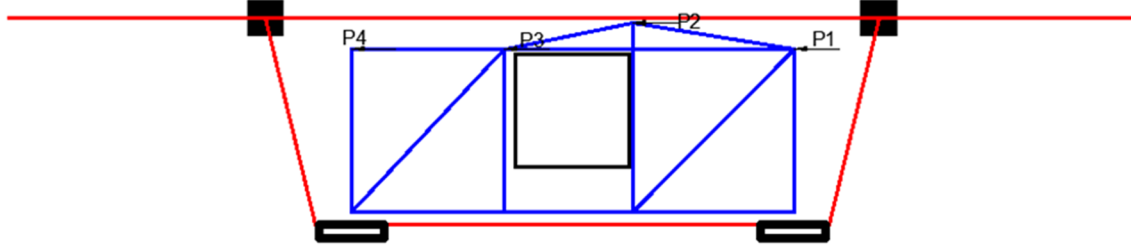


Figure 17-2: Simplified model of strut and tie

17.2 Load Generation

The load applied to the model is found from the shear flow coming from the torsional moment that is has computed before as 10741.5 kNm. Then simply by applying Bredt's formula the shear flow is found as follows:

$$M_t = 10741.5 \text{ kNm}$$

$$q = \frac{M_t}{\Omega} = \frac{10741.5 \text{ kNm}}{26.47 \text{ m}^2} = 405.86 \frac{\text{kN}}{\text{m}}$$

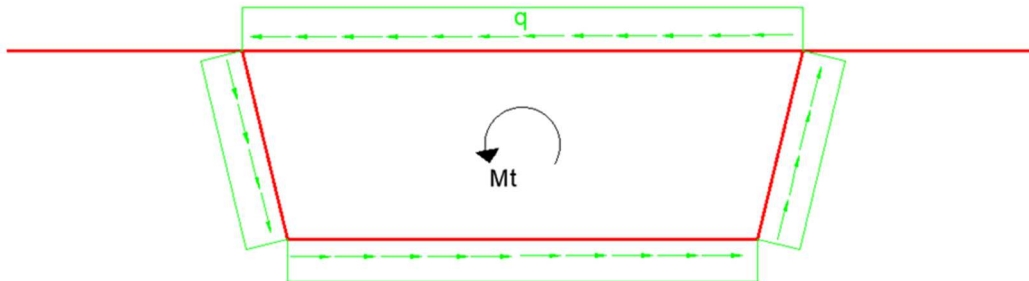


Figure 17-3: Shear flow distribution

Only the shear flow is considered in the top slab and neglected in the web and bottom slab because the support reaction force can resist, hence by taking the shear flow in the top slab, the nodal values are computed as follows:

$$P_1 = 373.4 \text{ kN}$$

$$P_2 = 660.6 \text{ kN}$$

$$P_3 = 586.47 \text{ kN}$$

$$P_4 = 660.43 \text{ kN}$$

The strut & tie of rigid diaphragm is modelled and analysed in Midas Civil software and the axial force in each strut and tie is obtained, then by using these member forces the reinforcement required in the diaphragm region is computed.

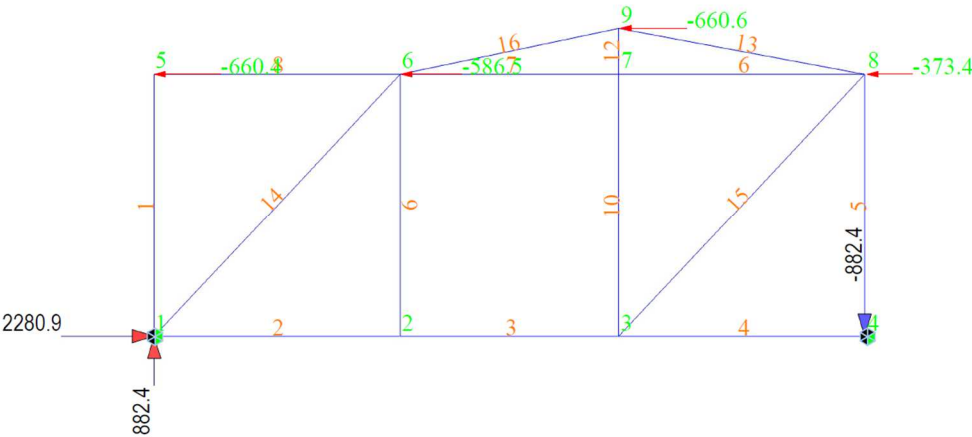


Figure 17-4: Strut and tie model in GEM Midas.

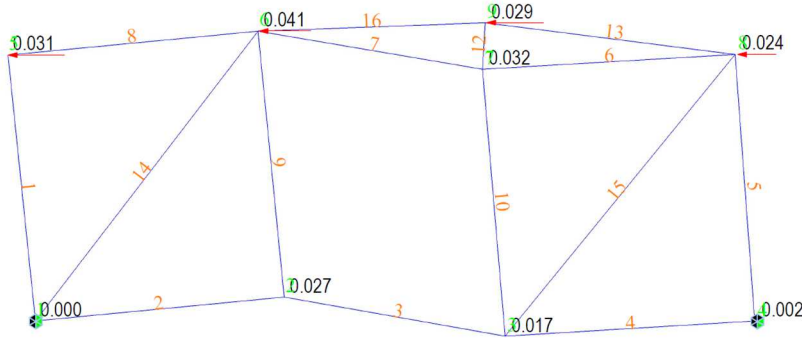


Figure 17-5: Deformed shape of truss.

Element	Load	Force-I (KN)
1	P	0.00
2	P	-1451.58
3	P	-1451.58
4	P	0.00
5	P	882.37
6	P	4656.48
7	P	4656.48
8	P	660.44
9	P	0.00
10	P	1544.36
12	P	1544.36
13	P	-3639.02
14	P	-1210.95
15	P	-2119.47
16	P	-4329.76

Figure 17-6: Truss forces from Midas GEN.

Area of reinforcement required at different regions of diaphragm upon applying the following generic formula:

$$A_{s,req} = \frac{F_{t,i}}{f_{yd}}, \quad F_{t,i} = \text{Tension force in tie no. } i, \quad f_{yd} = 435 \text{ MPa}$$

Region	Area of reinforcement $A_{s,req}$ (mm ²)
Bottom slab	3710.708
Top slab	11898.8
Left part, vertical reinforcement	3945.822
Left part, horizontal reinforcement	1686.685
Right part, vertical reinforcement	3945.822
Right part, horizontal reinforcement	11898.8

Figure 17-7: Required reinforcement in diaphragm.

18. Anchorage zone

The anchorage zone of a concrete member is the area in front of the pre-stress block out where force dispersion occurs. The analysis and design of the pre-tensioned anchorage zone have been studied since the early days of the pre-stressed concrete industry. The length over which the pre-stress dispersal takes place from the anchorage devices to the full effective section is referred to as the general zone. Sufficient reinforcement must be provided to handle force spreading in the general zone of the end anchorage, mainly including the bursting force and the spalling force.

18.1 Bursting reinforcement

The bursting force is the tensile force in the anchorage zone acting ahead of the anchorage device and transverse to the tendon axis. Bursting forces are caused by the lateral spreading of the prestressing forces concentrated at the anchorage, this leads to satisfactory behaviour at the service limit state by limiting the extent and opening of cracks and at the strength limit state by limiting the required amount of redistribution of forces in the anchorage zone.

Compressive stresses should be checked where geometry or loading discontinuities within or ahead of the anchorage zone may cause stress concentrations.

Resistance to bursting forces shall be provided by non-prestressed or prestressed reinforcement or in the form of spirals, closed hoops, or anchored transverse ties. This reinforcement shall resist the total bursting force. The following guidelines for the arrangement and anchorage of bursting reinforcement should apply:

- Reinforcement is extended over the full width of the member and anchored as close to the outer faces of the member as cover permits.

- Reinforcement is distributed ahead of the loaded surface along both sides of the tendon throughout a distance taken as the lesser of $2.5 d_{burst}$ for the plane considered and 1.5 times the corresponding lateral dimension of the section.
- The centroid of the bursting reinforcement coincides with the distance d_{burst} used for the design; and
- Spacing of reinforcement is not greater than either 24.0 bar diameters or 300 mm.

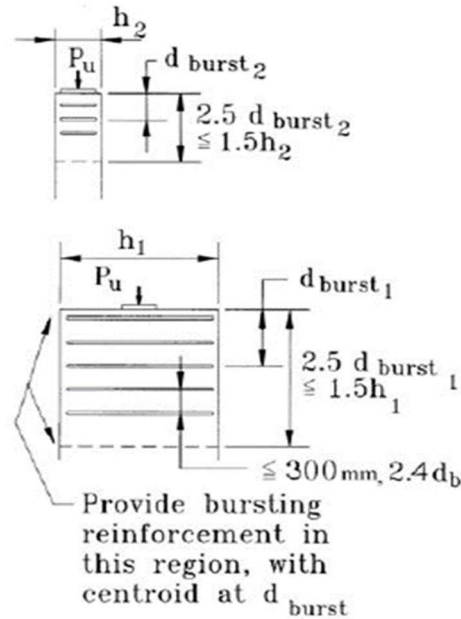


Figure 18-1: Bursting force and reinforcement layout

Unless a more detailed analysis is made, the minimum reinforcement specified herein to resist bursting force and edge tension force shall be provided. This reinforcement shall be anchored close to the faces of the slab with standard hooks bent around horizontal bars or equivalent. Minimum reinforcement should be two No. 10 bars per anchor located at a distance equal to one-half the slab thickness ahead of the anchor.

Reinforcement shall be provided to resist edge tension forces, between anchorages. Edge tension reinforcement shall be placed ahead of the anchors and shall effectively tie adjacent anchors together. Bursting reinforcement shall be distributed over the length of the anchorage zones.

$$T_1 = 0.10P_u(1 - a/s)$$

$$T_2 = 0.20P_u(1 - a/s)$$

Where:

T_1 = The edge tension force.

T_2 = The bursting force.

P_u = The factored tendon load on an individual anchor.

a = The anchor plate width.

s = The anchorage spacing.

Prism	Anchorage spacing	Anchor plate width	Pre-stressing force	Tensile stress function	Bursting Force	Area of reinforcement
	s [m]	a [m]	P [kN]	a/s	F_{burst} [kN]	A_s [mm ²]
1	0.4	0.3	4468.5	0.75	111.71	285.49
2	0.4	0.3	4468.5	0.75	111.71	285.49
3	0.33	0.3	4527.6	0.90	94.61	241.78
4	0.33	0.3	4527.6	0.90	94.61	241.78
5	1	0.3	4467	0.30	625.38	1598.21
6	1	0.3	4467	0.30	625.38	1598.21
7	0.4	0.3	4530.8	0.75	113.27	289.47
8	0.4	0.3	4530.8	0.75	113.27	289.47
9	0.33	0.3	4527.1	0.90	94.60	241.75
10	0.33	0.3	4527.1	0.90	94.60	241.75
11	1	0.3	4590.8	0.30	642.71	1642.50
12	1	0.3	4590.8	0.30	642.71	1642.50

Table 18-1: Bursting forces and reinforcement.

A spiral with 5 turns of $\emptyset$ 12mm for each anchorage.

18.2 Spalling reinforcement

For multiple anchorages with a centre-to-centre spacing of fewer than 0.4 times the depth of the section, the spalling force shall not be taken to be less than %2 of the total factored tendon force. For larger spacing's, the spalling forces shall be determined by analysis.

Edge tension forces are tensile forces in the anchorage zone acting parallel and close to the transverse edge and longitudinal edges of the member. The transverse edge is the surface loaded by the anchors. The tensile force along the transverse edge is referred to as spalling force. The tensile force along the longitudinal edge is referred to as the longitudinal edge tension force.

Resistance to edge tension forces shall be provided by reinforcement located close to the longitudinal and transverse edge of the concrete. Arrangement and anchorage of the edge tension reinforcement shall satisfy the following:

- Specified spalling reinforcement is extended over the full width of the member,
- Spalling reinforcement between multiple anchorage devices effectively ties the anchorage devices together, and
- Longitudinal edge tension reinforcement and spalling reinforcement for eccentric anchorage devices are continuous; the reinforcement extends along the tension face over the full length of the anchorage zone and along the loaded face from the longitudinal edge to the other side of the eccentric anchorage device or group of anchorage devices.

$$A_{s,spalling} = 0.03 \cdot 1.2 \frac{P}{f_{yd}} = 4988.6 \text{ mm}^2$$

Bibliography

- [1] J. Schlaich and H. Scheffer, Concrete Box-Girder Bridges, International Association for Bridge and Structural Engineering, 1982.

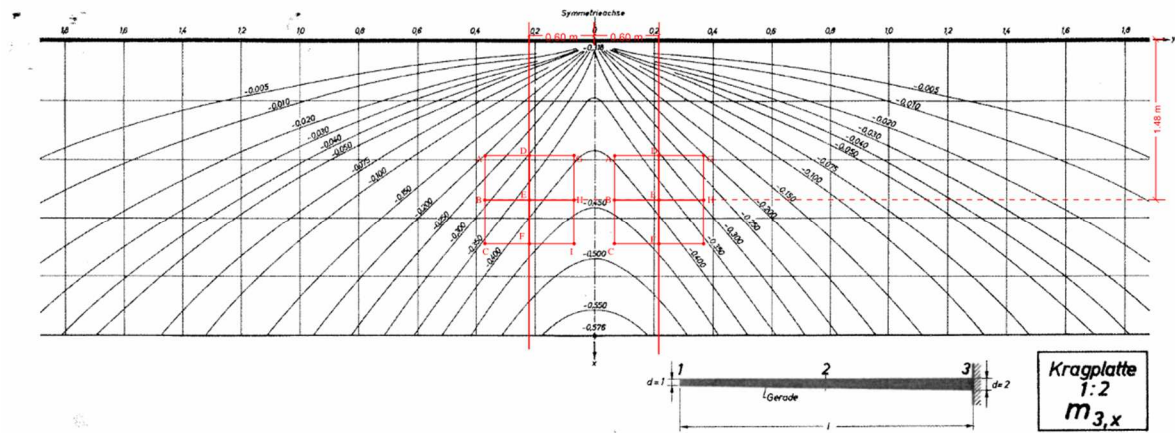
Appendices

- A. Homberg charts & Influence surfaces
- B. Flexural verifications, ULS
- C. Cross-section parameters from Autodesk Robot
- D. Internal forces in longitudinal scheme
- E. Cable paths
- F. Cable force diagrams
- G. ULS Shear and Torsional Verifications

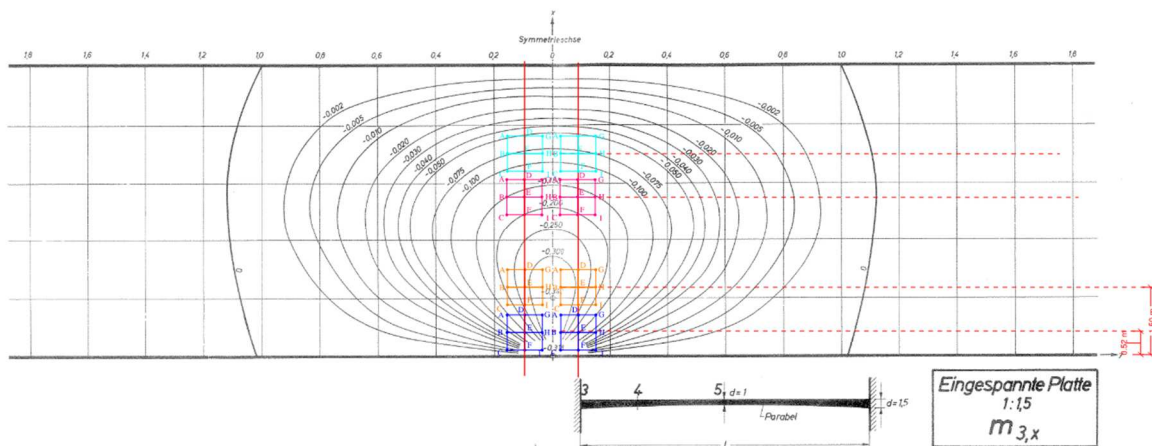
A. Homberg charts & Influence surfaces

The charts used for Homberg analysis are given in this section. We are showing only for live load case 1, but the same procedure is applied to the other cases.

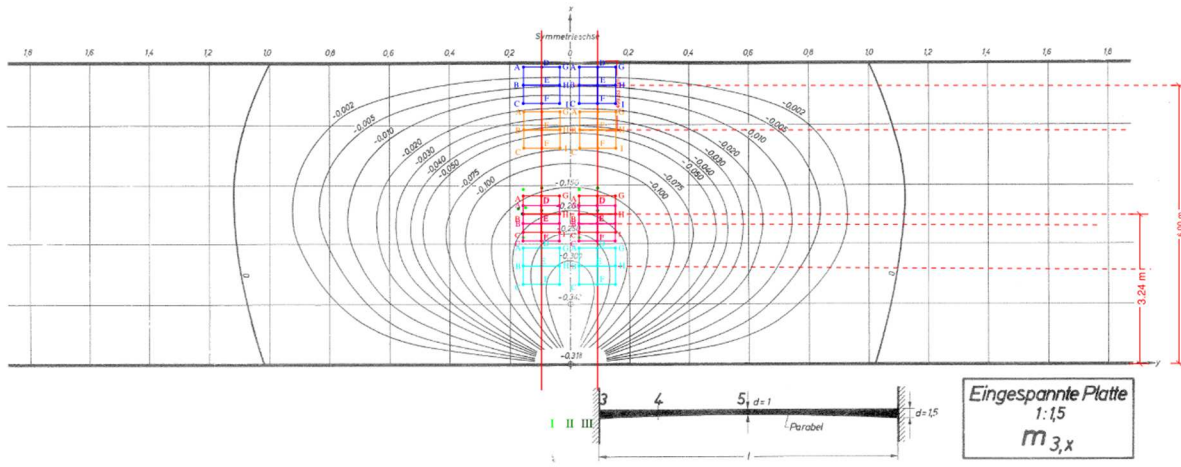
Live Load Case 1: $m_{3,x}$ (Cantilever)



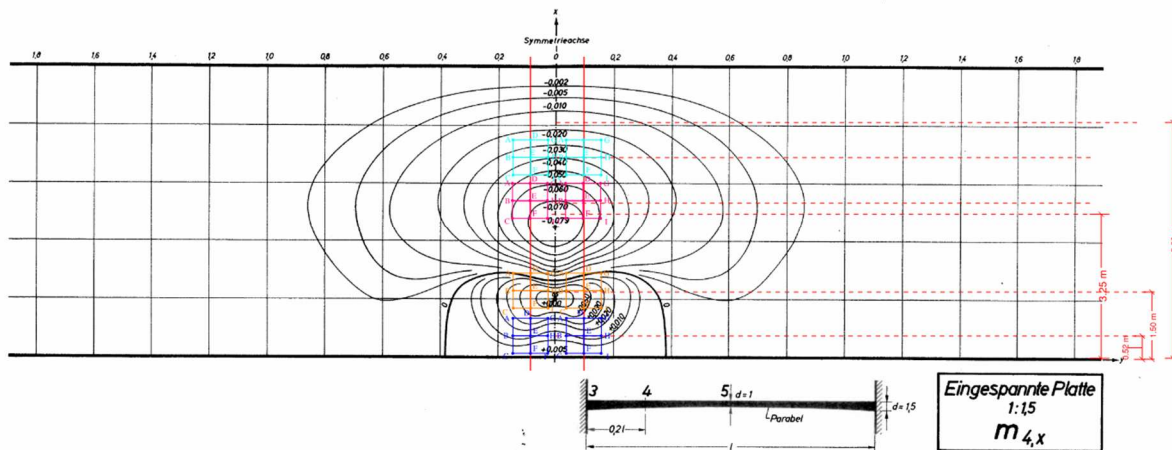
Live Load Case 1: $m_{3,x}$ (Clamped edges)



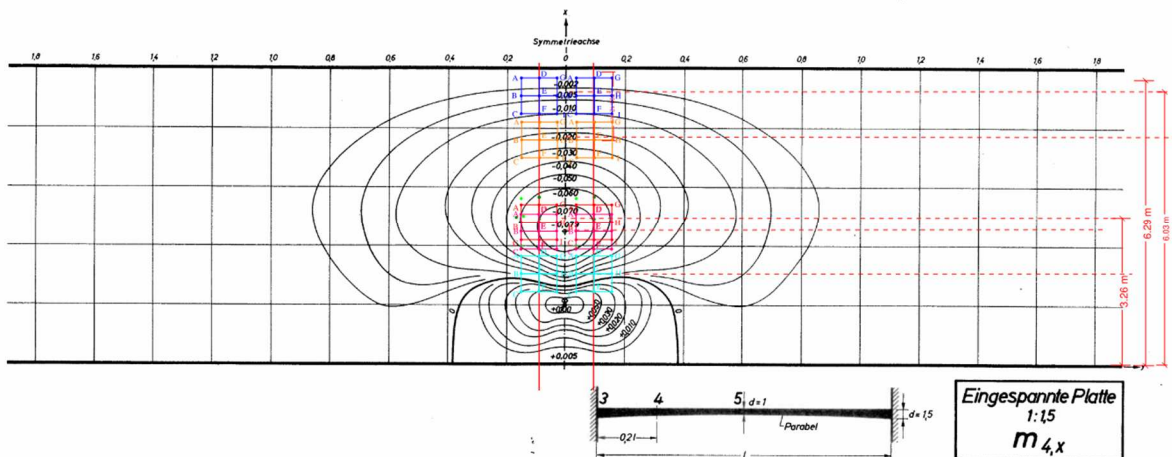
Live Load Case 1: $m_{3,x}$, symmetric (Clamped edges)



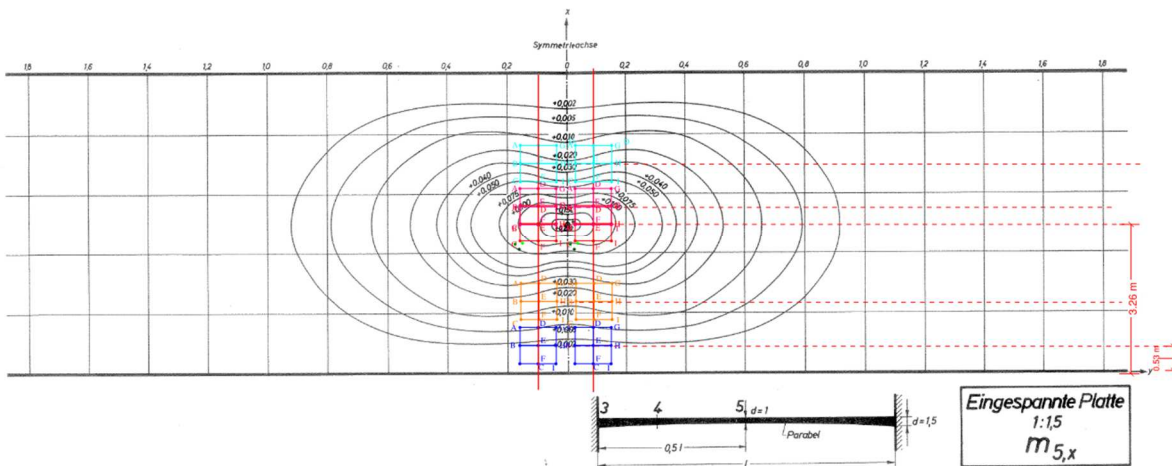
Live Load Case 1: $m_{4,x}$ (Clamped edges)



Live Load Case 1: $m_{4,x}$, symmetric (Clamped edges)



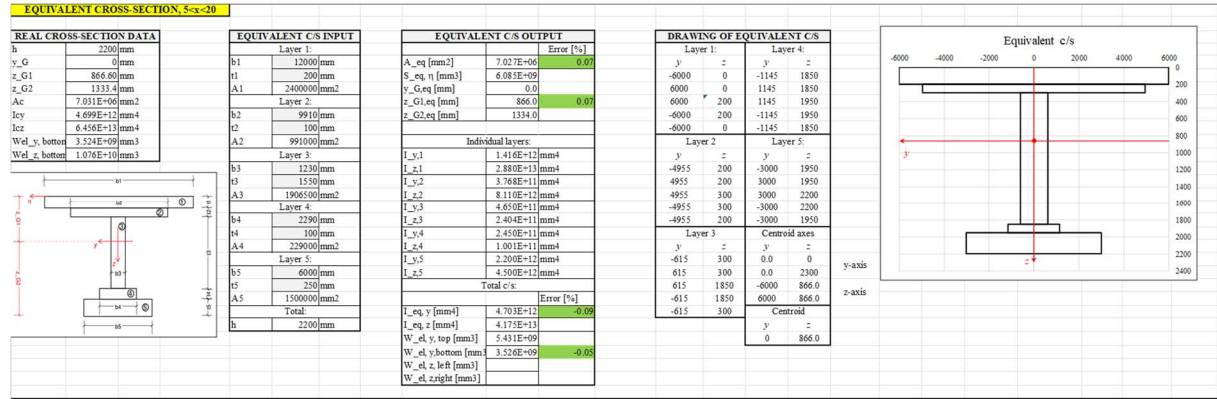
Live Load Case 1: $m_{5,x}$ (Clamped edges)



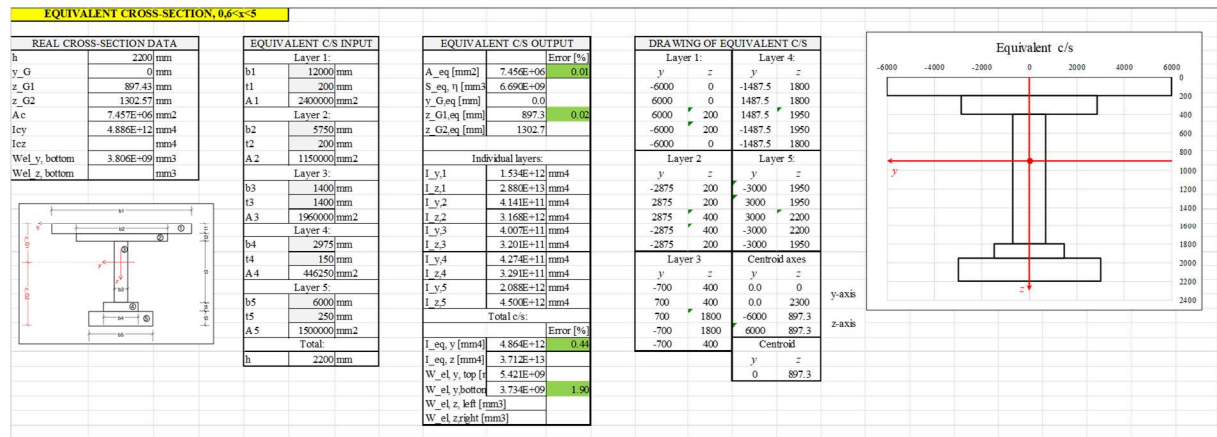
B. Flexural verifications, ULS

In this appendix, the equivalent cross-sections, and the flexural verifications in ULS are shown.

Firstly, the computation of equivalent section for $5 \text{ m} \leq x \leq 20 \text{ m}$:



Secondly, the computation of equivalent section for $0 \text{ m} \leq x \leq 5 \text{ m}$:



Below are the flexural verifications in ULS in four significant sections:

DESIGN SECTIONAL FORCE			
M _{Ed}	103686.9	kNm	
ASSUMED COMPRESSIVE ZONE			
x	237.08	mm	Neutral axis in layer 2
A _c		z _c	
[mm ²]		[mm]	
Zone 1	2.40E+06	765.98	
Zone 2	367500.00	647.44	
Zone 3	0.00	0.00	
Zone 4	0.00	0.00	
Zone 5	0.00	0.00	

SECTION I, x = 20 m

SECTION II, x = 10 m										
DESIGN SECTIONAL FORCE										
M _{Ed}	78035.1	kNm								
ASSUMED COMPRESSIVE ZONE										
x	237.08	mm	Neutral axis in layer 2							
	A _c	z _c								
	[mm ²]	[mm]								
Zone 1	2.40E+06	765.98								
Zone 2	367500.00	647.44								
Zone 3	0.00	0.00								
Zone 4	0.00	0.00								
Zone 5	0.00	0.00								
PRETENSION CABLES										
Layer:	F _{p,∞}	e _p	d _p	ε ₁	ε ₂	ε ₃	ε _p	σ _{pd}	A	
	[kN]	[m]	[m]	[%]	[%]	[%]	[%]	[MPa]	[mm ²]	
1a	7348.77	0.41	1.27	0.45363	0.005437	1.5294	1.988	1366.67	8100	
1b	7397.55	0.41	1.27	0.45664	0.005473	1.5294	1.991	1366.67	8100	
2a	7352.19	0.72	1.58	0.45384	0.007716	1.9881	2.450	1366.67	8100	
2b	7410.60	0.72	1.58	0.45744	0.007777	1.9881	2.453	1366.67	8100	
3a	7433.28	1.12	1.99	0.45884	0.012689	2.5840	3.056	1366.67	8100	
3b	7490.16	1.12	1.99	0.46236	0.012787	2.5840	3.059	1366.67	8100	
Total:	44432.55									
							Yielding	> ε _{pyd}		
							Elastic	< ε _{pyd}		
CHECK OF COMPRESSIVE ZONE*			MOMENT CAPACITY**							
F _c	66420.00	kN	M _c		49831.13		kNm			
F _p	66420.00	kN	M _p		49733.56		kNm			
Σ F	0.00	kN	M _{Rd}		99564.69		kNm		> M _{Ed}	
			Utilisation		78.4		%		< 100	
*Horizontal equilibrium			**Moment equilibrium around centroid							

SECTION III, x = 5 m										
DESIGN SECTIONAL FORCE										
M Ed	45565.5		kNm							
ASSUMED COMPRESSIVE ZONE										
x	237.08		mm		Neutral axis in layer 2					
	A c	z c								
	[mm ²]	[mm]								
Zone 1	2.40E+06	765.98								
Zone 2	367500.00	647.44								
Zone 3	0.00	0.00								
Zone 4	0.00	0.00								
Zone 5	0.00	0.00								
PRETENSION CABLES										
Layer:	F p,∞	e p	d p	ε 1	ε 2	ε 3	ε p	σ pd	A p	
	[kN]	[m]	[m]	[%]	[%]	[%]	[%]	[MPa]	[mm ²]	
1a	7249.97	0.02	0.89	0.44753	0.004302	0.9570	1.409	1366.67	8100	
1b	7298.23	0.02	0.89	0.45051	0.004330	0.9570	1.412	1366.67	8100	
2a	7246.90	0.35	1.22	0.44734	0.005103	1.4514	1.904	1366.67	8100	
2b	7304.65	0.35	1.22	0.45090	0.005144	1.4514	1.907	1366.67	8100	
3a	7369.14	1.03	1.89	0.45488	0.011251	2.4441	2.910	1366.67	8100	
3b	7425.63	1.03	1.89	0.45837	0.011337	2.4441	2.914	1366.67	8100	
Total:	43894.51									
							Yielding	> ε pyd		
							Elastic	< ε pyd		
HECK OF COMPRESSIVE ZONE				MOMENT CAPACITY**						
F c	66420.00		kN		M c	49831.13		kNm		
F p	66420.00		kN		M p	31001.81		kNm		
F	0.00		kN		M_Rd	80832.94		> M Ed		
					Utilisation	56.4		< 100		
*Horizontal equilibrium					**Moment equilibrium around centroid					

SECTION IV, x = 3 m									
DESIGN SECTIONAL FORCE									
M _{Ed}	28910.8 kNm								
ASSUMED COMPRESSIVE ZONE									
x	263.91 mm								
	A _c	z _c							
	[mm ²]	[mm]							
Zone 1	2.40E+06	797.26							
Zone 2	367500.00	665.31							
Zone 3	0.00	0.00							
Zone 4	0.00	0.00							
Zone 5	0.00	0.00							
PRETENSION CABLES									
Layer:	∞	e _p	d _p	ε ₁	ε ₂	ε ₃	ε _p	σ _{pd}	A _p
	[kN]	[m]	[m]	[%]	[%]	[%]	[%]	[MPa]	[mm ²]
1a	7168.99	-0.20	0.70	0.44253	0.004510	0.6766	1.124	1366.67	8100
1b	7255.35	-0.20	0.70	0.44786	0.004564	0.6766	1.129	1366.67	8100
2a	7156.97	0.14	1.04	0.44179	0.004367	1.1799	1.626	1366.67	8100
2b	7261.73	0.32	1.04	0.44826	0.004977	1.1799	1.633	1366.67	8100
3a	7278.00	1.00	1.84	0.44926	0.010704	2.3661	2.826	1366.67	8100
3b	7382.14	1.00	1.84	0.45569	0.010857	2.3661	2.833	1366.67	8100
Total:	43503.18								
							Yielding	> ε _{pyd}	
							Elastic	< ε _{pyd}	
CHECK OF COMPRESSIVE ZONE*			MOMENT CAPACITY**						
F _c	66420.00 kN		M _c	51790.365 kNm					
F _p	66420.00 kN		M _p	22683.68 kNm					
	0.00 kN		M _{Rd}	74474.04 kNm					
			Utilisation	38.8 %					
			> M _{Ed}						
			< 100						
*Horizontal equilibrium									
**Moment equilibrium around centroid									

C. Cross-section parameters from Autodesk Robot

SECTION ANALYSIS

Geometry description

Point no.	Y	Z
1	0.0 mm	0.0 mm
2	12000.0 mm	0.0 mm
3	12000.0 mm	-200.0 mm
4	9500.0 mm	-400.0 mm
5	9071.0 mm	-2000.0 mm
6	9018.0 mm	-2200.0 mm
7	2982.0 mm	-2200.0 mm
8	2929.0 mm	-2000.0 mm
9	2500.0 mm	-400.0 mm
10	0.0 mm	-200.0 mm
11	3121.2 mm	-400.0 mm
12	3621.2 mm	-250.0 mm
13	8378.8 mm	-250.0 mm
14	8878.8 mm	-400.0 mm
15	8503.7 mm	-1800.0 mm
16	8003.7 mm	-1950.0 mm
17	3996.3 mm	-1950.0 mm
18	3496.3 mm	-1800.0 mm

General results

Area	A	=	7031082.00 mm ²
Center of gravity	Yc	=	6000.0 mm
	Zc	=	-866.6 mm
Perimeter	S	=	27178.8 mm
Base material	ACCIA		
	E	=	206000.00 MPa
	den	=	7852.83 kg/m ³
	WU	=	55213.92 kG/m
Principal system			
Angle	alpha	=	0.0 Deg
Moments of inertia	Ix	=	0.00 mm ⁴
	Iy	=	4699003370461.74 mm ⁴
	Iz	=	64559522433262.10 mm ⁴
Radii of inertia	iy	=	817.5 mm
	iz	=	3030.2 mm
Shear areas			

	Ay	=	0.00 mm2		
	Az	=	0.00 mm2		
Elastic section moduli	Wely	=	3524042170.39 mm3		
	Welz	=	10759920405.54 mm3		
Shear rigidity factors	Wy	=	0.00 mm2		
	Wz	=	0.00 mm2		
Plastic section moduli	Wply	=	0.00 mm3		
	Wplz	=	0.00 mm3		
Maximum distances	Vy	=	6000.0 mm		
	Vpy	=	6000.0 mm		
	Vz	=	866.6 mm		
	Vpz	=	1333.4 mm		
Central system					
Moments of inertia	Iyc	=	4699003370461.74 mm4		
	Izc	=	64559522433262.10 mm4		
	Iyczc	=	-0.01 mm4		
Radii of inertia	iy	=	817.5 mm		
	iz	=	3030.2 mm		
Maximum distances	Vyc	=	6000.0 mm		
	Vpyc	=	6000.0 mm		
	Vzc	=	866.6 mm		
	Vpzc	=	1333.4 mm		
Arbitrary system					
System position	yc'	=	6000.0 mm	Angle	= 0.0 Deg
	zc'	=	-866.6 mm		
Moments of inertia	Iy'	=	4699003370461.74 mm4		
	Iz'	=	64559522433261.80 mm4		
	Iy'z'	=	-0.00 mm4		
Radii of inertia	iy'	=	817.5 mm		
	iz'	=	3030.2 mm		
First moments of area	Sy'	=	-0.00 mm3		
	Sz'	=	0.00 mm3		
Maximum distances	Vy'	=	6000.0 mm		
	Vpy'	=	6000.0 mm		
	Vz'	=	866.6 mm		
	Vpz'	=	1333.4 mm		

D. Internal forces in longitudinal scheme

INTERNAL FORCES										
Safety factors										
	1.35									
	1.5									
x	Vg1	Vg2	Vq	V max,Q	V ULS	Mg1	Mg2	Mq	M max,Q	M ULS
[m]	[kN]	[kN]	[kN]	[kN]	[kN]	[kNm]	[kNm]	[kNm]	[kNm]	[kNm]
0	3487.5	589.675	915	2364.00	10422.69	0.00	0.00	0.0	0	0.0
0.2	3452.6	583.77825	905.85	2352.00	10335.92	694.01	117.35	182.1	470.4	2074.1
0.4	3417.8	577.8815	896.7	2340.00	10249.15	1381.05	233.51	362.3	936	4127.2
0.6	3382.9	571.98475	887.55	2328.00	10162.39	2061.11	348.50	540.8	1396.8	6159.3
0.8	3348.0	566.088	878.4	2316.00	10075.62	2734.20	462.31	717.4	1852.8	8170.5
1	3313.1	560.19125	869.25	2304.00	9988.85	3400.31	574.93	892.1	2304	10160.8
1.2	3278.3	554.2945	860.1	2292.00	9902.09	4059.45	686.38	1065.1	2750.4	12130.1
1.4	3243.4	548.39775	850.95	2280.00	9815.32	4711.61	796.65	1236.2	3192	14078.4
1.6	3208.5	542.501	841.8	2268.00	9728.55	5356.80	905.74	1405.4	3628.8	16005.8
1.8	3173.6	536.60425	832.65	2256.00	9641.78	5995.01	1013.65	1572.9	4060.8	17912.2
2	3138.8	530.7075	823.5	2244.00	9555.02	6626.25	1120.38	1738.5	4488	19797.7
2.2	3103.9	524.81075	814.35	2232.00	9468.25	7250.51	1225.93	1902.3	4910.4	21662.2
2.4	3069.0	518.914	805.2	2220.00	9381.48	7867.80	1330.31	2064.2	5328	23505.8
2.6	3034.1	513.01725	796.05	2208.00	9294.72	8478.11	1433.50	2224.4	5740.8	25328.4
2.8	2999.3	507.1205	786.9	2196.00	9207.95	9081.45	1535.51	2382.7	6148.8	27130.1
3	2964.4	501.22375	777.75	2184.00	9121.18	9677.81	1636.35	2539.1	6552	28910.8
3.2	2929.5	495.327	768.6	2172.00	9034.42	10267.20	1736.00	2693.8	6950.4	30670.6
3.4	2894.6	489.43025	759.45	2160.00	8947.65	10849.61	1834.48	2846.6	7344	32409.4
3.6	2859.8	483.5335	750.3	2148.00	8860.88	11425.05	1931.78	2997.5	7732.8	34127.2
3.8	2824.9	477.63675	741.15	2136.00	8774.12	11993.51	2027.89	3146.7	8116.8	35824.1
4	2790.0	471.74	732	2124.00	8687.35	12555.00	2122.83	3294.0	8496	37500.1
4.2	2755.1	465.84325	722.85	2112.00	8600.58	13109.51	2216.59	3439.5	8870.4	39155.1
4.4	2720.3	459.9465	713.7	2100.00	8513.82	13657.05	2309.17	3583.1	9240	40789.1
4.6	2685.4	454.04975	704.55	2088.00	8427.05	14197.61	2400.57	3725.0	9604.8	42402.2
4.8	2650.5	448.153	695.4	2076.00	8340.28	14731.20	2490.79	3865.0	9964.8	43994.3
5	2615.6	442.25625	686.25	2064.00	8253.51	15257.81	2579.83	4003.1	10320	45565.5
5.2	2580.8	436.3595	677.1	2052.00	8166.75	15777.45	2667.69	4139.5	10670.4	47115.7
5.4	2545.9	430.46275	667.95	2040.00	8079.98	16290.11	2754.37	4274.0	11016	48645.0
5.6	2511.0	424.566	658.8	2028.00	7993.21	16795.80	2839.87	4406.6	11356.8	50153.3
5.8	2476.1	418.66925	649.65	2016.00	7906.45	17294.51	2924.20	4537.5	11692.8	51640.7
6	2441.3	412.7725	640.5	2004.00	7819.68	17786.25	3007.34	4666.5	12024	53107.1
6.2	2406.4	406.87575	631.35	1992.00	7732.91	18271.01	3089.31	4793.7	12350.4	54552.6
6.4	2371.5	400.979	622.2	1980.00	7646.15	18748.80	3170.09	4919.0	12672	55977.1
6.6	2336.6	395.08225	613.05	1968.00	7559.38	19219.61	3249.70	5042.6	12988.8	57380.6
6.8	2301.8	389.1855	603.9	1956.00	7472.61	19683.45	3328.13	5164.3	13300.8	58763.2
7	2266.9	383.28875	594.75	1944.00	7385.85	20140.31	3405.37	5284.1	13608	60124.9
7.2	2232.0	377.392	585.6	1932.00	7299.08	20590.20	3481.44	5402.2	13910.4	61465.6
7.4	2197.1	371.49525	576.45	1920.00	7212.31	21033.11	3556.33	5518.4	14208	62785.3
7.6	2162.3	365.5985	567.3	1908.00	7125.55	21469.05	3630.04	5632.7	14500.8	64084.1
7.8	2127.4	359.70175	558.15	1896.00	7038.78	21898.01	3702.57	5745.3	14788.8	65361.9
8	2092.5	353.805	549	1884.00	6952.01	22320.00	3773.92	5856.0	15072	66618.8
8.2	2057.6	347.90825	539.85	1872.00	6865.24	22735.01	3844.09	5964.9	15350.4	67854.7
8.4	2022.8	342.0115	530.7	1860.00	6778.48	23143.05	3913.08	6071.9	15624	69069.7
8.6	1987.9	336.11475	521.55	1848.00	6691.71	23544.11	3980.90	6177.2	15892.8	70263.7
8.8	1953.0	330.218	512.4	1836.00	6604.94	23938.20	4047.53	6280.6	16156.8	71436.8
9	1918.1	324.32125	503.25	1824.00	6518.18	24325.31	4112.98	6382.1	16416	72588.9
9.2	1883.3	318.4245	494.1	1812.00	6431.41	24705.45	4177.26	6481.9	16670.4	73720.0
9.4	1848.4	312.52775	484.95	1800.00	6344.64	25078.61	4240.35	6579.8	16920	74830.3
9.6	1813.5	306.631	475.8	1788.00	6257.88	25444.80	4302.27	6675.8	17164.8	75919.5
9.8	1778.6	300.73425	466.65	1776.00	6171.11	25804.01	4363.01	6770.1	17404.8	76987.8
10	1743.8	294.8375	457.5	1764.00	6084.34	26156.25	4422.56	6862.5	17640	78035.1

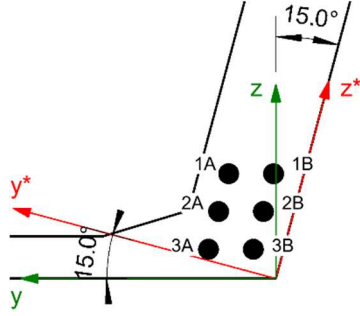
10.2	1708.9	288.94075	448.35	1752.00	5997.58	26501.51	4480.94	6953.1	17870.4	79061.5
10.4	1674.0	283.044	439.2	1740.00	5910.81	26839.80	4538.14	7041.8	18096	80067.0
10.6	1639.1	277.14725	430.05	1728.00	5824.04	27171.11	4594.16	7128.8	18316.8	81051.5
10.8	1604.3	271.2505	420.9	1716.00	5737.28	27495.45	4649.00	7213.9	18532.8	82015.0
11	1569.4	265.35375	411.75	1704.00	5650.51	27812.81	4702.66	7297.1	18744	82957.6
11.2	1534.5	259.457	402.6	1692.00	5563.74	28123.20	4755.14	7378.6	18950.4	83879.2
11.4	1499.6	253.56025	393.45	1680.00	5476.98	28426.61	4806.44	7458.2	19152	84779.9
11.6	1464.8	247.6635	384.3	1668.00	5390.21	28723.05	4856.56	7535.9	19348.8	85659.6
11.8	1429.9	241.76675	375.15	1656.00	5303.44	29012.51	4905.51	7611.9	19540.8	86518.4
12	1395.0	235.87	366	1644.00	5216.67	29295.00	4953.27	7686.0	19728	87356.2
12.2	1360.1	229.97325	356.85	1632.00	5129.91	29570.51	4999.85	7758.3	19910.4	88173.0
12.4	1325.3	224.0765	347.7	1620.00	5043.14	29839.05	5045.26	7828.7	20088	88968.9
12.6	1290.4	218.17975	338.55	1608.00	4956.37	30100.61	5089.48	7897.4	20260.8	89743.9
12.8	1255.5	212.283	329.4	1596.00	4869.61	30355.20	5132.53	7964.2	20428.8	90497.9
13	1220.6	206.38625	320.25	1584.00	4782.84	30602.81	5174.40	8029.1	20592	91230.9
13.2	1185.8	200.4895	311.1	1572.00	4696.07	30843.45	5215.09	8092.3	20750.4	91943.0
13.4	1150.9	194.59275	301.95	1560.00	4609.31	31077.11	5254.59	8153.6	20904	92634.2
13.6	1116.0	188.696	292.8	1548.00	4522.54	31303.80	5292.92	8213.0	21052.8	93304.3
13.8	1081.1	182.79925	283.65	1536.00	4435.77	31523.51	5330.07	8270.7	21196.8	93953.6
14	1046.3	176.9025	274.5	1524.00	4349.01	31736.25	5366.04	8326.5	21336	94581.8
14.2	1011.4	171.00575	265.35	1512.00	4262.24	31942.01	5400.83	8380.5	21470.4	95189.2
14.4	976.5	165.109	256.2	1500.00	4175.47	32140.80	5434.44	8432.6	21600	95775.5
14.6	941.6	159.21225	247.05	1488.00	4088.71	32332.61	5466.88	8483.0	21724.8	96341.0
14.8	906.8	153.3155	237.9	1476.00	4001.94	32517.45	5498.13	8531.5	21844.8	96885.4
15	871.9	147.41875	228.75	1464.00	3915.17	32695.31	5528.20	8578.1	21960	97408.9
15.2	837.0	141.522	219.6	1452.00	3828.40	32866.20	5557.10	8623.0	22070.4	97911.5
15.4	802.1	135.62525	210.45	1440.00	3741.64	33030.11	5584.81	8666.0	22176	98393.1
15.6	767.3	129.7285	201.3	1428.00	3654.87	33187.05	5611.35	8707.1	22276.8	98853.7
15.8	732.4	123.83175	192.15	1416.00	3568.10	33337.01	5636.70	8746.5	22372.8	99293.4
16	697.5	117.935	183	1404.00	3481.34	33480.00	5660.88	8784.0	22464	99712.2
16.2	662.6	112.03825	173.85	1392.00	3394.57	33616.01	5683.88	8819.7	22550.4	100110.0
16.4	627.8	106.1415	164.7	1380.00	3307.80	33745.05	5705.70	8853.5	22632	100486.8
16.6	592.9	100.24475	155.55	1368.00	3221.04	33867.11	5726.33	8885.6	22708.8	100842.7
16.8	558.0	94.348	146.4	1356.00	3134.27	33982.20	5745.79	8915.8	22780.8	101177.6
17	523.1	88.45125	137.25	1344.00	3047.50	34090.31	5764.07	8944.1	22848	101491.6
17.2	488.3	82.5545	128.1	1332.00	2960.74	34191.45	5781.17	8970.7	22910.4	101784.6
17.4	453.4	76.65775	118.95	1320.00	2873.97	34285.61	5797.09	8995.4	22968	102056.7
17.6	418.5	70.761	109.8	1308.00	2787.20	34372.80	5811.84	9018.2	23020.8	102307.8
17.8	383.6	64.86425	100.65	1296.00	2700.44	34453.01	5825.40	9039.3	23068.8	102538.0
18	348.8	58.9675	91.5	1284.00	2613.67	34526.25	5837.78	9058.5	23112	102747.2
18.2	313.9	53.07075	82.35	1272.00	2526.90	34592.51	5848.99	9075.9	23150.4	102935.5
18.4	279.0	47.174	73.2	1260.00	2440.13	34651.80	5859.01	9091.4	23184	103102.8
18.6	244.1	41.27725	64.05	1248.00	2353.37	34704.11	5867.86	9105.2	23212.8	103249.1
18.8	209.3	35.3805	54.9	1236.00	2266.60	34749.45	5875.52	9117.1	23236.8	103374.5
19	174.4	29.48375	45.75	1224.00	2179.83	34787.81	5882.01	9127.1	23256	103478.9
19.2	139.5	23.587	36.6	1212.00	2093.07	34819.20	5887.32	9135.4	23270.4	103562.4
19.4	104.6	17.69025	27.45	1200.00	2006.30	34843.61	5891.44	9141.8	23280	103625.0
19.6	69.7	11.7935	18.3	1188.00	1919.53	34861.05	5894.39	9146.3	23284.8	103666.6
19.8	34.9	5.89675	9.15	1176.00	1832.77	34871.51	5896.16	9149.1	23284.8	103687.2
20	0.0	0	0	1164.00	1746.00	34875.00	5896.75	9150.0	23280	103686.9

20.2	-34.9	-5.89675	-9.15	-1176.00	-1832.77	34871.51	5896.16	9149.1	23284.8	103687.2
20.4	-69.7	-11.7935	-18.3	-1188.00	-1919.53	34861.05	5894.39	9146.3	23284.8	103666.6
20.6	-104.6	-17.69025	-27.45	-1200.00	-2006.30	34843.61	5891.44	9141.8	23280	103625.0
20.8	-139.5	-23.587	-36.6	-1212.00	-2093.07	34819.20	5887.32	9135.4	23270.4	103562.4
21	-174.4	-29.48375	-45.75	-1224.00	-2179.83	34787.81	5882.01	9127.1	23256	103478.9
21.2	-209.3	-35.3805	-54.9	-1236.00	-2266.60	34749.45	5875.52	9117.1	23236.8	103374.5
21.4	-244.1	-41.27725	-64.05	-1248.00	-2353.37	34704.11	5867.86	9105.2	23212.8	103249.1
21.6	-279.0	-47.174	-73.2	-1260.00	-2440.13	34651.80	5859.01	9091.4	23184	103102.8
21.8	-313.9	-53.07075	-82.35	-1272.00	-2526.90	34592.51	5848.99	9075.9	23150.4	102935.5
22	-348.8	-58.9675	-91.5	-1284.00	-2613.67	34526.25	5837.78	9058.5	23112	102747.2
22.2	-383.6	-64.86425	-100.65	-1296.00	-2700.44	34453.01	5825.40	9039.3	23068.8	102538.0
22.4	-418.5	-70.761	-109.8	-1308.00	-2787.20	34372.80	5811.84	9018.2	23020.8	102307.8
22.6	-453.4	-76.65775	-118.95	-1320.00	-2873.97	34285.61	5797.09	8995.4	22968	102056.7
22.8	-488.3	-82.5545	-128.1	-1332.00	-2960.74	34191.45	5781.17	8970.7	22910.4	101784.6
23	-523.1	-88.45125	-137.25	-1344.00	-3047.50	34090.31	5764.07	8944.1	22848	101491.6
23.2	-558.0	-94.348	-146.4	-1356.00	-3134.27	33982.20	5745.79	8915.8	22780.8	101177.6
23.4	-592.9	-100.2448	-155.55	-1368.00	-3221.04	33867.11	5726.33	8885.6	22708.8	100842.7
23.6	-627.8	-106.1415	-164.7	-1380.00	-3307.80	33745.05	5705.70	8853.5	22632	100486.8
23.8	-662.6	-112.0383	-173.85	-1392.00	-3394.57	33616.01	5683.88	8819.7	22550.4	100110.0
24	-697.5	-117.935	-183	-1404.00	-3481.34	33480.00	5660.88	8784.0	22464	99712.2
24.2	-732.4	-123.8318	-192.15	-1416.00	-3568.10	33337.01	5636.70	8746.5	22372.8	99293.4
24.4	-767.3	-129.7285	-201.3	-1428.00	-3654.87	33187.05	5611.35	8707.1	22276.8	98853.7
24.6	-802.1	-135.6253	-210.45	-1440.00	-3741.64	33030.11	5584.81	8666.0	22176	98393.1
24.8	-837.0	-141.522	-219.6	-1452.00	-3828.40	32866.20	5557.10	8623.0	22070.4	97911.5
25	-871.9	-147.4188	-228.75	-1464.00	-3915.17	32695.31	5528.20	8578.1	21960	97408.9
25.2	-906.8	-153.3155	-237.9	-1476.00	-4001.94	32517.45	5498.13	8531.5	21844.8	96885.4
25.4	-941.6	-159.2123	-247.05	-1488.00	-4088.71	32332.61	5466.88	8483.0	21724.8	96341.0
25.6	-976.5	-165.109	-256.2	-1500.00	-4175.47	32140.80	5434.44	8432.6	21600	95775.5
25.8	-1011.4	-171.0058	-265.35	-1512.00	-4262.24	31942.01	5400.83	8380.5	21470.4	95189.2
26	-1046.3	-176.9025	-274.5	-1524.00	-4349.01	31736.25	5366.04	8326.5	21336	94581.8
26.2	-1081.1	-182.7993	-283.65	-1536.00	-4435.77	31523.51	5330.07	8270.7	21196.8	93953.6
26.4	-1116.0	-188.696	-292.8	-1548.00	-4522.54	31303.80	5292.92	8213.0	21052.8	93304.3
26.6	-1150.9	-194.5928	-301.95	-1560.00	-4609.31	31077.11	5254.59	8153.6	20904	92634.2
26.8	-1185.8	-200.4895	-311.1	-1572.00	-4696.07	30843.45	5215.09	8092.3	20750.4	91943.0
27	-1220.6	-206.3863	-320.25	-1584.00	-4782.84	30602.81	5174.40	8029.1	20592	91230.9
27.2	-1255.5	-212.283	-329.4	-1596.00	-4869.61	30355.20	5132.53	7964.2	20428.8	90497.9
27.4	-1290.4	-218.1798	-338.55	-1608.00	-4956.37	30100.61	5089.48	7897.4	20260.8	89743.9
27.6	-1325.3	-224.0765	-347.7	-1620.00	-5043.14	29839.05	5045.26	7828.7	20088	88968.9
27.8	-1360.1	-229.9733	-356.85	-1632.00	-5129.91	29570.51	4999.85	7758.3	19910.4	88173.0
28	-1395.0	-235.87	-366	-1644.00	-5216.67	29295.00	4953.27	7686.0	19728	87356.2
28.2	-1429.9	-241.7668	-375.15	-1656.00	-5303.44	29012.51	4905.51	7611.9	19540.8	86518.4
28.4	-1464.8	-247.6635	-384.3	-1668.00	-5390.21	28723.05	4856.56	7535.9	19348.8	85659.6
28.6	-1499.6	-253.5603	-393.45	-1680.00	-5476.98	28426.61	4806.44	7458.2	19152	84779.9
28.8	-1534.5	-259.457	-402.6	-1692.00	-5563.74	28123.20	4755.14	7378.6	18950.4	83879.2
29	-1569.4	-265.3538	-411.75	-1704.00	-5650.51	27812.81	4702.66	7297.1	18744	82957.6
29.2	-1604.3	-271.2505	-420.9	-1716.00	-5737.28	27495.45	4649.00	7213.9	18532.8	82015.0
29.4	-1639.1	-277.1473	-430.05	-1728.00	-5824.04	27171.11	4594.16	7128.8	18316.8	81051.5
29.6	-1674.0	-283.044	-439.2	-1740.00	-5910.81	26839.80	4538.14	7041.8	18096	80067.0
29.8	-1708.9	-288.9408	-448.35	-1752.00	-5997.58	26501.51	4480.94	6953.1	17870.4	79061.5
30	-1743.8	-294.8375	-457.5	-1764.00	-6084.34	26156.25	4422.56	6862.5	17640	78035.1

30.2	-1778.6	-300.7343	-466.65	-1776.00	-6171.11	25804.01	4363.01	6770.1	17404.8	76987.8
30.4	-1813.5	-306.631	-475.8	-1788.00	-6257.88	25444.80	4302.27	6675.8	17164.8	75919.5
30.6	-1848.4	-312.5278	-484.95	-1800.00	-6344.64	25078.61	4240.35	6579.8	16920	74830.3
30.8	-1883.3	-318.4245	-494.1	-1812.00	-6431.41	24705.45	4177.26	6481.9	16670.4	73720.0
31	-1918.1	-324.3213	-503.25	-1824.00	-6518.18	24325.31	4112.98	6382.1	16416	72588.9
31.2	-1953.0	-330.218	-512.4	-1836.00	-6604.94	23938.20	4047.53	6280.6	16156.8	71436.8
31.4	-1987.9	-336.1148	-521.55	-1848.00	-6691.71	23544.11	3980.90	6177.2	15892.8	70263.7
31.6	-2022.8	-342.0115	-530.7	-1860.00	-6778.48	23143.05	3913.08	6071.9	15624	69069.7
31.8	-2057.6	-347.9083	-539.85	-1872.00	-6865.24	22735.01	3844.09	5964.9	15350.4	67854.7
32	-2092.5	-353.805	-549	-1884.00	-6952.01	22320.00	3773.92	5856.0	15072	66618.8
32.2	-2127.4	-359.7018	-558.15	-1896.00	-7038.78	21898.01	3702.57	5745.3	14788.8	65361.9
32.4	-2162.3	-365.5985	-567.3	-1908.00	-7125.55	21469.05	3630.04	5632.7	14500.8	64084.1
32.6	-2197.1	-371.4953	-576.45	-1920.00	-7212.31	21033.11	3556.33	5518.4	14208	62785.3
32.8	-2232.0	-377.392	-585.6	-1932.00	-7299.08	20590.20	3481.44	5402.2	13910.4	61465.6
33	-2266.9	-383.2888	-594.75	-1944.00	-7385.85	20140.31	3405.37	5284.1	13608	60124.9
33.2	-2301.8	-389.1855	-603.9	-1956.00	-7472.61	19683.45	3328.13	5164.3	13300.8	58763.2
33.4	-2336.6	-395.0823	-613.05	-1968.00	-7559.38	19219.61	3249.70	5042.6	12988.8	57380.6
33.6	-2371.5	-400.979	-622.2	-1980.00	-7646.15	18748.80	3170.09	4919.0	12672	55977.1
33.8	-2406.4	-406.8758	-631.35	-1992.00	-7732.91	18271.01	3089.31	4793.7	12350.4	54552.6
34	-2441.3	-412.7725	-640.5	-2004.00	-7819.68	17786.25	3007.34	4666.5	12024	53107.1
34.2	-2476.1	-418.6693	-649.65	-2016.00	-7906.45	17294.51	2924.20	4537.5	11692.8	51640.7
34.4	-2511.0	-424.566	-658.8	-2028.00	-7993.21	16795.80	2839.87	4406.6	11356.8	50153.3
34.6	-2545.9	-430.4628	-667.95	-2040.00	-8079.98	16290.11	2754.37	4274.0	11016	48645.0
34.8	-2580.8	-436.3595	-677.1	-2052.00	-8166.75	15777.45	2667.69	4139.5	10670.4	47115.7
35	-2615.6	-442.2563	-686.25	-2064.00	-8253.51	15257.81	2579.83	4003.1	10320	45565.5
35.2	-2650.5	-448.153	-695.4	-2076.00	-8340.28	14731.20	2490.79	3865.0	9964.8	43994.3
35.4	-2685.4	-454.0498	-704.55	-2088.00	-8427.05	14197.61	2400.57	3725.0	9604.8	42402.2
35.6	-2720.3	-459.9465	-713.7	-2100.00	-8513.82	13657.05	2309.17	3583.1	9240	40789.1
35.8	-2755.1	-465.8433	-722.85	-2112.00	-8600.58	13109.51	2216.59	3439.5	8870.4	39155.1
36	-2790.0	-471.74	-732	-2124.00	-8687.35	12555.00	2122.83	3294.0	8496	37500.1
36.2	-2824.9	-477.6368	-741.15	-2136.00	-8774.12	11993.51	2027.89	3146.7	8116.8	35824.1
36.4	-2859.8	-483.5335	-750.3	-2148.00	-8860.88	11425.05	1931.78	2997.5	7732.8	34127.2
36.6	-2894.6	-489.4303	-759.45	-2160.00	-8947.65	10849.61	1834.48	2846.6	7344	32409.4
36.8	-2929.5	-495.327	-768.6	-2172.00	-9034.42	10267.20	1736.00	2693.8	6950.4	30670.6
37	-2964.4	-501.2238	-777.75	-2184.00	-9121.18	9677.81	1636.35	2539.1	6552	28910.8
37.2	-2999.3	-507.1205	-786.9	-2196.00	-9207.95	9081.45	1535.51	2382.7	6148.8	27130.1
37.4	-3034.1	-513.0173	-796.05	-2208.00	-9294.72	8478.11	1433.50	2224.4	5740.8	25328.4
37.6	-3069.0	-518.914	-805.2	-2220.00	-9381.48	7867.80	1330.31	2064.2	5328	23505.8
37.8	-3103.9	-524.8108	-814.35	-2232.00	-9468.25	7250.51	1225.93	1902.3	4910.4	21662.2
38	-3138.8	-530.7075	-823.5	-2244.00	-9555.02	6626.25	1120.38	1738.5	4488	19797.7
38.2	-3173.6	-536.6043	-832.65	-2256.00	-9641.78	5995.01	1013.65	1572.9	4060.8	17912.2
38.4	-3208.5	-542.501	-841.8	-2268.00	-9728.55	5356.80	905.74	1405.4	3628.8	16005.8
38.6	-3243.4	-548.3978	-850.95	-2280.00	-9815.32	4711.61	796.65	1236.2	3192	14078.4
38.8	-3278.3	-554.2945	-860.1	-2292.00	-9902.09	4059.45	686.38	1065.1	2750.4	12130.1
39	-3313.1	-560.1913	-869.25	-2304.00	-9988.85	3400.31	574.93	892.1	2304	10160.8
39.2	-3348.0	-566.088	-878.4	-2316.00	-10075.62	2734.20	462.31	717.4	1852.8	8170.5
39.4	-3382.9	-571.9848	-887.55	-2328.00	-10162.39	2061.11	348.50	540.8	1396.8	6159.3
39.6	-3417.8	-577.8815	-896.7	-2340.00	-10249.15	1381.05	233.51	362.3	936	4127.2
39.8	-3452.6	-583.7783	-905.85	-2352.00	-10335.92	694.01	117.35	182.1	470.4	2074.1
40	-3487.5	-589.675	-915	-2364.00	-10422.69	0.00	0.00	0.0	0	0.0

E. Cable paths

The following appendix refers to the following frames of reference and transformation rules:



$$y = y^* \cdot \cos(15^\circ) - z^* \cdot \sin(15^\circ)$$

$$z = y^* \cdot \sin(15^\circ) + z^* \cdot \cos(15^\circ)$$

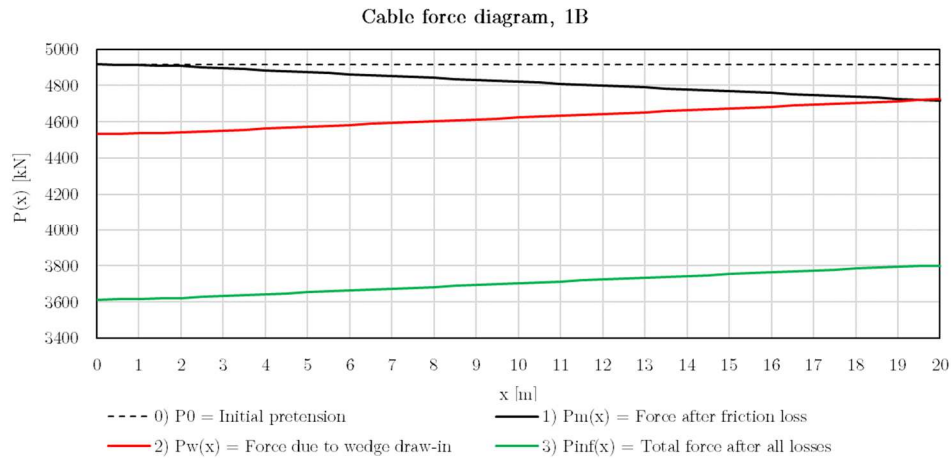
The following shows only half the bridge due to symmetry, units are [m]:

CABLE 1A					CABLE 1B				
x	y*	z*	y	z	x	y*	z*	y	z
0	0.6500	1.6893	0.1906	1.8000	0	0.2500	1.7965	-0.2235	1.8000
0.5	0.6182	1.6467	0.1709	1.7506	0.5	0.2389	1.7483	-0.2217	1.7506
1	0.5864	1.6041	0.1513	1.7012	1	0.2279	1.7002	-0.2199	1.7012
1.5	0.5546	1.5615	0.1316	1.6518	1.5	0.2168	1.6520	-0.2182	1.6518
2	0.5229	1.5189	0.1119	1.6024	2	0.2057	1.6038	-0.2164	1.6024
2.5	0.4937	1.4758	0.0949	1.5533	2.5	0.1956	1.5557	-0.2137	1.5533
3	0.4699	1.4318	0.0833	1.5046	3	0.1873	1.5075	-0.2093	1.5046
3.5	0.4513	1.3868	0.0770	1.4564	3.5	0.1808	1.4593	-0.2031	1.4564
4	0.4381	1.3410	0.0761	1.4087	4	0.1762	1.4112	-0.1950	1.4087
4.5	0.4301	1.2942	0.0805	1.3614	4.5	0.1734	1.3630	-0.1853	1.3614
5	0.4275	1.2465	0.0903	1.3147	5	0.1725	1.3148	-0.1737	1.3147
5.5	0.4275	1.1991	0.1026	1.2689	5.5	0.1725	1.2675	-0.1614	1.2689
6	0.4275	1.1534	0.1144	1.2247	6	0.1725	1.2217	-0.1496	1.2247
6.5	0.4275	1.1092	0.1258	1.1821	6.5	0.1725	1.1775	-0.1381	1.1821
7	0.4275	1.0667	0.1369	1.1410	7	0.1725	1.1350	-0.1271	1.1410
7.5	0.4275	1.0257	0.1475	1.1014	7.5	0.1725	1.0941	-0.1165	1.1014
8	0.4275	0.9864	0.1576	1.0634	8	0.1725	1.0547	-0.1064	1.0634
8.5	0.4275	0.9487	0.1674	1.0270	8.5	0.1725	1.0170	-0.0966	1.0270
9	0.4275	0.9125	0.1768	0.9921	9	0.1725	0.9809	-0.0872	0.9921
9.5	0.4275	0.8780	0.1857	0.9587	9.5	0.1725	0.9463	-0.0783	0.9587
10	0.4275	0.8451	0.1942	0.9269	10	0.1725	0.9134	-0.0698	0.9269
10.5	0.4275	0.8138	0.2023	0.8967	10.5	0.1725	0.8821	-0.0617	0.8967
11	0.4275	0.7841	0.2100	0.8680	11	0.1725	0.8524	-0.0540	0.8680
11.5	0.4275	0.7560	0.2173	0.8409	11.5	0.1725	0.8243	-0.0467	0.8409
12	0.4275	0.7295	0.2241	0.8153	12	0.1725	0.7978	-0.0399	0.8153
12.5	0.4275	0.7046	0.2306	0.7912	12.5	0.1725	0.7729	-0.0334	0.7912
13	0.4275	0.6813	0.2366	0.7688	13	0.1725	0.7497	-0.0274	0.7688
13.5	0.4275	0.6596	0.2422	0.7478	13.5	0.1725	0.7280	-0.0218	0.7478
14	0.4275	0.6396	0.2474	0.7284	14	0.1725	0.7079	-0.0166	0.7284
14.5	0.4275	0.6211	0.2522	0.7106	14.5	0.1725	0.6894	-0.0118	0.7106
15	0.4275	0.6043	0.2565	0.6943	15	0.1725	0.6726	-0.0075	0.6943
15.5	0.4275	0.5890	0.2605	0.6796	15.5	0.1725	0.6573	-0.0035	0.6796
16	0.4275	0.5754	0.2640	0.6664	16	0.1725	0.6437	0.0000	0.6664
16.5	0.4275	0.5633	0.2671	0.6548	16.5	0.1725	0.6316	0.0031	0.6548
17	0.4275	0.5529	0.2698	0.6447	17	0.1725	0.6212	0.0058	0.6447
17.5	0.4275	0.5440	0.2721	0.6362	17.5	0.1725	0.6124	0.0081	0.6362
18	0.4275	0.5368	0.2740	0.6292	18	0.1725	0.6051	0.0100	0.6292
18.5	0.4275	0.5312	0.2754	0.6237	18.5	0.1725	0.5995	0.0115	0.6237
19	0.4275	0.5272	0.2765	0.6199	19	0.1725	0.5955	0.0125	0.6199
19.5	0.4275	0.5248	0.2771	0.6175	19.5	0.1725	0.5931	0.0131	0.6175
20	0.4275	0.5240	0.2773	0.6168	20	0.1725	0.5923	0.0133	0.6168

CABLE 2A					CABLE 2B				
x	y*	z*	y	z	x	y*	z*	y	z
0	0.6500	1.3270	0.2844	1.4500	0	0.2500	1.4342	-0.1297	1.4500
0.5	0.6182	1.2859	0.2643	1.4021	0.5	0.2389	1.3876	-0.1283	1.4021
1	0.5864	1.2449	0.2442	1.3542	1	0.2279	1.3410	-0.1270	1.3542
1.5	0.5546	1.2038	0.2242	1.3064	1.5	0.2168	1.2944	-0.1256	1.3064
2	0.5229	1.1628	0.2041	1.2585	2	0.2057	1.2478	-0.1242	1.2585
2.5	0.4937	1.1213	0.1867	1.2108	2.5	0.1956	1.2011	-0.1220	1.2108
3	0.4699	1.0788	0.1747	1.1637	3	0.1873	1.1545	-0.1179	1.1637
3.5	0.4513	1.0355	0.1680	1.1170	3.5	0.1808	1.1079	-0.1121	1.1170
4	0.4381	0.9912	0.1666	1.0708	4	0.1762	1.0613	-0.1045	1.0708
4.5	0.4301	0.9459	0.1707	1.0250	4.5	0.1734	1.0147	-0.0951	1.0250
5	0.4275	0.8998	0.1800	0.9798	5	0.1725	0.9681	-0.0839	0.9798
5.5	0.4275	0.8541	0.1919	0.9356	5.5	0.1725	0.9224	-0.0721	0.9356
6	0.4275	0.8102	0.2032	0.8932	6	0.1725	0.8785	-0.0608	0.8932
6.5	0.4275	0.7681	0.2141	0.8525	6.5	0.1725	0.8364	-0.0499	0.8525
7	0.4275	0.7277	0.2246	0.8136	7	0.1725	0.7961	-0.0394	0.8136
7.5	0.4275	0.6892	0.2346	0.7764	7.5	0.1725	0.7575	-0.0294	0.7764
8	0.4275	0.6525	0.2441	0.7409	8	0.1725	0.7208	-0.0199	0.7409
8.5	0.4275	0.6175	0.2531	0.7071	8.5	0.1725	0.6858	-0.0109	0.7071
9	0.4275	0.5843	0.2617	0.6751	9	0.1725	0.6527	-0.0023	0.6751
9.5	0.4275	0.5530	0.2698	0.6448	9.5	0.1725	0.6213	0.0058	0.6448
10	0.4275	0.5234	0.2775	0.6162	10	0.1725	0.5917	0.0135	0.6162
10.5	0.4275	0.4956	0.2847	0.5894	10.5	0.1725	0.5639	0.0207	0.5894
11	0.4275	0.4696	0.2914	0.5643	11	0.1725	0.5380	0.0274	0.5643
11.5	0.4275	0.4454	0.2976	0.5409	11.5	0.1725	0.5138	0.0337	0.5409
12	0.4275	0.4230	0.3034	0.5193	12	0.1725	0.4914	0.0395	0.5193
12.5	0.4275	0.4024	0.3088	0.4993	12.5	0.1725	0.4707	0.0448	0.4993
13	0.4275	0.3836	0.3137	0.4812	13	0.1725	0.4519	0.0497	0.4812
13.5	0.4275	0.3666	0.3181	0.4647	13.5	0.1725	0.4349	0.0541	0.4647
14	0.4275	0.3513	0.3220	0.4500	14	0.1725	0.4197	0.0580	0.4500
14.5	0.4275	0.3379	0.3255	0.4370	14.5	0.1725	0.4062	0.0615	0.4370
15	0.4275	0.3262	0.3285	0.4258	15	0.1725	0.3946	0.0645	0.4258
15.5	0.4275	0.3164	0.3310	0.4162	15.5	0.1725	0.3847	0.0671	0.4162
16	0.4275	0.3083	0.3331	0.4085	16	0.1725	0.3766	0.0691	0.4085
16.5	0.4275	0.3020	0.3348	0.4024	16.5	0.1725	0.3704	0.0708	0.4024
17	0.4275	0.2976	0.3359	0.3981	17	0.1725	0.3659	0.0719	0.3981
17.5	0.4275	0.2949	0.3366	0.3955	17.5	0.1725	0.3632	0.0726	0.3955
18	0.4275	0.2940	0.3368	0.3946	18	0.1725	0.3623	0.0729	0.3946
18.5	0.4275	0.2940	0.3368	0.3946	18.5	0.1725	0.3623	0.0729	0.3946
19	0.4275	0.2940	0.3368	0.3946	19	0.1725	0.3623	0.0729	0.3946
19.5	0.4275	0.2940	0.3368	0.3946	19.5	0.1725	0.3623	0.0729	0.3946
20	0.4275	0.2940	0.3368	0.3946	20	0.1725	0.3623	0.0729	0.3946

CABLE 3A					CABLE 3B				
x	y*	z*	y	z	x	y*	z*	y	z
0	0.6500	0.2917	0.5524	0.4500	0	0.2500	0.3989	0.1382	0.4500
0.5	0.6182	0.2846	0.5235	0.4349	0.5	0.2389	0.3862	0.1308	0.4349
1	0.5864	0.2774	0.4946	0.4197	1	0.2279	0.3735	0.1234	0.4197
1.5	0.5546	0.2703	0.4658	0.4046	1.5	0.2168	0.3608	0.1160	0.4046
2	0.5229	0.2631	0.4369	0.3895	2	0.2057	0.3481	0.1086	0.3895
2.5	0.4937	0.2555	0.4108	0.3746	2.5	0.1956	0.3354	0.1021	0.3746
3	0.4699	0.2470	0.3899	0.3602	3	0.1873	0.3227	0.0974	0.3602
3.5	0.4513	0.2375	0.3745	0.3463	3.5	0.1808	0.3100	0.0944	0.3463
4	0.4381	0.2272	0.3644	0.3328	4	0.1762	0.2973	0.0932	0.3328
4.5	0.4301	0.2158	0.3596	0.3198	4.5	0.1734	0.2846	0.0938	0.3198
5	0.4275	0.2036	0.3602	0.3073	5	0.1725	0.2719	0.0962	0.3073
5.5	0.4275	0.1912	0.3634	0.2953	5.5	0.1725	0.2595	0.0994	0.2953
6	0.4275	0.1794	0.3665	0.2839	6	0.1725	0.2477	0.1025	0.2839
6.5	0.4275	0.1681	0.3694	0.2730	6.5	0.1725	0.2365	0.1054	0.2730
7	0.4275	0.1575	0.3722	0.2627	7	0.1725	0.2258	0.1082	0.2627
7.5	0.4275	0.1474	0.3748	0.2530	7.5	0.1725	0.2157	0.1108	0.2530
8	0.4275	0.1378	0.3773	0.2438	8	0.1725	0.2062	0.1133	0.2438
8.5	0.4275	0.1289	0.3796	0.2351	8.5	0.1725	0.1972	0.1156	0.2351
9	0.4275	0.1205	0.3817	0.2271	9	0.1725	0.1888	0.1177	0.2271
9.5	0.4275	0.1127	0.3838	0.2195	9.5	0.1725	0.1811	0.1198	0.2195
10	0.4275	0.1055	0.3856	0.2126	10	0.1725	0.1738	0.1216	0.2126
10.5	0.4275	0.0989	0.3873	0.2062	10.5	0.1725	0.1672	0.1233	0.2062
11	0.4275	0.0928	0.3889	0.2003	11	0.1725	0.1612	0.1249	0.2003
11.5	0.4275	0.0873	0.3903	0.1950	11.5	0.1725	0.1557	0.1263	0.1950
12	0.4275	0.0824	0.3916	0.1903	12	0.1725	0.1508	0.1276	0.1903
12.5	0.4275	0.0781	0.3927	0.1861	12.5	0.1725	0.1464	0.1287	0.1861
13	0.4275	0.0744	0.3937	0.1825	13	0.1725	0.1427	0.1297	0.1825
13.5	0.4275	0.0712	0.3945	0.1794	13.5	0.1725	0.1395	0.1305	0.1794
14	0.4275	0.0686	0.3952	0.1769	14	0.1725	0.1369	0.1312	0.1769
14.5	0.4275	0.0666	0.3957	0.1749	14.5	0.1725	0.1349	0.1317	0.1749
15	0.4275	0.0651	0.3961	0.1736	15	0.1725	0.1335	0.1321	0.1736
15.5	0.4275	0.0643	0.3963	0.1727	15.5	0.1725	0.1326	0.1323	0.1727
16	0.4275	0.0640	0.3964	0.1724	16	0.1725	0.1323	0.1324	0.1724
16.5	0.4275	0.0640	0.3964	0.1724	16.5	0.1725	0.1323	0.1324	0.1724
17	0.4275	0.0640	0.3964	0.1724	17	0.1725	0.1323	0.1324	0.1724
17.5	0.4275	0.0640	0.3964	0.1724	17.5	0.1725	0.1323	0.1324	0.1724
18	0.4275	0.0640	0.3964	0.1724	18	0.1725	0.1323	0.1324	0.1724
18.5	0.4275	0.0640	0.3964	0.1724	18.5	0.1725	0.1323	0.1324	0.1724
19	0.4275	0.0640	0.3964	0.1724	19	0.1725	0.1323	0.1324	0.1724
19.5	0.4275	0.0640	0.3964	0.1724	19.5	0.1725	0.1323	0.1324	0.1724
20	0.4275	0.0640	0.3964	0.1724	20	0.1725	0.1323	0.1324	0.1724

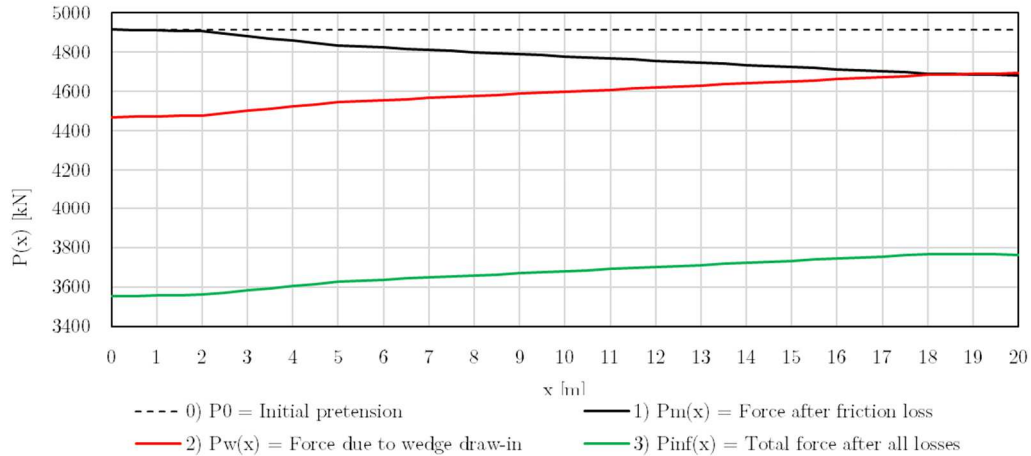
F. Cable force diagrams



Cable 1B														
x	y^{*}	z^{*}	$atan(y^{*})$	$atan(z^{*})$	θ_y	θ_z	ΔP_{μ}	$P_{\mu}(x)$	$P_w(x)$	$P_0(x)$	$\frac{Loss}{ratio, t_0}$	ΔP_{∞}	$P_{\infty}(x)$	$\frac{Loss}{ratio, t_{\infty}}$
[m]	[-]	[-]	[rad]	[rad]	[rad]	[rad]	[kN]	[kN]	[kN]	[kN]	[%]	[kN]	[kN]	[%]
0	-0.02214	-0.0963	0.0221	0.09604	0	0	0.00	4916.7	4532.2	4532.2		919.25	3612.90	
0.5	-0.02214	-0.0963	0.0221	0.09604	0	0	2.33	4914.3	4534.3	4534.3		919.25	3615.06	
1	-0.02214	-0.0963	0.0221	0.09604	0	0	4.67	4912.0	4536.5	4536.5		919.25	3617.21	
1.5	-0.02214	-0.0963	0.0221	0.09604	0	0	7.00	4909.7	4538.6	4538.6		919.25	3619.37	
2	-0.02214	-0.0963	0.0221	0.09604	0	0	9.33	4907.3	4540.8	4540.8		919.25	3621.52	
2.5	-0.01845	-0.0963	0.0185	0.09604	0.0037	0	15.10	4901.6	4546.1	4546.1		919.25	3626.87	
3	-0.01476	-0.0963	0.0148	0.09604	0.0074	0	20.86	4895.8	4551.5	4551.5		919.25	3632.22	
3.5	-0.01107	-0.0963	0.0111	0.09604	0.0111	0	26.62	4890.1	4556.8	4556.8		919.25	3637.57	
4	-0.00738	-0.0963	0.0074	0.09604	0.0148	0	32.36	4884.3	4562.2	4562.2		919.25	3642.93	
4.5	-0.00369	-0.0963	0.0037	0.09604	0.0184	0	38.10	4878.6	4567.6	4567.6		919.25	3648.30	
5	0	-0.0963	0	0.09604	0.0221	0	43.84	4872.8	4572.9	4572.9		919.25	3653.68	
5.5	0	-0.0931	0	0.09286	0.0221	0.00318	49.10	4867.6	4577.9	4577.9		919.25	3658.62	
6	0	-0.0899	0	0.08967	0.0221	0.00637	54.35	4862.3	4582.8	4582.8		919.25	3663.57	
6.5	0	-0.0867	0	0.08649	0.0221	0.00955	59.60	4857.1	4587.8	4587.8		919.25	3668.52	
7	0	-0.0835	0	0.08330	0.0221	0.01274	64.85	4851.8	4592.7	4592.7		919.25	3673.48	
7.5	0	-0.0803	0	0.08011	0.0221	0.01593	70.09	4846.6	4597.7	4597.7		919.25	3678.45	
8	0	-0.0771	0	0.07692	0.0221	0.01912	75.33	4841.3	4602.7	4602.7		919.25	3683.42	
8.5	0	-0.0739	0	0.07372	0.0221	0.02232	80.56	4836.1	4607.7	4607.7		919.25	3688.40	
9	0	-0.0706	0	0.07053	0.0221	0.02551	85.79	4830.9	4612.6	4612.6		919.25	3693.39	
9.5	0	-0.0674	0	0.06733	0.0221	0.02871	91.02	4825.6	4617.6	4617.6		919.25	3698.39	
10	0	-0.0642	0	0.06414	0.0221	0.03190	96.24	4820.4	4622.6	4622.6		919.25	3703.39	
10.5	0	-0.0610	0	0.06094	0.0221	0.03510	101.46	4815.2	4627.7	4627.7		919.25	3708.40	
11	0	-0.0578	0	0.05774	0.0221	0.03830	106.67	4810.0	4632.7	4632.7		919.25	3713.41	
11.5	0	-0.0546	0	0.05454	0.0221	0.04150	111.88	4804.8	4637.7	4637.7		919.25	3718.43	
12	0	-0.0514	0	0.05133	0.0221	0.04471	117.08	4799.6	4642.7	4642.7		919.25	3723.46	
12.5	0	-0.0482	0	0.04813	0.0221	0.04791	122.28	4794.4	4647.7	4647.7		919.25	3728.49	
13	0	-0.0450	0	0.04493	0.0221	0.05111	127.47	4789.2	4652.8	4652.8		919.25	3733.53	
13.5	0	-0.0417	0	0.04172	0.0221	0.05432	132.66	4784.0	4657.8	4657.8		919.25	3738.58	
14	0	-0.0385	0	0.03852	0.0221	0.05752	137.84	4778.8	4662.9	4662.9		919.25	3743.63	
14.5	0	-0.0353	0	0.03531	0.0221	0.06073	143.02	4773.6	4667.9	4667.9		919.25	3748.69	
15	0	-0.0321	0	0.03210	0.0221	0.06394	148.19	4768.5	4673.0	4673.0		919.25	3753.76	
15.5	0	-0.0289	0	0.02889	0.0221	0.06715	153.36	4763.3	4678.1	4678.1		919.25	3758.83	
16	0	-0.0257	0	0.02568	0.0221	0.07036	158.53	4758.1	4683.2	4683.2		919.25	3763.90	
16.5	0	-0.0225	0	0.02247	0.0221	0.07357	163.69	4753.0	4688.2	4688.2		919.25	3768.99	
17	0	-0.0193	0	0.01926	0.0221	0.07678	168.84	4747.8	4693.3	4693.3		919.25	3774.08	
17.5	0	-0.0161	0	0.01605	0.0221	0.07999	173.99	4742.7	4698.4	4698.4		919.25	3779.17	
18	0	-0.0128	0	0.01284	0.0221	0.08320	179.13	4737.5	4703.5	4703.5		919.25	3784.27	
18.5	0	-0.0096	0	0.00963	0.0221	0.08641	184.27	4732.4	4708.6	4708.6		919.25	3789.38	
19	0	-0.0064	0	0.00642	0.0221	0.08962	189.40	4727.3	4713.7	4713.7		919.25	3794.49	
19.5	0	-0.0032	0	0.00321	0.0221	0.09283	194.53	4722.1	4718.9	4718.9		919.25	3799.61	
20	0	0.0000	0	0.00000	0.0221	0.09604	199.65	4717.0	4724.0	4717.0	4.061	919.25	3797.76	22.76

Computing actual W _{int} for Lc			
Lc	19.66 m	Corrected such that the wedge loss area is sufficiently close to the exact value W _{int} (error < 2%).	
Pc	4720.50 kN		
P' 0	4532.16 kN		
P' 2	4540.78 kN		
P' 5	4572.93 kN		
P 5 - P' 5	299.89 kN		
P 2 - P' 2	366.55 kN		
P 0 - P' 0	384.51 kN		
W _{int} approx	3948.95 kNm	Area of 2 trapezoids and a triangle	
Error	0.20 kNm	< 0.02*W _{int}	79.0 OK

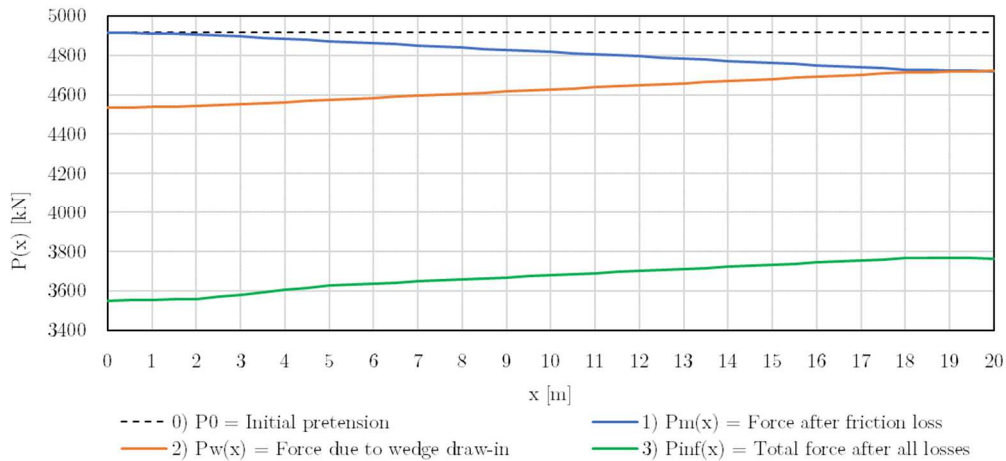
Cable force diagram, 2A



Cable 2A															
x [m]	y^{*} [-]	z^{*} [-]	$atan(y^{*})$ [rad]	$atan(z^{*})$ [rad]	θ_y [rad]	θ_z [rad]	ΔP_{μ} [kN]	P_{μ} [kN]	ΔP_w [kN]	P_w [kN]	$P_{0(x)}$ [kN]	Loss ratio [%]	ΔP_{∞} [kN]	$P_{\infty(x)}$ [kN]	Loss ratio t_{∞} [%]
0	-0.06357	-0.0932	0.0635	0.09294	0	0	0.00	4916.7	0.0	4470.1	4470.1		919.25	3550.86	
0.5	-0.06357	-0.0932	0.0635	0.09294	0.0000	0	2.33	4914.3	-2.1	4472.2	4472.2		919.25	3552.98	
1	-0.06357	-0.0932	0.0635	0.09294	0.0000	0	4.67	4912.0	-4.2	4474.4	4474.4		919.25	3555.10	
1.5	-0.06357	-0.0932	0.0635	0.09294	0.0000	0	7.00	4909.7	-6.4	4476.5	4476.5		919.25	3557.23	
2	-0.06357	-0.0932	0.0635	0.09294	0.0000	0	9.33	4907.3	-8.5	4478.6	4478.6		919.25	3559.36	
2.5	-0.05298	-0.0932	0.0529	0.09294	0.0106	0	21.49	4895.2	-19.6	4489.7	4489.7		919.25	3570.48	
3	-0.04238	-0.0932	0.0424	0.09294	0.0211	0	33.64	4883.0	-30.8	4500.9	4500.9		919.25	3581.65	
3.5	-0.03179	-0.0932	0.0318	0.09294	0.0317	0	45.76	4870.9	-42.0	4512.1	4512.1		919.25	3592.85	
4	-0.02119	-0.0932	0.0212	0.09294	0.0423	0	57.85	4858.8	-53.2	4523.3	4523.3		919.25	3604.08	
4.5	-0.01060	-0.0932	0.0106	0.09294	0.0529	0	69.93	4846.7	-64.5	4534.6	4534.6		919.25	3615.35	
5	0	-0.0932	0.0000	0.09294	0.0635	0	81.97	4834.7	-75.8	4545.9	4545.9		919.25	3626.64	
5.5	0	-0.0896	0.0000	0.08938	0.0635	0.00356	87.53	4829.1	-81.0	4551.1	4551.1		919.25	3631.88	
6	0	-0.0860	0.0000	0.08582	0.0635	0.00711	93.08	4823.6	-86.3	4556.4	4556.4		919.25	3637.12	
6.5	0	-0.0825	0.0000	0.08226	0.0635	0.01067	98.63	4818.0	-91.5	4561.6	4561.6		919.25	3642.37	
7	0	-0.0789	0.0000	0.07870	0.0635	0.01423	104.18	4812.5	-96.8	4566.9	4566.9		919.25	3647.62	
7.5	0	-0.0753	0.0000	0.07514	0.0635	0.01780	109.72	4806.9	-102.0	4572.1	4572.1		919.25	3652.89	
8	0	-0.0717	0.0000	0.07157	0.0635	0.02136	115.26	4801.4	-107.3	4577.4	4577.4		919.25	3658.16	
8.5	0	-0.0681	0.0000	0.06801	0.0635	0.02493	120.79	4795.9	-112.6	4582.7	4582.7		919.25	3663.44	
9	0	-0.0645	0.0000	0.06444	0.0635	0.02850	126.32	4790.4	-117.9	4588.0	4588.0		919.25	3668.73	
9.5	0	-0.0609	0.0000	0.06087	0.0635	0.03207	131.84	4784.8	-123.2	4593.3	4593.3		919.25	3674.02	
10	0	-0.0574	0.0000	0.05729	0.0635	0.03564	137.36	4779.3	-128.5	4598.6	4598.6		919.25	3679.33	
10.5	0	-0.0538	0.0000	0.05372	0.0635	0.03922	142.87	4773.8	-133.8	4603.9	4603.9		919.25	3684.64	
11	0	-0.0502	0.0000	0.05015	0.0635	0.04279	148.38	4768.3	-139.1	4609.2	4609.2		919.25	3689.95	
11.5	0	-0.0466	0.0000	0.04657	0.0635	0.04637	153.88	4762.8	-144.4	4614.5	4614.5		919.25	3695.28	
12	0	-0.0430	0.0000	0.04299	0.0635	0.04995	159.37	4757.3	-149.8	4619.9	4619.9		919.25	3700.61	
12.5	0	-0.0394	0.0000	0.03941	0.0635	0.05352	164.87	4751.8	-155.1	4625.2	4625.2		919.25	3705.95	
13	0	-0.0358	0.0000	0.03583	0.0635	0.05710	170.35	4746.3	-160.4	4630.5	4630.5		919.25	3711.29	
13.5	0	-0.0323	0.0000	0.03225	0.0635	0.06068	175.83	4740.8	-165.8	4635.9	4635.9		919.25	3716.65	
14	0	-0.0287	0.0000	0.02867	0.0635	0.06427	181.31	4735.4	-171.2	4641.3	4641.3		919.25	3722.01	
14.5	0	-0.0251	0.0000	0.02509	0.0635	0.06785	186.78	4729.9	-176.5	4646.6	4646.6		919.25	3727.37	
15	0	-0.0215	0.0000	0.02151	0.0635	0.07143	192.24	4724.4	-181.9	4652.0	4652.0		919.25	3732.75	
15.5	0	-0.0179	0.0000	0.01792	0.0635	0.07501	197.70	4719.0	-187.3	4657.4	4657.4		919.25	3738.13	
16	0	-0.0143	0.0000	0.01434	0.0635	0.07860	203.15	4713.5	-192.7	4662.8	4662.8		919.25	3743.51	
16.5	0	-0.0108	0.0000	0.01075	0.0635	0.08218	208.59	4708.1	-198.1	4668.2	4668.2		919.25	3748.91	
17	0	-0.0072	0.0000	0.00717	0.0635	0.08577	214.03	4702.6	-203.5	4673.6	4673.6		919.25	3754.31	
17.5	0	-0.0036	0.0000	0.00358	0.0635	0.08935	219.47	4697.2	-208.9	4679.0	4679.0		919.25	3759.71	
18	0	0.0000	0.0000	0	0.0635	0.09294	224.89	4691.8	-214.3	4684.4	4684.4		919.25	3765.12	
18.5	0	0.0000	0.0000	0	0.0635	0.09294	227.12	4689.5	-216.5	4686.6	4686.6		919.25	3767.35	
19	0	0.0000	0.0000	0	0.0635	0.09294	229.35	4687.3	-218.7	4688.8	4688.8		919.25	3768.06	
19.5	0	0.0000	0.0000	0	0.0635	0.09294	231.58	4685.1	-220.9	4691.1	4685.1		919.25	3765.84	
20	0	0.0000	0.0000	0	0.0635	0.09294	233.80	4682.9	-223.2	4693.3	4682.9	4.755	919.25	3763.61	23.45

Computing actual W_{int} for L_c			
L_c	18.83 m	Corrected such that the wedge loss area is sufficiently close to the exact value W_{int} (error < 2%).	
P_c	4688.07 kN		
P'_0	4470.11 kN		
P'_2	4478.61 kN		
P'_5	4545.90 kN		
$P_5 - P'_5$	288.80 kN		
$P_2 - P'_2$	428.72 kN	Area of 2 trapezoids and a triangle	
$P_0 - P'_0$	446.56 kN		
$W_{int,ap}$	3948.60 kNm		
Error	-0.15 kNm	< 0.02* W_{int}	79.0 OK

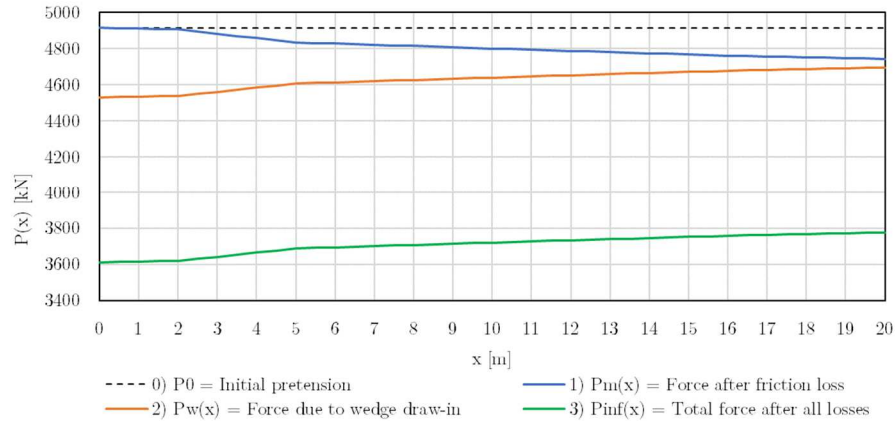
Cable force diagram, 2B



Cable 2B															
x	y**	z**	atan(y**)	atan(z**)	θ_y	θ_z	ΔP_{μ}	P_{μ}	ΔP_w	P_w	$P_{\theta(x)}$	Loss ratio	ΔP_{∞}	$P_{\infty(x)}$	Loss ratio, t_{∞}
[m]	[-]	[-]	[rad]	[rad]	[rad]	[rad]	[kN]	[kN]	[kN]	[kN]	[kN]	[%]	[kN]	[kN]	[%]
0	-0.02214	-0.0932	0.0221	0.09294	0	0	0.00	4916.67	0.0	4533.1	4533.1		919.25	3613.84	
0.5	-0.02214	-0.0932	0.0221	0.09294	0.0000	0	2.33	4914.33	-2.2	4535.3	4535.3		919.25	3616.00	
1	-0.02214	-0.0932	0.0221	0.09294	0.0000	0	4.67	4912.00	-4.3	4537.4	4537.4		919.25	3618.15	
1.5	-0.02214	-0.0932	0.0221	0.09294	0.0000	0	7.00	4909.67	-6.5	4539.6	4539.6		919.25	3620.31	
2	-0.02214	-0.0932	0.0221	0.09294	0.0000	0	9.33	4907.33	-8.6	4541.7	4541.7		919.25	3622.46	
2.5	-0.01845	-0.0932	0.0185	0.09294	0.0037	0	15.10	4901.57	-14.0	4547.1	4547.1		919.25	3627.81	
3	-0.01476	-0.0932	0.0148	0.09294	0.0074	0	20.86	4895.81	-19.3	4552.4	4552.4		919.25	3633.16	
3.5	-0.01107	-0.0932	0.0111	0.09294	0.0111	0	26.62	4890.05	-24.7	4557.8	4557.8		919.25	3638.51	
4	-0.00738	-0.0932	0.0074	0.09294	0.0148	0	32.36	4884.30	-30.0	4563.1	4563.1		919.25	3643.88	
4.5	-0.00369	-0.0932	0.0037	0.09294	0.0184	0	38.10	4878.56	-35.4	4568.5	4568.5		919.25	3649.25	
5	0	-0.0932	0.0000	0.09294	0.0221	0	43.84	4872.83	-40.8	4573.9	4573.9		919.25	3654.63	
5.5	0	-0.0896	0.0000	0.08938	0.0221	0.003555	49.44	4867.22	-46.0	4579.1	4579.1		919.25	3659.89	
6	0	-0.0860	0.0000	0.08582	0.0221	0.007112	55.04	4861.63	-51.3	4584.4	4584.4		919.25	3665.16	
6.5	0	-0.0825	0.0000	0.08226	0.0221	0.010672	60.63	4856.03	-56.6	4589.7	4589.7		919.25	3670.44	
7	0	-0.0789	0.0000	0.07870	0.0221	0.014234	66.22	4850.44	-61.9	4595.0	4595.0		919.25	3675.73	
7.5	0	-0.0753	0.0000	0.07514	0.0221	0.017797	71.81	4844.86	-67.2	4600.3	4600.3		919.25	3681.03	
8	0	-0.0717	0.0000	0.07157	0.0221	0.021363	77.39	4839.28	-72.5	4605.6	4605.6		919.25	3686.34	
8.5	0	-0.0681	0.0000	0.06801	0.0221	0.024930	82.96	4833.70	-77.8	4610.9	4610.9		919.25	3691.65	
9	0	-0.0645	0.0000	0.06444	0.0221	0.028499	88.54	4828.13	-83.1	4616.2	4616.2		919.25	3696.97	
9.5	0	-0.0609	0.0000	0.06087	0.0221	0.032070	94.10	4822.57	-88.5	4621.6	4621.6		919.25	3702.30	
10	0	-0.0574	0.0000	0.05729	0.0221	0.035643	99.66	4817.00	-93.8	4626.9	4626.9		919.25	3707.63	
10.5	0	-0.0538	0.0000	0.05372	0.0221	0.039216	105.22	4811.45	-99.1	4632.2	4632.2		919.25	3712.97	
11	0	-0.0502	0.0000	0.05015	0.0221	0.042791	110.77	4805.90	-104.5	4637.6	4637.6		919.25	3718.32	
11.5	0	-0.0466	0.0000	0.04657	0.0221	0.046368	116.31	4800.35	-109.8	4642.9	4642.9		919.25	3723.68	
12	0	-0.0430	0.0000	0.04299	0.0221	0.049946	121.85	4794.81	-115.2	4648.3	4648.3		919.25	3729.04	
12.5	0	-0.0394	0.0000	0.03941	0.0221	0.053524	127.39	4789.28	-120.6	4653.7	4653.7		919.25	3734.42	
13	0	-0.0358	0.0000	0.03583	0.0221	0.057104	132.92	4783.75	-126.0	4659.1	4659.1		919.25	3739.80	
13.5	0	-0.0323	0.0000	0.03225	0.0221	0.060685	138.44	4778.23	-131.3	4664.4	4664.4		919.25	3745.18	
14	0	-0.0287	0.0000	0.02867	0.0221	0.064266	143.96	4772.71	-136.7	4669.8	4669.8		919.25	3750.57	
14.5	0	-0.0251	0.0000	0.02509	0.0221	0.067848	149.47	4767.20	-142.1	4675.2	4675.2		919.25	3755.97	
15	0	-0.0215	0.0000	0.02151	0.0221	0.071431	154.98	4761.69	-147.5	4680.6	4680.6		919.25	3761.38	
15.5	0	-0.0179	0.0000	0.01792	0.0221	0.075015	160.48	4756.19	-153.0	4686.0	4686.0		919.25	3766.79	
16	0	-0.0143	0.0000	0.01434	0.0221	0.078599	165.97	4750.69	-158.4	4691.5	4691.5		919.25	3772.21	
16.5	0	-0.0108	0.0000	0.01075	0.0221	0.082183	171.46	4745.20	-163.8	4696.9	4696.9		919.25	3777.64	
17	0	-0.0072	0.0000	0.00717	0.0221	0.085767	176.94	4739.72	-169.2	4702.3	4702.3		919.25	3783.07	
17.5	0	-0.0036	0.0000	0.00358	0.0221	0.089352	182.42	4734.25	-174.7	4707.8	4707.8		919.25	3788.51	
18	0	0.0000	0.0000	0	0.0221	0.092937	187.89	4728.78	-180.1	4713.2	4713.2		919.25	3793.96	
18.5	0	0.0000	0.0000	0	0.0221	0.092937	190.14	4726.53	-182.4	4715.5	4715.5		919.25	3796.20	
19	0	0.0000	0.0000	0	0.0221	0.092937	192.38	4724.29	-184.6	4717.7	4717.7		919.25	3798.44	
19.5	0	0.0000	0.0000	0	0.0221	0.092937	194.62	4722.04	-186.8	4719.9	4719.9		919.25	3800.68	
20	0	0.0000	0.0000	0	0.0221	0.092937	196.87	4719.80	-189.1	4722.2	4719.8	4.004	919.25	3800.54	22.70

Computing actual W _{int} for Lc		
Lc	19.74 m	Corrected such that the wedge loss area is sufficiently close to the exact value W _{int} (error < 2%).
Pc	4720.99 kN	
P' 0	4533.10 kN	
P' 2	4541.72 kN	
P' 5	4573.88 kN	
P 5 - P' 0	298.95 kN	
P 2 - P' 0	365.62 kN	
P 0 - P' 0	383.57 kN	
W _{int ap}	3948.53 kNm	Area of 2 trapezoids and a triangle
Error	-0.22 kNm	< 0.02*W _{int} 79.0 OK

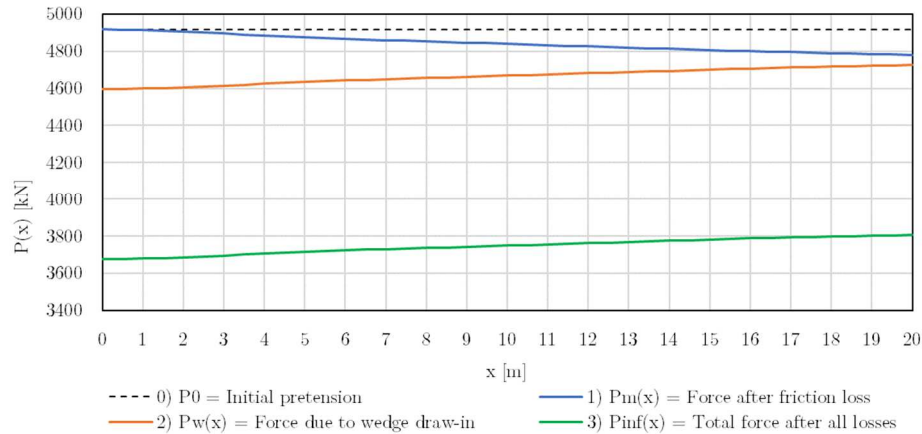
Cable force diagram, 3A



CABLE 3A															
x [m]	y^{**} [-]	z^{**} [-]	$atan(y^{**})$ [rad]	$atan(z^{**})$ [rad]	θ_y [rad]	θ_z [rad]	ΔP_μ [kN]	P_μ [kN]	ΔP_w [kN]	P_w [kN]	$P_0(x)$ [kN]	Loss ratio [%]	ΔP_∞ [kN]	$P_\infty(x)$ [kN]	Loss ratio, $t \rightarrow \infty$ [%]
0	-0.06357	-0.0254	0.0635	0.02538	0	0	0.00	4916.67	0.0	4530.6	4530.6		919.25	3611.36	
0.5	-0.06357	-0.0254	0.0635	0.02538	0.0000	0	2.33	4914.33	-2.2	4532.8	4532.8		919.25	3613.51	
1	-0.06357	-0.0254	0.0635	0.02538	0.0000	0	4.67	4912.00	-4.3	4534.9	4534.9		919.25	3615.67	
1.5	-0.06357	-0.0254	0.0635	0.02538	0.0000	0	7.00	4909.67	-6.5	4537.1	4537.1		919.25	3617.82	
2	-0.06357	-0.0254	0.0635	0.02538	0.0000	0	9.33	4907.33	-8.6	4539.2	4539.2		919.25	3619.98	
2.5	-0.05298	-0.0254	0.0529	0.02538	0.0106	0	21.49	4895.17	-19.9	4550.5	4550.5		919.25	3631.25	
3	-0.04238	-0.0254	0.0424	0.02538	0.0211	0	33.64	4883.03	-31.2	4561.8	4561.8		919.25	3642.57	
3.5	-0.03179	-0.0254	0.0318	0.02538	0.0317	0	45.76	4870.91	-42.6	4573.2	4573.2		919.25	3653.92	
4	-0.02119	-0.0254	0.0212	0.02538	0.0423	0	57.85	4858.81	-53.9	4584.6	4584.6		919.25	3665.31	
4.5	-0.0106	-0.0254	0.0106	0.02538	0.0529	0	69.93	4846.74	-65.4	4596.0	4596.0		919.25	3676.73	
5	0	-0.0254	0.0000	0.02538	0.0635	0	81.97	4834.70	-76.8	4607.4	4607.4		919.25	3688.18	
5.5	0	-0.0242	0.0000	0.02423	0.0635	0.00115	85.32	4831.34	-80.0	4610.6	4610.6		919.25	3691.37	
6	0	-0.0231	0.0000	0.02308	0.0635	0.00231	88.68	4827.99	-83.2	4613.8	4613.8		919.25	3694.58	
6.5	0	-0.0219	0.0000	0.02192	0.0635	0.00346	92.03	4824.64	-86.4	4617.0	4617.0		919.25	3697.78	
7	0	-0.0208	0.0000	0.02077	0.0635	0.00461	95.38	4821.29	-89.6	4620.2	4620.2		919.25	3700.99	
7.5	0	-0.0196	0.0000	0.01962	0.0635	0.00577	98.72	4817.95	-92.8	4623.4	4623.4		919.25	3704.19	
8	0	-0.0185	0.0000	0.01846	0.0635	0.00692	102.06	4814.60	-96.0	4626.7	4626.7		919.25	3707.41	
8.5	0	-0.0173	0.0000	0.01731	0.0635	0.00807	105.41	4811.26	-99.3	4629.9	4629.9		919.25	3710.62	
9	0	-0.0162	0.0000	0.01616	0.0635	0.00923	108.74	4807.92	-102.5	4633.1	4633.1		919.25	3713.83	
9.5	0	-0.0150	0.0000	0.01500	0.0635	0.01038	112.08	4804.59	-105.7	4636.3	4636.3		919.25	3717.05	
10	0	-0.0138	0.0000	0.01385	0.0635	0.01154	115.42	4801.25	-108.9	4639.5	4639.5		919.25	3720.27	
10.5	0	-0.0127	0.0000	0.01269	0.0635	0.01269	118.75	4797.92	-112.1	4642.7	4642.7		919.25	3723.49	
11	0	-0.0115	0.0000	0.01154	0.0635	0.01384	122.08	4794.59	-115.4	4646.0	4646.0		919.25	3726.72	
11.5	0	-0.0104	0.0000	0.01039	0.0635	0.01500	125.40	4791.26	-118.6	4649.2	4649.2		919.25	3729.94	
12	0	-0.0092	0.0000	0.00923	0.0635	0.01615	128.73	4787.94	-121.8	4652.4	4652.4		919.25	3733.17	
12.5	0	-0.0081	0.0000	0.00808	0.0635	0.01731	132.05	4784.61	-125.0	4655.7	4655.7		919.25	3736.40	
13	0	-0.0069	0.0000	0.00692	0.0635	0.01846	135.37	4781.29	-128.3	4658.9	4658.9		919.25	3739.64	
13.5	0	-0.0058	0.0000	0.00577	0.0635	0.01961	138.69	4777.98	-131.5	4662.1	4662.1		919.25	3742.87	
14	0	-0.0046	0.0000	0.00462	0.0635	0.02077	142.01	4774.66	-134.7	4665.4	4665.4		919.25	3746.11	
14.5	0	-0.0035	0.0000	0.00346	0.0635	0.02192	145.32	4771.35	-138.0	4668.6	4668.6		919.25	3749.35	
15	0	-0.0023	0.0000	0.00231	0.0635	0.02308	148.63	4768.03	-141.2	4671.8	4671.8		919.25	3752.59	
15.5	0	-0.0012	0.0000	0.00115	0.0635	0.02423	151.94	4764.72	-144.5	4675.1	4675.1		919.25	3755.84	
16	0	0.0000	0.0000	0	0.0635	0.02538	155.25	4761.42	-147.7	4678.3	4678.3		919.25	3759.08	
16.5	0	0.0000	0.0000	0	0.0635	0.02538	157.51	4759.16	-149.9	4680.6	4680.6		919.25	3761.31	
17	0	0.0000	0.0000	0	0.0635	0.02538	159.77	4756.90	-152.2	4682.8	4682.8		919.25	3763.53	
17.5	0	0.0000	0.0000	0	0.0635	0.02538	162.03	4754.64	-154.4	4685.0	4685.0		919.25	3765.76	
18	0	0.0000	0.0000	0	0.0635	0.02538	164.29	4752.38	-156.6	4687.2	4687.2		919.25	3767.98	
18.5	0	0.0000	0.0000	0	0.0635	0.02538	166.54	4750.12	-158.8	4689.5	4689.5		919.25	3770.21	
19	0	0.0000	0.0000	0	0.0635	0.02538	168.80	4747.87	-161.1	4691.7	4691.7		919.25	3772.44	
19.5	0	0.0000	0.0000	0	0.0635	0.02538	171.05	4745.61	-163.3	4693.9	4693.9		919.25	3774.67	
20	0	0.0000	0.0000	0	0.0635	0.02538	173.31	4743.36	-165.5	4696.2	4696.2	3.525	919.25	3776.90	23.18

Computing actual W_{int} for L_c		
L_c	25.25 m	Corrected such that the wedge loss area is sufficiently close to the exact value W_{int} (error < 2%).
P_c	4719.70 kN	
P'_0	4530.62 kN	
P'_2	4539.23 kN	
P'_5	4607.43 kN	
$P_5 - P'_5$	227.27 kN	
$P_2 - P'_2$	368.10 kN	
$P_0 - P'_0$	386.05 kN	
$W_{int app}$	3948.28 kNm	Area of 2 trapezoids and a triangle
Error	-0.47 kNm	< 0.02* W_{int} 79.0 OK

Cable force diagram, 3A



CABLE 3B																
x	y^{**}	z^{**}	$atan(y^{**})$	$atan(z^{**})$	θ_y	θ_z	ΔP_μ	P_μ	ΔP_w	P_w	$P_{0(x)}$	Loss ratio	ΔP_∞	$P_\infty(x)$	Loss ratio,	
[m]	[-]	[-]	[rad]	[rad]	[rad]	[rad]	[kN]	[kN]	[kN]	[kN]	[kN]	[%]	[kN]	[kN]	[%]	
0	-0.02214	-0.0254	0.0221	0.025384	0	0	0.00	4916.7	0.0	4594.3	4594.3		919.25	3675.07		
0.5	-0.02214	-0.0254	0.0221	0.025384	0.0000	0	2.33	4914.3	-2.2	4596.5	4596.5		919.25	3677.25		
1	-0.02214	-0.0254	0.0221	0.025384	0.0000	0	4.67	4912.0	-4.4	4598.7	4598.7		919.25	3679.43		
1.5	-0.02214	-0.0254	0.0221	0.025384	0.0000	0	7.00	4909.7	-6.6	4600.9	4600.9		919.25	3681.62		
2	-0.02214	-0.0254	0.0221	0.025384	0.0000	0	9.33	4907.3	-8.7	4603.1	4603.1		919.25	3683.81		
2.5	-0.01845	-0.0254	0.0185	0.025384	0.0037	0	15.10	4901.6	-14.2	4608.5	4608.5		919.25	3689.22		
3	-0.01476	-0.0254	0.0148	0.025384	0.0074	0	20.86	4895.8	-19.6	4613.9	4613.9		919.25	3694.64		
3.5	-0.01107	-0.0254	0.0111	0.025384	0.0111	0	26.62	4890.1	-25.0	4619.3	4619.3		919.25	3700.07		
4	-0.00738	-0.0254	0.0074	0.025384	0.0148	0	32.36	4884.3	-30.4	4624.8	4624.8		919.25	3705.51		
4.5	-0.00369	-0.0254	0.0037	0.025384	0.0184	0	38.10	4878.6	-35.9	4630.2	4630.2		919.25	3710.95		
5	0	-0.0254	0.0000	0.025384	0.0221	0	43.84	4872.8	-41.3	4635.7	4635.7		919.25	3716.40		
5.5	0	-0.0242	0.0000	0.024230	0.0221	0.001153	47.22	4869.4	-44.6	4638.9	4638.9		919.25	3719.62		
6	0	-0.0231	0.0000	0.023077	0.0221	0.002307	50.60	4866.1	-47.8	4642.1	4642.1		919.25	3722.84		
6.5	0	-0.0219	0.0000	0.021924	0.0221	0.003460	53.98	4862.7	-51.0	4645.3	4645.3		919.25	3726.07		
7	0	-0.0208	0.0000	0.020770	0.0221	0.004614	57.35	4859.3	-54.2	4648.5	4648.5		919.25	3729.29		
7.5	0	-0.0196	0.0000	0.019616	0.0221	0.005767	60.72	4855.9	-57.5	4651.8	4651.8		919.25	3732.52		
8	0	-0.0185	0.0000	0.018463	0.0221	0.006921	64.09	4852.6	-60.7	4655.0	4655.0		919.25	3735.75		
8.5	0	-0.0173	0.0000	0.017309	0.0221	0.008075	67.46	4849.2	-63.9	4658.2	4658.2		919.25	3738.98		
9	0	-0.0162	0.0000	0.016155	0.0221	0.009228	70.83	4845.8	-67.1	4661.5	4661.5		919.25	3742.22		
9.5	0	-0.0150	0.0000	0.015002	0.0221	0.010382	74.19	4842.5	-70.4	4664.7	4664.7		919.25	3745.45		
10	0	-0.0138	0.0000	0.013848	0.0221	0.011536	77.55	4839.1	-73.6	4667.9	4667.9		919.25	3748.69		
10.5	0	-0.0127	0.0000	0.012694	0.0221	0.012690	80.91	4835.8	-76.9	4671.2	4671.2		919.25	3751.94		
11	0	-0.0115	0.0000	0.011540	0.0221	0.013844	84.26	4832.4	-80.1	4674.4	4674.4		919.25	3755.18		
11.5	0	-0.0104	0.0000	0.010386	0.0221	0.014998	87.62	4829.1	-83.4	4677.7	4677.7		919.25	3758.43		
12	0	-0.0092	0.0000	0.009232	0.0221	0.016152	90.97	4825.7	-86.6	4680.9	4680.9		919.25	3761.67		
12.5	0	-0.0081	0.0000	0.008078	0.0221	0.017306	94.32	4822.3	-89.9	4684.2	4684.2		919.25	3764.93		
13	0	-0.0069	0.0000	0.006924	0.0221	0.018460	97.66	4819.0	-93.1	4687.4	4687.4		919.25	3768.18		
13.5	0	-0.0058	0.0000	0.005770	0.0221	0.019614	101.01	4815.7	-96.4	4690.7	4690.7		919.25	3771.43		
14	0	-0.0046	0.0000	0.004616	0.0221	0.020768	104.35	4812.3	-99.6	4693.9	4693.9		919.25	3774.69		
14.5	0	-0.0035	0.0000	0.003462	0.0221	0.021922	107.69	4809.0	-102.9	4697.2	4697.2		919.25	3777.95		
15	0	-0.0023	0.0000	0.002308	0.0221	0.023076	111.03	4805.6	-106.1	4700.5	4700.5		919.25	3781.21		
15.5	0	-0.0012	0.0000	0.001154	0.0221	0.024230	114.36	4802.3	-109.4	4703.7	4703.7		919.25	3784.48		
16	0	0.0000	0.0000	0	0.0221	0.025384	117.70	4799.0	-112.7	4707.0	4707.0		919.25	3787.74		
16.5	0	0.0000	0.0000	0	0.0221	0.025384	119.98	4796.7	-114.9	4709.2	4709.2		919.25	3789.98		
17	0	0.0000	0.0000	0	0.0221	0.025384	122.25	4794.4	-117.2	4711.5	4711.5		919.25	3792.22		
17.5	0	0.0000	0.0000	0	0.0221	0.025384	124.53	4792.1	-119.4	4713.7	4713.7		919.25	3794.46		
18	0	0.0000	0.0000	0	0.0221	0.025384	126.81	4789.9	-121.6	4716.0	4716.0		919.25	3796.70		
18.5	0	0.0000	0.0000	0	0.0221	0.025384	129.08	4787.6	-123.9	4718.2	4718.2		919.25	3798.94		
19	0	0.0000	0.0000	0	0.0221	0.025384	131.35	4785.3	-126.1	4720.4	4720.4		919.25	3801.18		
19.5	0	0.0000	0.0000	0	0.0221	0.025384	133.63	4783.0	-128.4	4722.7	4722.7		919.25	3803.42		
20	0	0.0000	0.0000	0	0.0221	0.025384	135.90	4780.8	-130.6	4724.9	4724.9	2.764	919.25	3805.67	22.60	

Computing actual W_{int} for L_c			
L_c	26.17	m	Corrected such that the wedge loss area is sufficiently close to the exact value W_{int} (error < 2%).
P_c	4752.76	kN	
P'_0	4594.32	kN	
P'_2	4603.06	kN	
P'_5	4635.66	kN	
$P_5 - P'_5$	237.17	kN	
$P_2 - P'_2$	304.27	kN	
$P_0 - P'_0$	322.34	kN	
$W_{int, app}$	3948.65	kNm	Area of 2 trapezoids and a triangle < 0.02* W_{int} 79.0 OK
Error	-0.10	kNm	

G. ULS Shear and Torsional Verifications

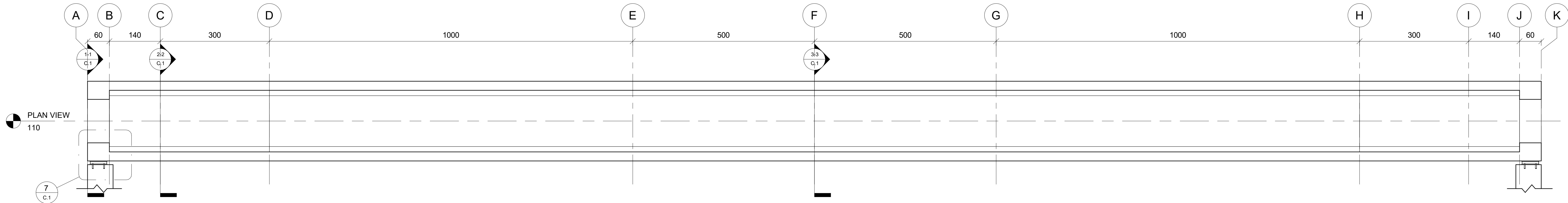
1) Contribution from cables:

x [m]	Cable layer 1												Cable layer 2											
	Layer 1a						Layer 1b						Layer 2a						Layer 2b					
	$\alpha_{y,1a}$ [rad]	$\alpha_{z,1a}$ [rad]	P_{1a} [kN]	$\Delta V_{y,1a}$ [kN]	$\Delta V_{z,1a}$ [kN]		$\alpha_{y,1b}$ [rad]	$\alpha_{z,1b}$ [rad]	P_{1b} [kN]	$\Delta V_{y,1b}$ [kN]	$\Delta V_{z,1b}$ [kN]		$\alpha_{y,2a}$ [rad]	$\alpha_{z,2a}$ [rad]	P_{2a} [kN]	$\Delta V_{y,2a}$ [kN]	$\Delta V_{z,2a}$ [kN]	$\alpha_{y,2b}$ [rad]	$\alpha_{z,2b}$ [rad]	P_{2b} [kN]	$\Delta V_{y,2b}$ [kN]	$\Delta V_{z,2b}$ [kN]		
0	0.06	0.10	7107.37	450.92	681.55		0.02	0.10	7225.81	159.96	692.90		0.06	0.09	3550.86	225.28	329.53	0.02	0.09	3613.84	80.00	335.38		
0.5	0.06	0.10	7111.62	451.18	681.95		0.02	0.10	7230.12	160.06	693.32		0.06	0.09	3552.98	225.41	329.73	0.02	0.09	3616.00	80.05	335.58		
1	0.06	0.10	7115.87	451.45	682.36		0.02	0.10	7234.42	160.15	693.73		0.06	0.09	3555.10	225.55	329.93	0.02	0.09	3618.15	80.10	335.78		
1.5	0.06	0.10	7120.12	451.72	682.77		0.02	0.10	7238.74	160.25	694.14		0.06	0.09	3557.23	225.68	330.12	0.02	0.09	3620.31	80.14	335.98		
2	0.06	0.10	7124.38	451.99	683.18		0.02	0.10	7243.05	160.34	694.56		0.06	0.09	3559.36	225.82	330.32	0.02	0.09	3622.46	80.19	336.18		
2.5	0.05	0.10	7146.65	378.07	685.31		0.02	0.10	7253.73	133.83	695.58		0.05	0.09	3570.48	188.89	331.35	0.02	0.09	3627.81	66.93	336.67		
3	0.04	0.10	7168.99	303.56	687.45		0.01	0.10	7264.43	107.23	696.61		0.04	0.09	3581.65	151.66	332.39	0.01	0.09	3633.16	53.63	337.17		
3.5	0.03	0.10	7191.40	228.47	689.60		0.01	0.10	7275.14	80.54	697.63		0.03	0.09	3592.85	114.14	333.43	0.01	0.09	3638.51	40.28	337.67		
4	0.02	0.10	7213.89	152.83	691.76		0.01	0.10	7285.87	53.78	698.66		0.02	0.09	3604.08	76.36	334.47	0.01	0.09	3643.88	26.89	338.16		
4.5	0.01	0.10	7236.43	76.67	693.92		0.00	0.10	7296.61	26.93	699.69		0.01	0.09	3615.35	38.30	335.52	0.00	0.09	3649.25	13.47	338.66		
5	0.00	0.10	7259.04	0.00	696.09		0.00	0.10	7307.36	0.00	700.72		0.00	0.09	3626.64	0.00	336.56	0.00	0.09	3654.63	0.00	339.16		
5.5	0.00	0.09	7268.87	0.00	674.00		0.00	0.09	7317.24	0.00	678.49		0.00	0.09	3631.88	0.00	324.19	0.00	0.09	3659.89	0.00	326.69		
6	0.00	0.09	7278.71	0.00	651.83		0.00	0.09	7327.13	0.00	656.17		0.00	0.09	3637.12	0.00	311.77	0.00	0.09	3665.16	0.00	314.17		
6.5	0.00	0.09	7288.57	0.00	629.58		0.00	0.09	7337.04	0.00	633.77		0.00	0.08	3642.37	0.00	299.30	0.00	0.08	3670.44	0.00	301.61		
7	0.00	0.08	7298.43	0.00	607.25		0.00	0.08	7346.96	0.00	611.28		0.00	0.08	3647.62	0.00	286.78	0.00	0.08	3675.73	0.00	288.99		
7.5	0.00	0.08	7308.32	0.00	584.83		0.00	0.08	7356.90	0.00	588.72		0.00	0.08	3652.89	0.00	274.22	0.00	0.08	3681.03	0.00	276.33		
8	0.00	0.08	7318.21	0.00	562.34		0.00	0.08	7366.85	0.00	566.08		0.00	0.07	3658.16	0.00	261.61	0.00	0.07	3686.34	0.00	263.62		
8.5	0.00	0.07	7328.12	0.00	539.77		0.00	0.07	7376.81	0.00	543.36		0.00	0.07	3663.44	0.00	248.95	0.00	0.07	3691.65	0.00	250.86		
9	0.00	0.07	7338.05	0.00	517.12		0.00	0.07	7386.79	0.00	520.56		0.00	0.06	3668.73	0.00	236.24	0.00	0.06	3696.97	0.00	238.06		
9.5	0.00	0.07	7347.98	0.00	494.39		0.00	0.07	7396.78	0.00	497.68		0.00	0.06	3674.02	0.00	223.49	0.00	0.06	3702.30	0.00	225.21		
10	0.00	0.06	7357.93	0.00	471.59		0.00	0.06	7406.78	0.00	474.72		0.00	0.06	3679.33	0.00	210.69	0.00	0.06	3707.63	0.00	221.31		
10.5	0.00	0.06	7367.90	0.00	448.71		0.00	0.06	7416.79	0.00	451.68		0.00	0.05	3684.64	0.00	197.85	0.00	0.05	3712.97	0.00	199.37		
11	0.00	0.06	7377.87	0.00	425.75		0.00	0.06	7426.82	0.00	428.57		0.00	0.05	3689.95	0.00	184.96	0.00	0.05	3718.32	0.00	186.38		
11.5	0.00	0.05	7387.86	0.00	402.71		0.00	0.05	7436.86	0.00	405.38		0.00	0.05	3695.28	0.00	172.02	0.00	0.05	3723.68	0.00	173.35		
12	0.00	0.05	7397.86	0.00	379.60		0.00	0.05	7446.92	0.00	382.12		0.00	0.04	3700.61	0.00	159.05	0.00	0.04	3729.04	0.00	160.27		
12.5	0.00	0.05	7407.88	0.00	356.41		0.00	0.05	7456.99	0.00	358.78		0.00	0.04	3705.95	0.00	146.02	0.00	0.04	3734.42	0.00	147.15		
13	0.00	0.04	7417.90	0.00	333.15		0.00	0.04	7467.07	0.00	335.96		0.00	0.04	3711.29	0.00	132.96	0.00	0.04	3739.80	0.00	133.98		
13.5	0.00	0.04	7427.94	0.00	309.82		0.00	0.04	7477.16	0.00	311.87		0.00	0.03	3716.65	0.00	119.85	0.00	0.03	3745.19	0.00	120.77		
14	0.00	0.04	7437.99	0.00	286.41		0.00	0.04	7487.26	0.00	288.31		0.00	0.03	3722.01	0.00	106.70	0.00	0.03	3750.57	0.00	107.52		
14.5	0.00	0.04	7448.06	0.00	262.93		0.00	0.04	7497.38	0.00	264.67		0.00	0.03	3727.37	0.00	93.50	0.00	0.03	3755.97	0.00	94.22		
15	0.00	0.03	7458.14	0.00	239.37		0.00	0.03	7507.51	0.00	240.96		0.00	0.02	3732.75	0.00	80.27	0.00	0.02	3761.38	0.00	80.88		
15.5	0.00	0.03	7468.22	0.00	215.75		0.00	0.03	7517.65	0.00	217.18		0.00	0.02	3738.13	0.00	66.99	0.00	0.02	3766.79	0.00	67.51		
16	0.00	0.03	7478.32	0.00	192.05		0.00	0.03	7527.81	0.00	193.32		0.00	0.01	3743.51	0.00	53.67	0.00	0.01	3772.21	0.00	54.09		
16.5	0.00	0.02	7488.44	0.00	168.29		0.00	0.02	7537.97	0.00	169.40		0.00	0.01	3748.91	0.00	40.32	0.00	0.01	3777.64	0.00	40.62		
17	0.00	0.02	7498.56	0.00	144.45		0.00	0.02	7548.15	0.00	145.41		0.00	0.01	3754.31	0.00	26.92	0.00	0.01	3783.07	0.00	27.12		
17.5	0.00	0.02	7508.70	0.00	120.55		0.00	0.02	7558.34	0.00	121.94		0.00	0.00	3759.71	0.00	13.48	0.00	0.00	3788.51	0.00	13.58		
18	0.00	0.01	7518.84	0.00	96.57		0.00	0.01	7568.54	0.00	97.21		0.00	0.00	3765.12	0.00	0.00	0.00	0.00	3793.96	0.00	0.00		
18.5	0.00	0.01	7529.00	0.00	72.53		0.00	0.01	7578.75	0.00	73.01		0.00	0.00	3767.35	0.00	0.00	0.00	0.00	3796.20	0.00	0.00		
19	0.00	0.01	7539.17	0.00	48.42		0.00	0.01	7588.98	0.00	48.74		0.00	0.00	3768.06	0.00	0.00	0.00	0.00	3798.44	0.00	0.00		
19.5	0.00	0.00	7531.87	0.00	24.19		0.00	0.00	7599.21	0.00	24.40		0.00	0.00	3765.84	0.00	0.00	0.00	0.00	3800.68	0.00	0.00		
20	0.00	0.00	7521.70	0.00	0.00		0.00	0.00	7595.53	0.00	0.00		0.00	0.00	3763.61	0.00	0.00	0.00	0.00	3800.54	0.00	0.00		

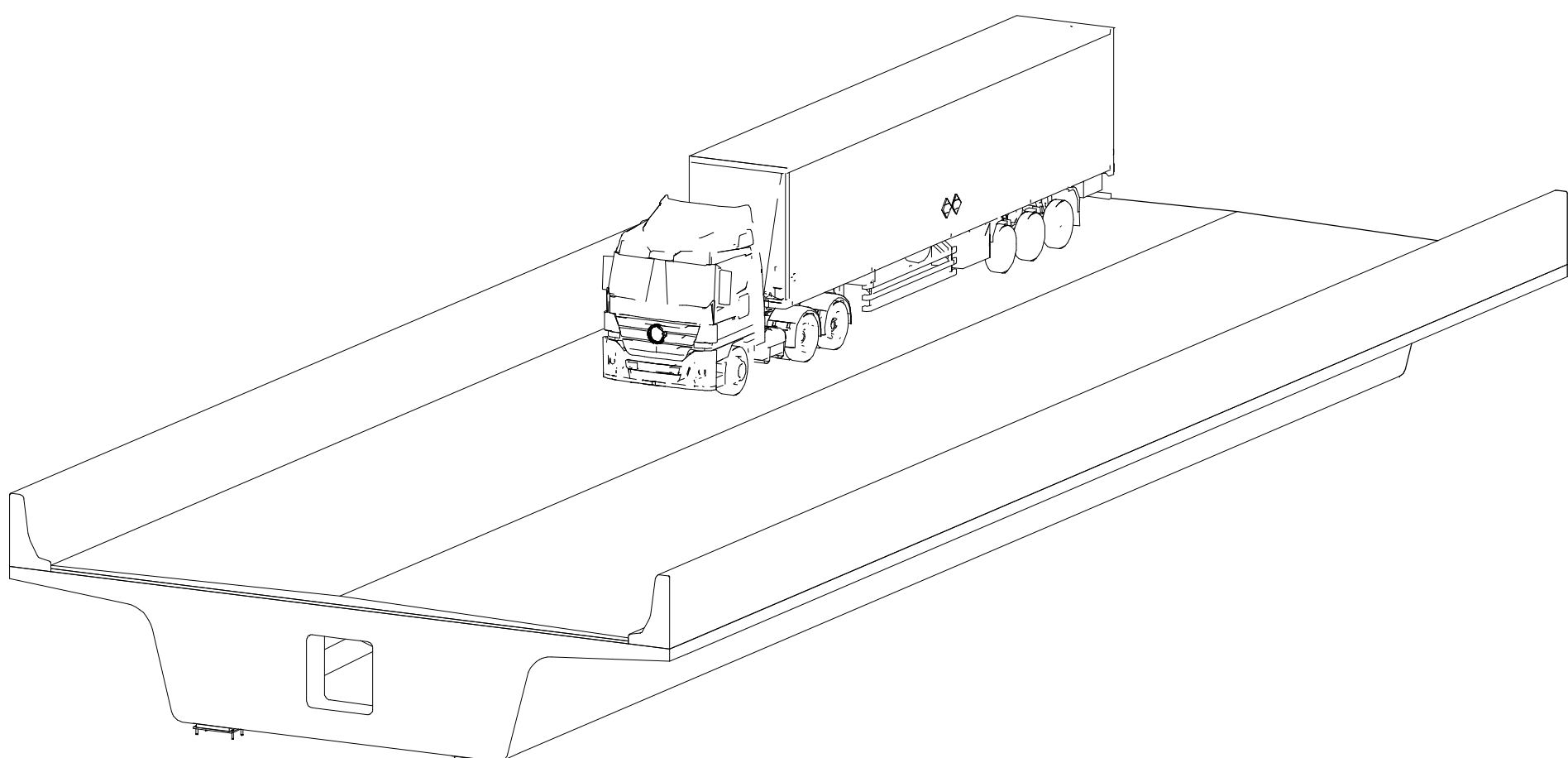
Cable layer 3										Sum	
Layer 3a					Layer 3b					$\Sigma \Delta V_y$	$\Sigma \Delta V_z$
$\alpha_{y,3a}$	$\alpha_{z,3a}$	P_{3a}	$\Delta V_{y,3a}$	$\Delta V_{z,3a}$	$\alpha_{y,3b}$	$\alpha_{z,3b}$	P_{3b}	$\Delta V_{y,3b}$	$\Delta V_{z,3b}$		
[rad]	[rad]	[kN]	[kN]	[kN]	[rad]	[rad]	[kN]	[kN]	[kN]		[kN]
0.0635	0.025383783	3611.36	229.12	91.66	0.0221	0.0254	3675.07	81.357	93.27710625		2124.29
0.0635	0.025383783	3613.51	229.25	91.715	0.0221	0.0254	3677.25	81.405	93.33250857		2225.62
0.0635	0.025383783	3615.67	229.39	91.769	0.0221	0.0254	3679.43	81.453	93.38793721		2226.95
0.0635	0.025383783	3617.82	229.53	91.824	0.0221	0.0254	3681.62	81.502	93.44339219		2228.28
0.0635	0.025383783	3619.98	229.66	91.879	0.0221	0.0254	3683.81	81.55	93.49887351		2229.61
0.0529	0.025383783	3631.25	192.1	92.165	0.0185	0.0254	3689.22	68.063	93.63633545		2234.72
0.0424	0.025383783	3642.57	154.24	92.452	0.0148	0.0254	3694.64	54.534	93.77397029		2239.85
0.0318	0.025383783	3653.92	116.08	92.74	0.0111	0.0254	3700.07	40.963	93.91177604		2244.98
0.0212	0.025383783	3665.31	77.652	93.029	0.0074	0.0254	3705.51	27.349	94.04975069		2250.14
0.0106	0.025383783	3676.73	38.954	93.319	0.0037	0.0254	3710.95	13.695	94.18789219		2255.30
0	0.025383783	3688.18	0	93.61	0	0.0254	3716.40	0	94.32619851		2260.47
0	0.024230437	3691.37	0	89.435	0	0.0242	3719.62	0	90.1920823		2182.93
0	0.023077027	3694.58	0	85.252	0	0.0231	3722.84	0	85.90448817		2105.10
0	0.021923555	3697.78	0	81.062	0	0.0219	3726.07	0	81.68204655		2027.00
0	0.020770024	3700.99	0	76.864	0	0.0208	3729.29	0	77.45189167		1948.63
0	0.019616439	3704.19	0	72.658	0	0.0196	3732.52	0	73.21403188		1869.98
0	0.018462801	3707.41	0	68.445	0	0.0185	3735.75	0	68.96847562		1791.06
0	0.017309114	3710.62	0	64.224	0	0.0173	3738.98	0	64.71523139		1711.88
0	0.016155381	3713.83	0	59.996	0	0.0162	3742.22	0	60.45430775		1632.43
0	0.015001605	3717.05	0	55.76	0	0.015	3745.45	0	56.18571335		1552.71
0	0.013847789	3720.27	0	51.516	0	0.0138	3748.69	0	51.90945689		1472.73
0	0.012693936	3723.49	0	47.265	0	0.0127	3751.94	0	47.62554714		1392.49
0	0.01154005	3726.72	0	43.006	0	0.0115	3755.18	0	43.33399294		1311.99
0	0.010386132	3729.94	0	38.739	0	0.0104	3758.43	0	39.03480321		1231.23
0	0.009232187	3733.17	0	34.465	0	0.0092	3761.67	0	34.72798692		1150.22
0	0.008078218	3736.40	0	30.183	0	0.0081	3764.93	0	30.41355312		1068.95
0	0.006924227	3739.64	0	25.894	0	0.0069	3768.18	0	26.09151092		987.43
0	0.005770217	3742.87	0	21.597	0	0.0058	3771.43	0	21.7618695		905.66
0	0.004616192	3746.11	0	17.293	0	0.0046	3774.69	0	17.4246381		823.65
0	0.003462155	3749.35	0	12.981	0	0.0035	3777.95	0	13.07982604		741.38
0	0.002308108	3752.59	0	8.6614	0	0.0023	3781.21	0	8.727442692		658.88
0	0.001154056	3755.84	0	4.3344	0	0.0012	3784.48	0	4.367497509		576.13
0	0	3759.08	0	0	0	0	3787.74	0	0		493.14
0	0	3761.31	0	0	0	0	3789.98	0	0		418.63
0	0	3763.53	0	0	0	0	3792.22	0	0		343.90
0	0	3765.76	0	0	0	0	3794.46	0	0		268.95
0	0	3767.98	0	0	0	0	3796.70	0	0		193.78
0	0	3770.21	0	0	0	0	3798.94	0	0		145.54
0	0	3772.44	0	0	0	0	3801.18	0	0		97.16
0	0	3774.67	0	0	0	0	3803.42	0	0		48.59
0	0	3776.90	0	0	0	0	3805.67	0	0		0

ELEVATION

1 : 50

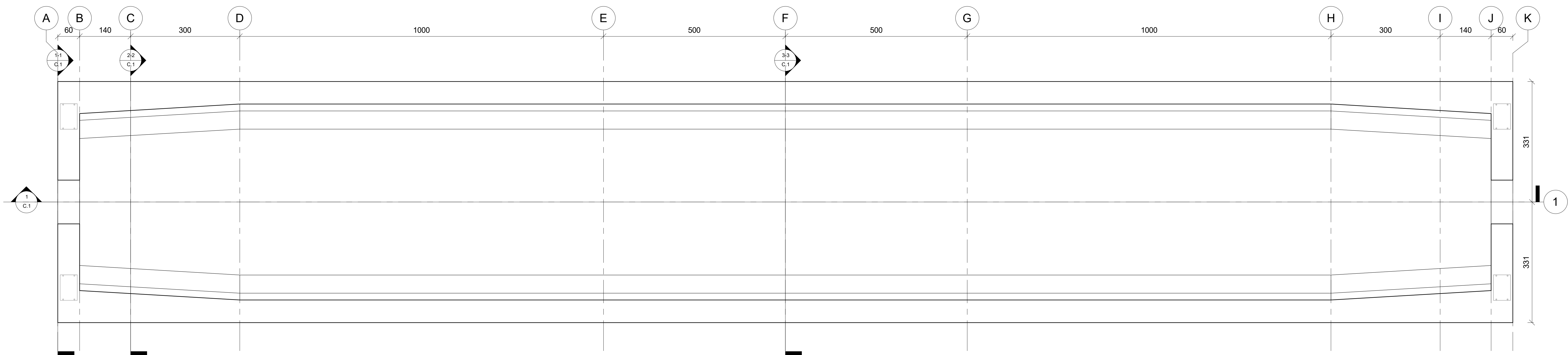


3D ISOMETRY



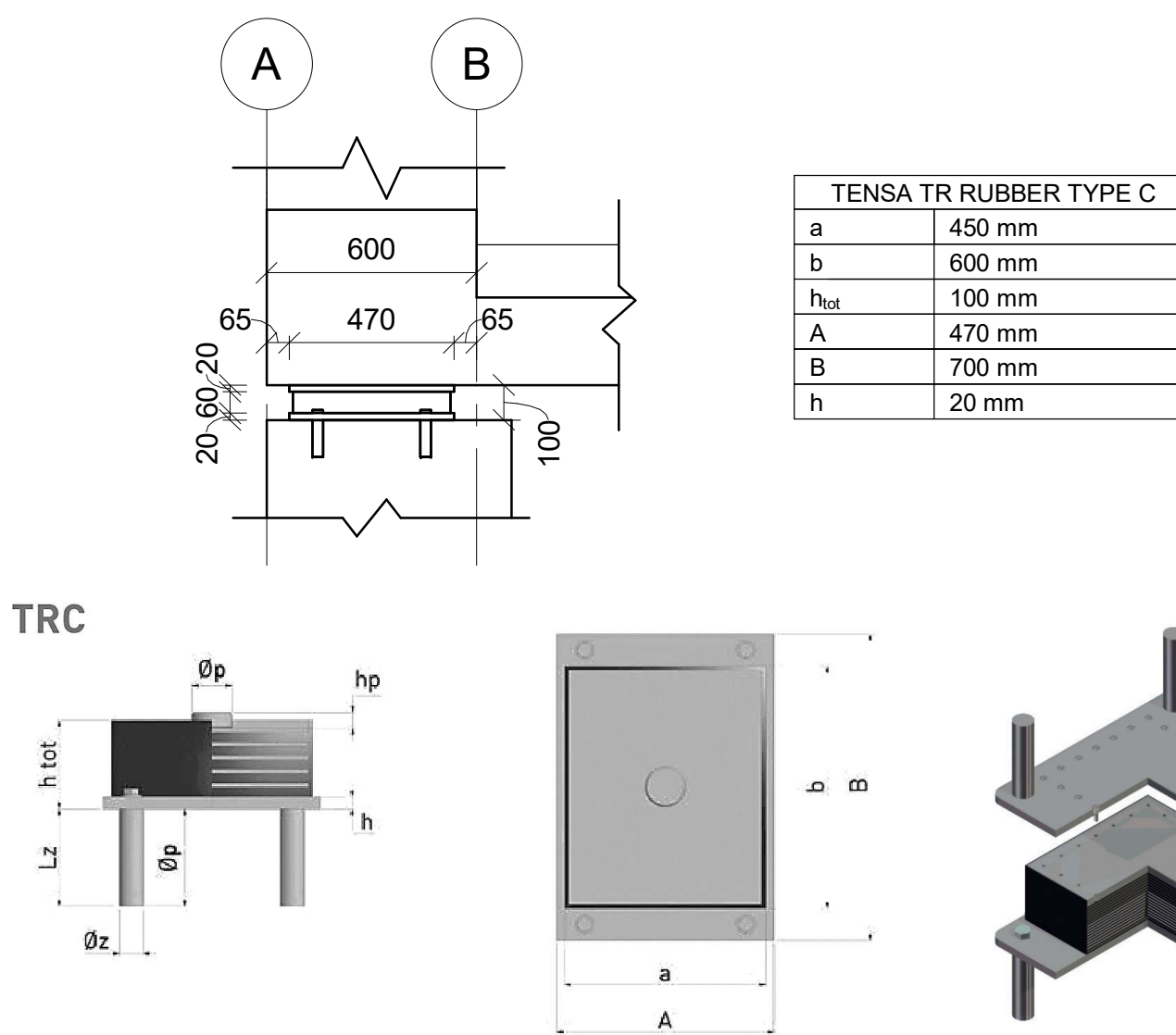
PLAN VIEW

1 : 50



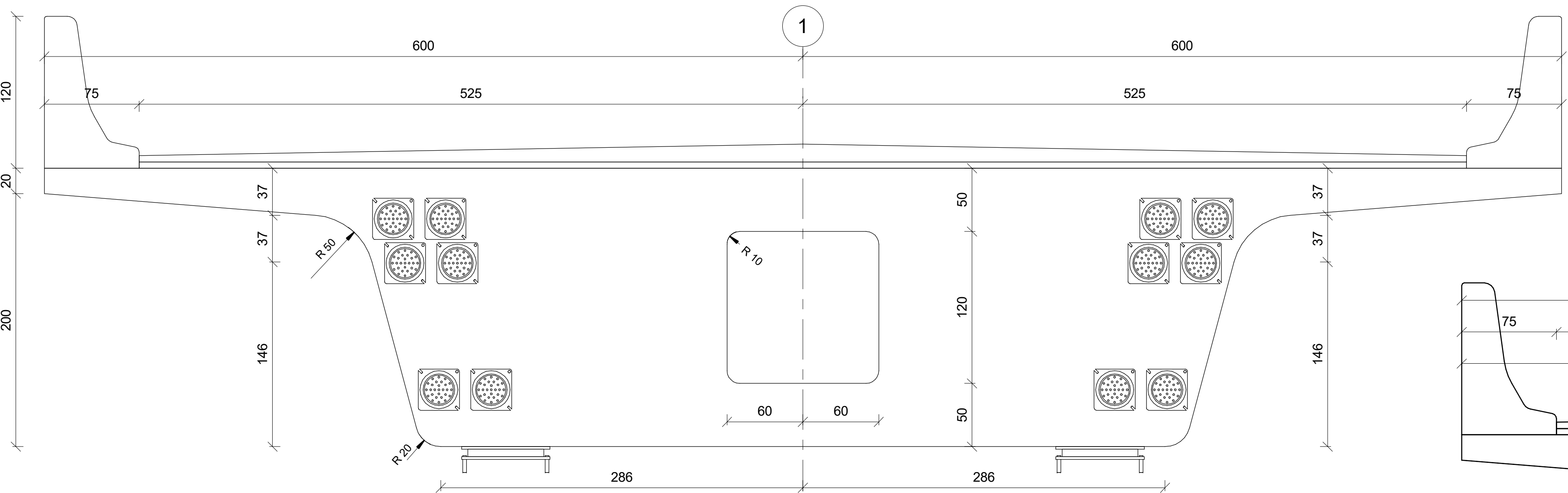
BEARING DEVICE

1 : 20



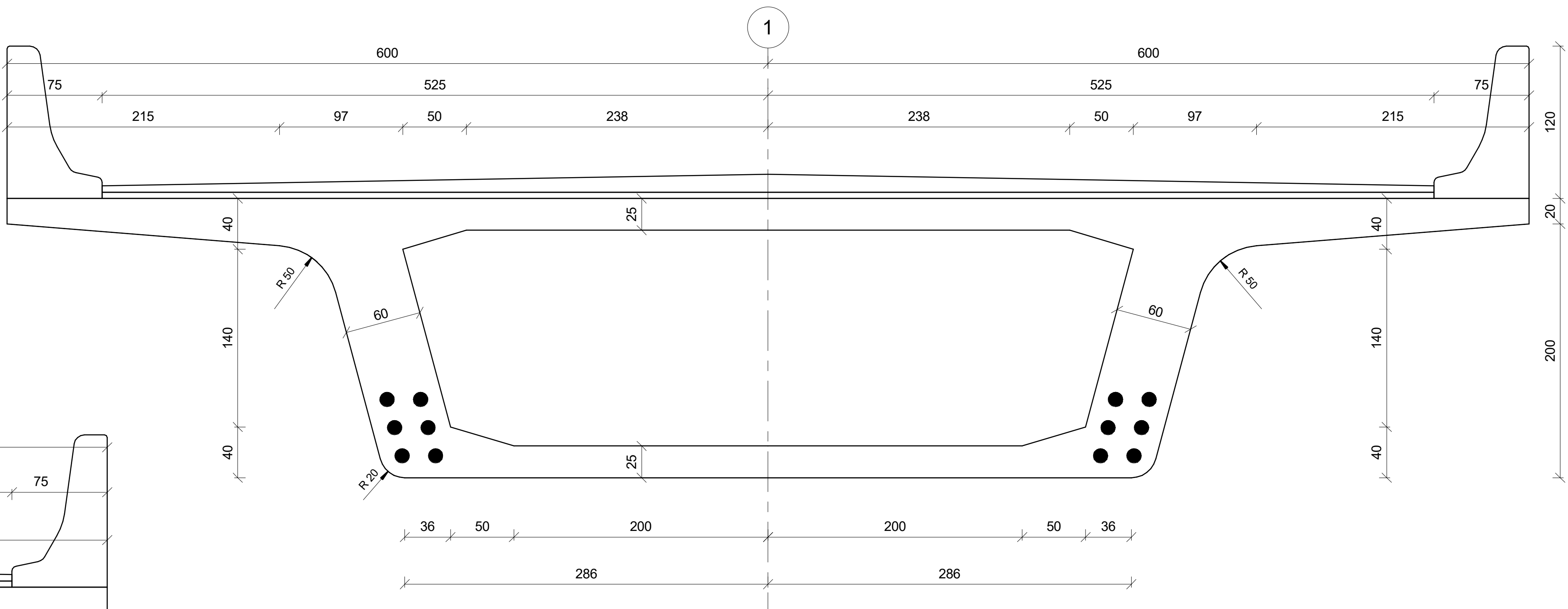
SECTION 1-1

1 : 25



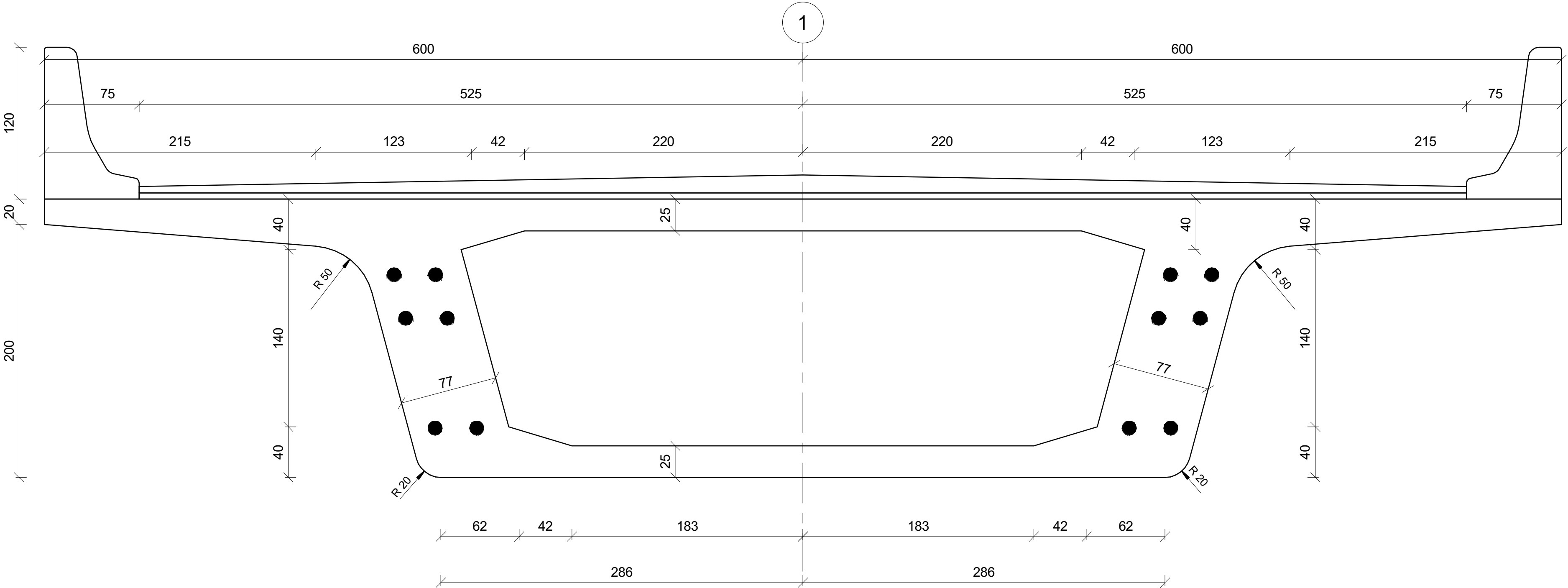
SECTION 3-3

1 : 25



SECTION 2-2

1 : 25



MATERIAL CHARACTERISTICS	
CONCRETE	
Strength class	C45/55, Min. strength: f _{ck} = 45 MPa
Min. cover, c (mm)	45
Exposure class	XD3
PRESTRESSING CABLES	
Type	Tensa, 27 strands, Strand diameter = 15,7 mm
Duct size, d (mm)	115
Ultimate strength, f _{pk} (MPa)	1860
0.1% strength, f _{p0.1k} (MPa)	1640
Total pre-stress (kN)	59000 (4917 per tendon)
Elastic modulus, E _p (MPa)	190000
ORDINARY REINFORCEMENT	
Class	B450C
Min. yield strength, f _{yk} (MPa)	450
PRESTRESSING DEVICES	
Type	Tensa, MTAI System

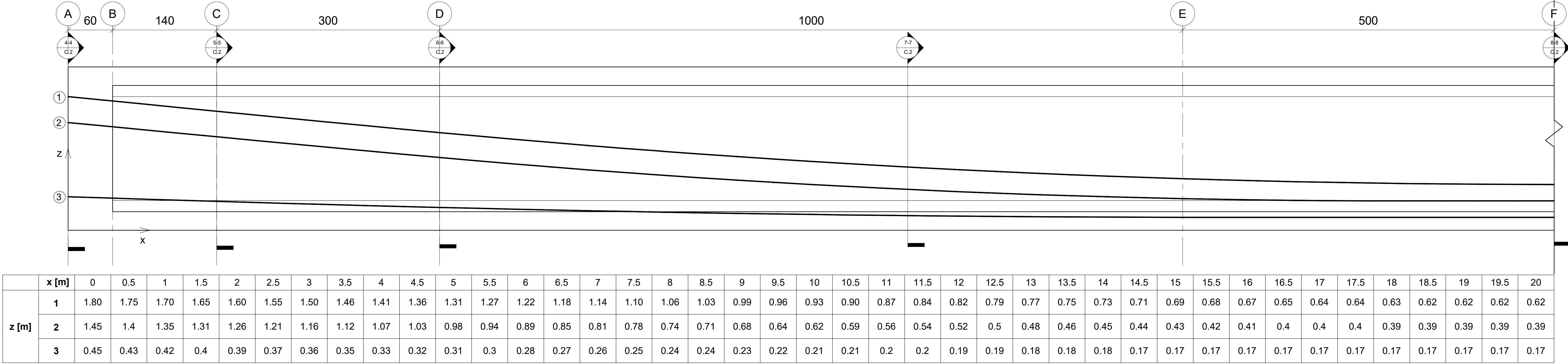
NOTE

All dimensions are given in [cm].
All rebar spacings are given in [mm].

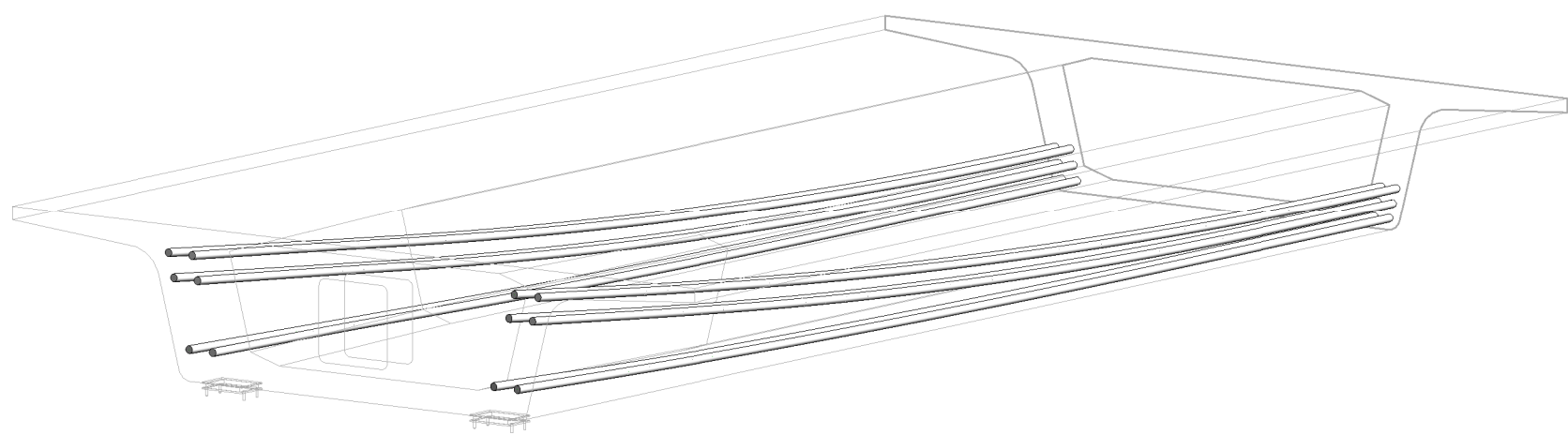
Sponsor				
Program				
SCHOOL OF CIVIL, ENVIRONMENTAL, AND LAND MANAGEMENT ENGINEERING GRADUATE PROGRAM IN CIVIL ENGINEERING A.A. 2023-2024				
Course				
BRIDGE THEORY AND DESIGN				
Project				
DESIGN OF A PRESTRESSED BOX GIRDER BRIDGE				
Title		Sheet	Scale	Size
GENERAL DRAWINGS		1	1:100	A0
Teacher		Team:		
Prof. Ing. Fabio Biondini		# 19		
Assistant		1 - Alexander Nariswari, 216341 2 - Advait Chhatman Sathie, 221727 3 - Dhruvi Wadhwa, 222359 4 - Haseeb Ali, 222634 5 - Nisa Nur Sahin, 216433		
File		general_drawings.dwg		
Notes				
Update	Date	Drawn by	Checked by	Status
Descriptions			Approved by	

CABLE ELEVATION

1 : 25

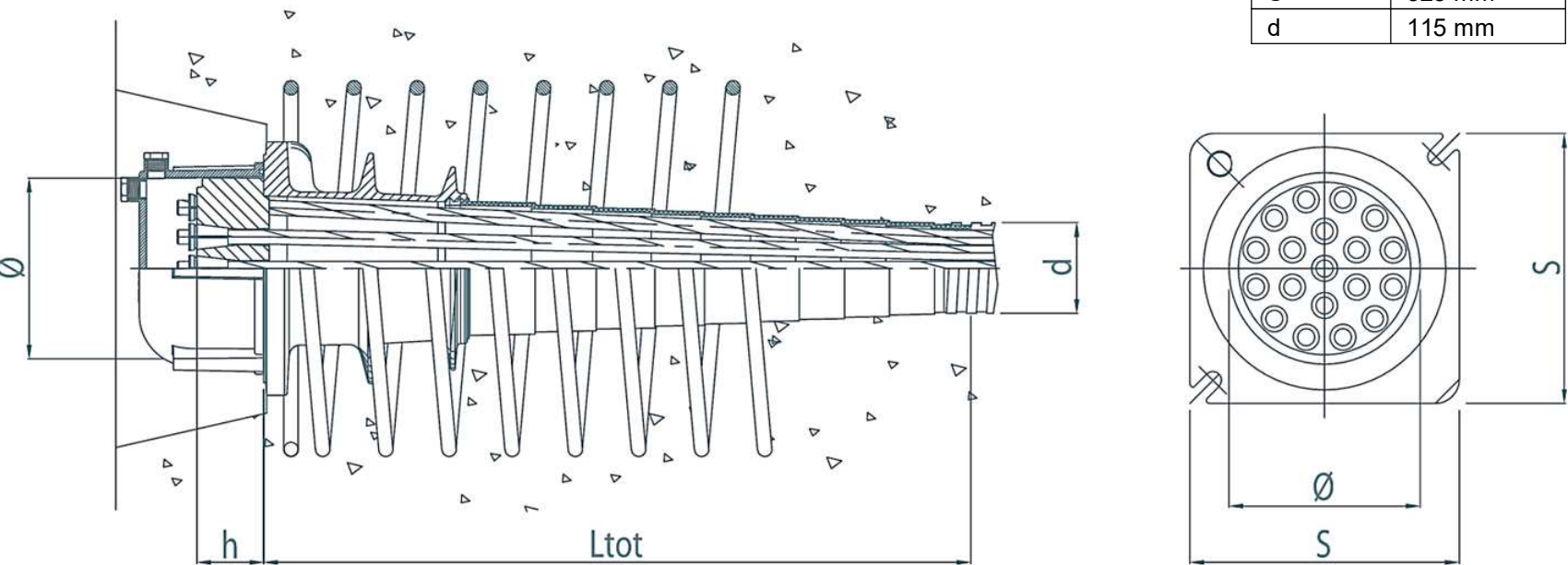


CABLE ISOMETRY



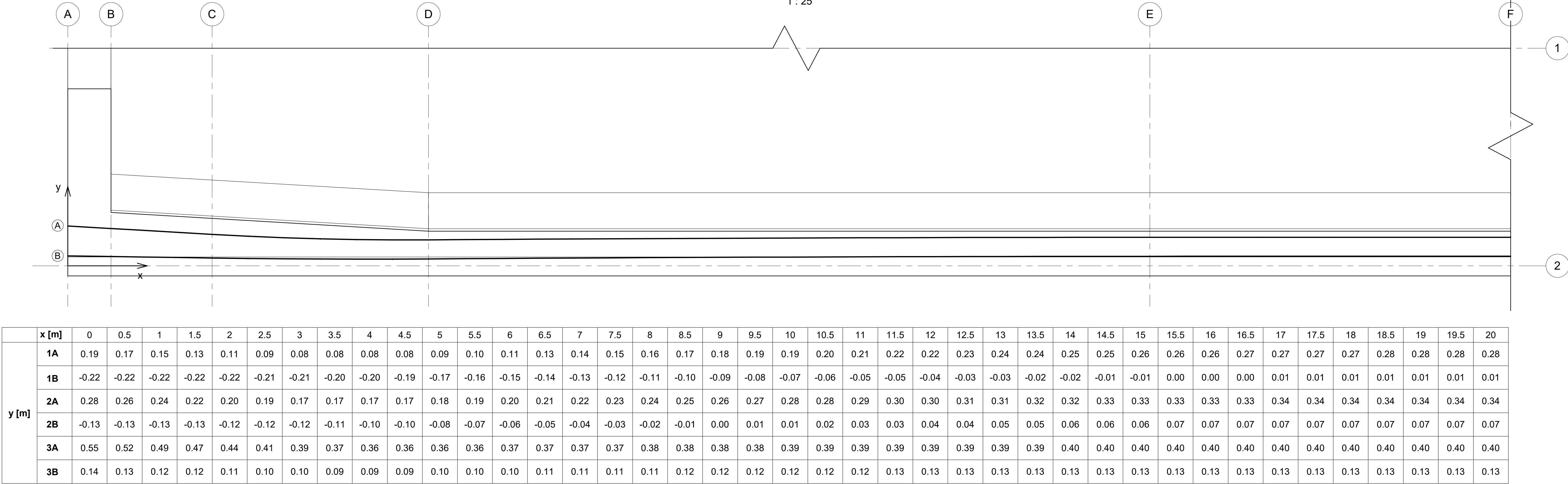
ANCHORAGE DEVICE

1 : 10



CABLE PLAN

1 : 25

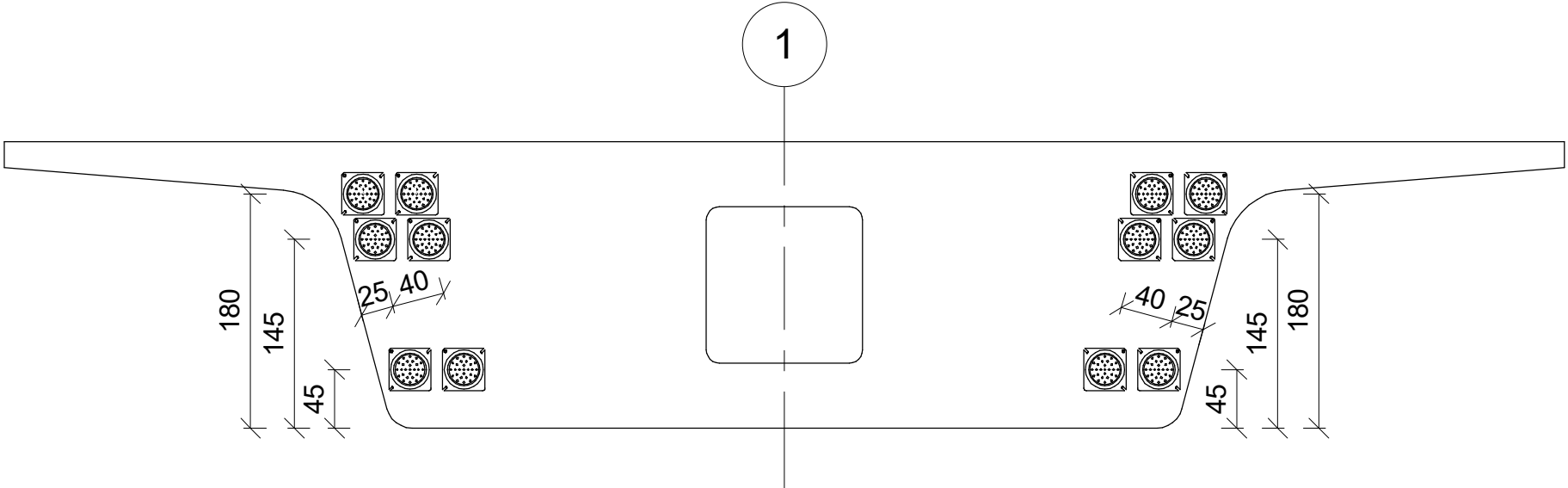


MATERIAL CHARACTERISTICS	
CONCRETE	
Strength class	C45/55, Min. strength: f _{ck} = 45 MPa
Min. cover, c (mm)	45
Exposure class	XD3
PRESTRESSING CABLES	
Type	Tensa, 27 strands, Strand diameter = 15,7 mm
Duct size, d (mm)	115
Ultimate strength, f _{pk} (MPa)	1860
0.1% strength, f _{p0.1k} (MPa)	1640
Total pre-stress (kN)	59000 (4917 per tendon)
Elastic modulus, E _p (MPa)	190000
ORDINARY REINFORCEMENT	
Class	B450C
Min. yield strength, f _{yk} (MPa)	450
PRESTRESSING DEVICES	
Type	Tensa, MTAI System

NOTE
All dimensions are given in [cm].
All rebar spacings are given in [mm].

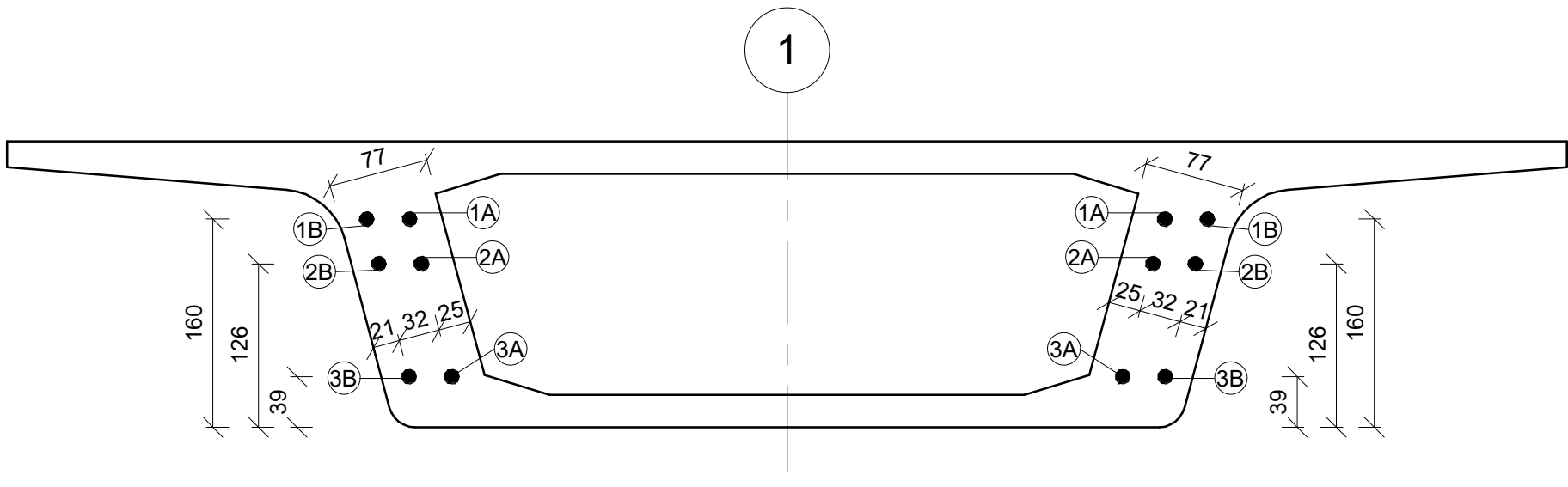
SECTION 4-4

1 : 50



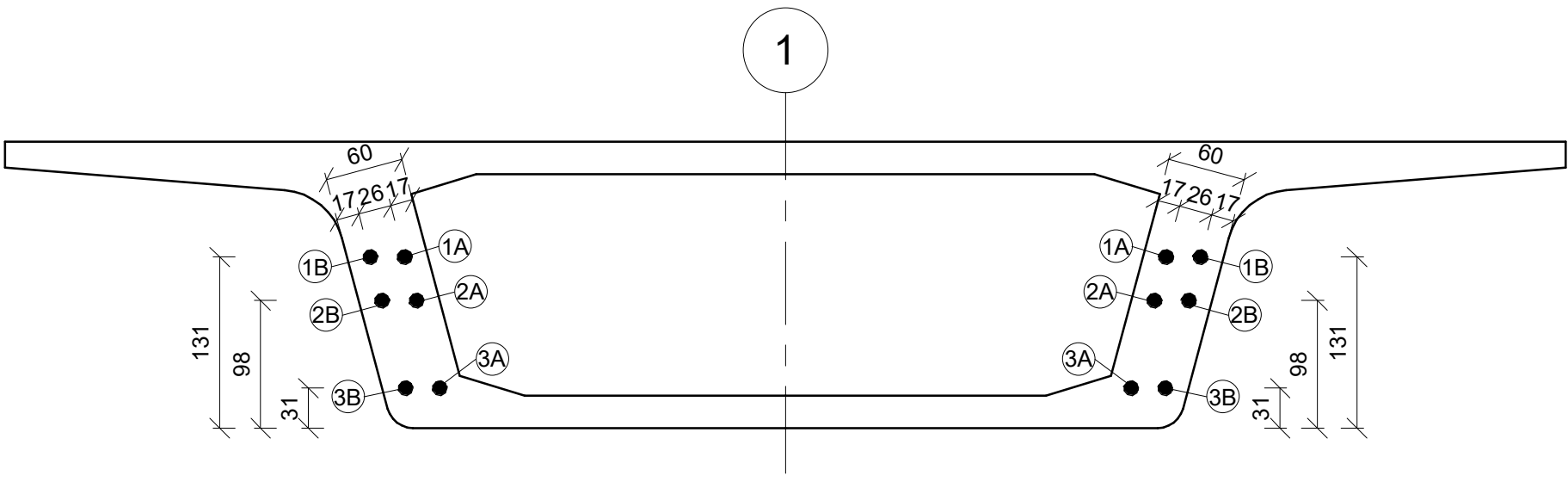
SECTION 5-5

1 : 50



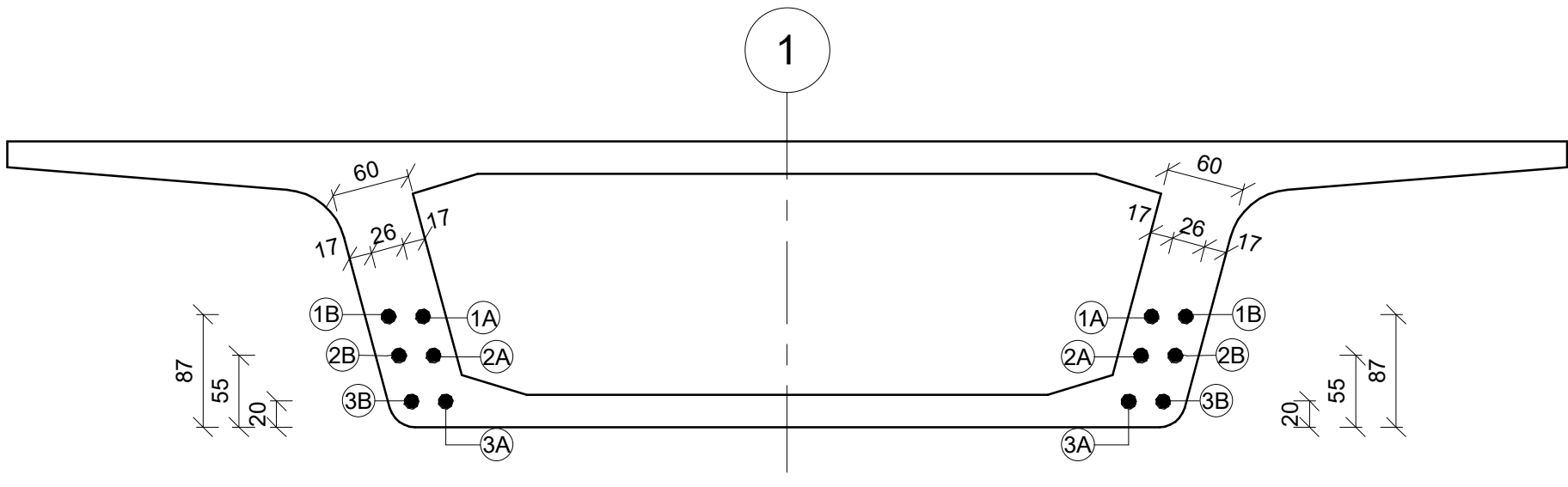
SECTION 6-6

1 : 50



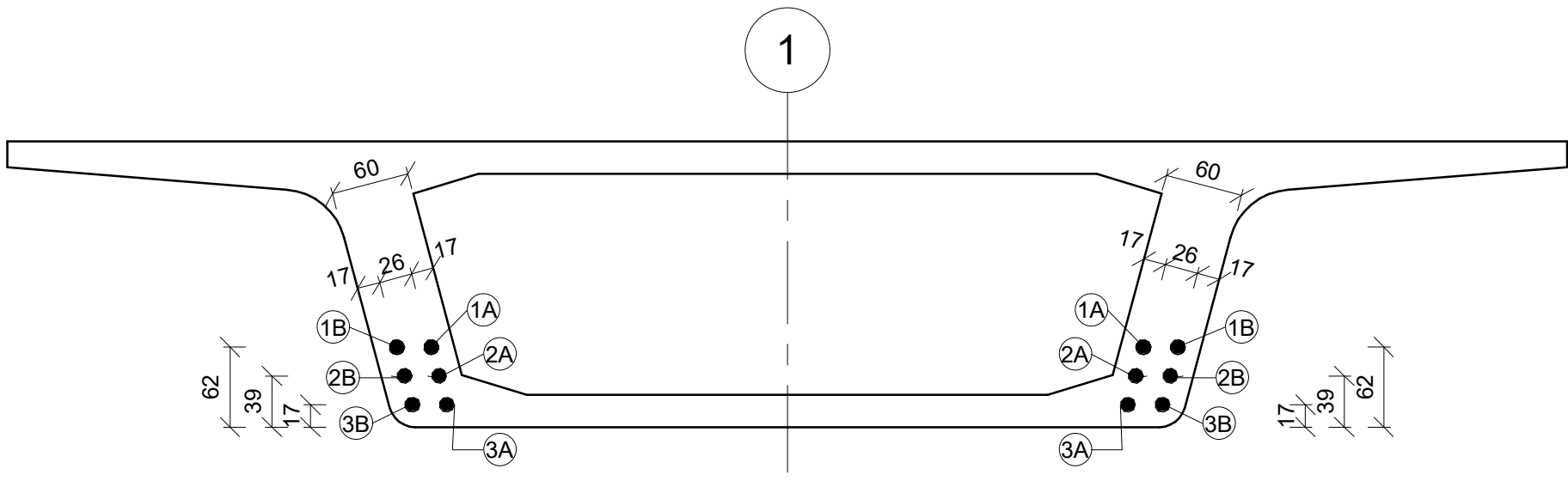
SECTION 7-7

1 : 50



SECTION 8-8

1 : 50



Sponsor				
Program				
SCHOOL OF CIVIL, ENVIRONMENTAL, AND LAND MANAGEMENT ENGINEERING				
GRADUATE PROGRAM IN CIVIL ENGINEERING				
A.A. 2023-2024				
Course				
BRIDGE THEORY AND DESIGN				
Project				
DESIGN OF A PRESTRESSED BOX GIRDER BRIDGE				
Title		Sheet	Scale	Size
CABLE LAYOUT		2	1:100	A0
Teacher	Prof. Ing. Fabio Biondini	Team:		
		# 19		
Assistant	1 - Alexander Nariswari, 216341 2 - Adhikar Chaitanya Sathya, 221727 3 - Dhruvi Wadhwa, 222359 4 - Haseeb Ali, 222634 5 - Nisa Nur Sahin, 216433			
File	general_drawings.dwg			
Notes				
Update	Date	Drawn by	Checked by	Status
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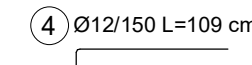
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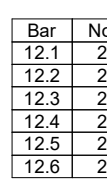
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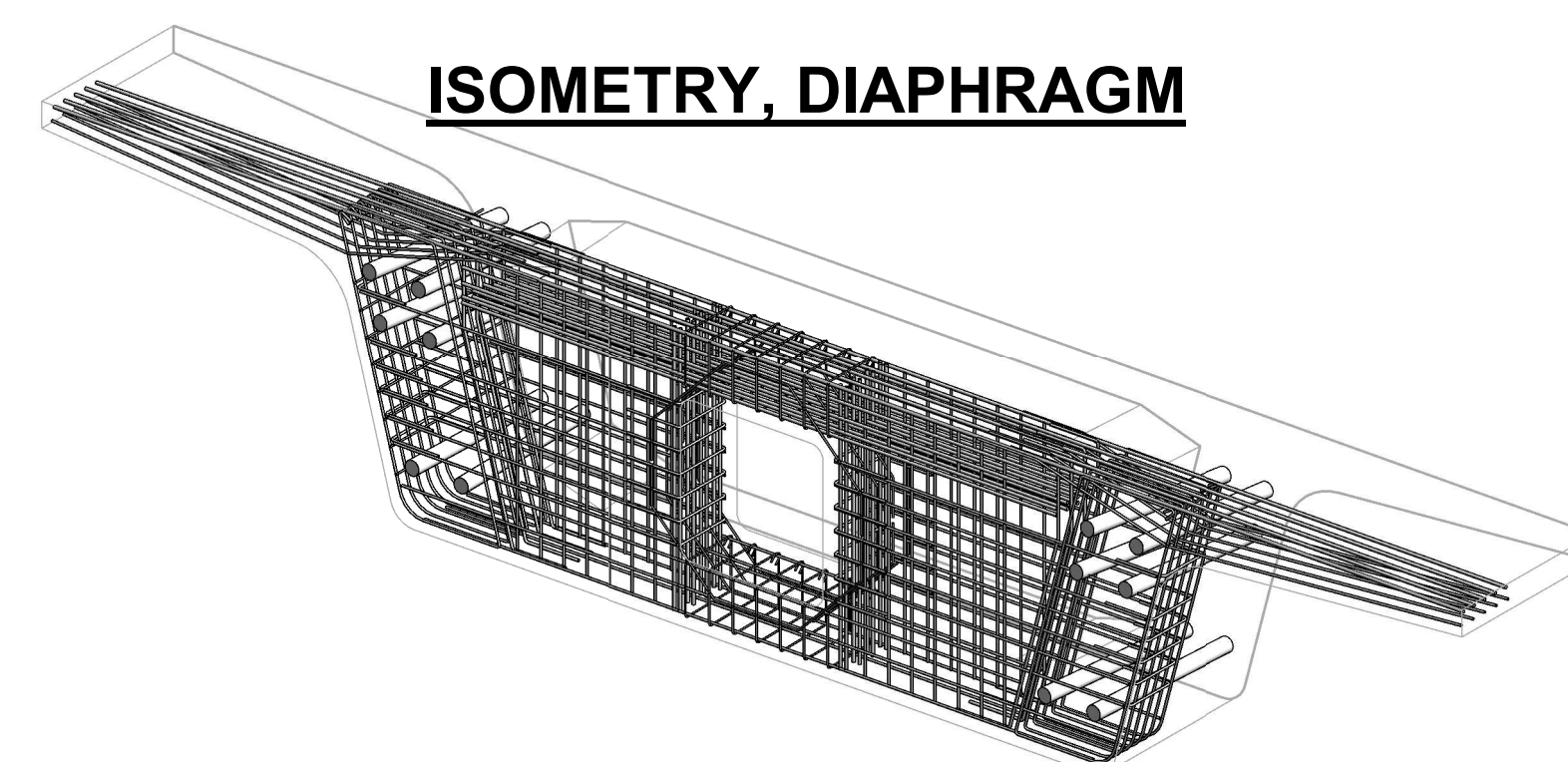
1 : 25



1 : 25



ISOMETRY, SLABS



NOTE

All dimensions are given in [cm].
All rebar spacings are given in [mm].

Sponsor

Program	SCHOOL OF CIVIL, ENVIRONMENTAL, AND LAND MANAGEMENT ENGINEERING GRADUATE PROGRAM IN CIVIL ENGINEERING A.A. 2023-2024
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BRIDGE THEORY AND DESIGN

DESIGN OF A PRESTRESSED BOX GIRDER BRIDGE

Title ORDINARY REINFORCEMENT		Sheet 3	Scale 1:100	Size A0
Teacher Prof. Ing. Fabio Biondini		Team: # 19		
Assistant		1 - Alexander Nariiswar, 216341 2 - Adwait Chintaman Sathe, 221727 3 - Dhruv Wadhwa, 222359 4 - Haseeb Ali, 222834 5 - Nise Nur Sahin, 216433		
File general_drawings.dwg				

Notes						
Update	Date	Drawn by	Checked by	Status	Descriptions	Approved by